

Project Data Page:



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Boiler
(A150NRJ10EQ001)

DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\Boiler.pvdb

Date of Analysis : Sep 17,2026 1:50pm

PV Elite 28 U2, April 2026

Vessel Design Summary:

ASME Code, Section VIII Division 2, 2025

Diameter Spec : 2508.000 x 2300.000 x 2000.000 mm. OD
Vessel Design Length, Tangent to Tangent 10600.00 mm.

Distance of Bottom Tangent above Grade 7500.00 mm.
Distance of Base above Grade 7500.00 mm.
Specified Datum Line Distance -50.00 mm.

Shell Side Design Temperature 286 °C
Channel Side Design Temperature 286 °C

Shell Side Design Pressure 51.000 bars
Channel Side Design Pressure 191.000 bars

Shop Shell Side Test Pressure 27.380 bars
Shop Channel Side Test Pressure 27.197 bars

Wind Design Code ASCE/SEI 7-22
Earthquake Design Code ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-387 11	2	...	K11789	
Head	SA-387 11	2	...	K11789	
Cone	SA-387 11	2	...	K11789	
Nozzle	SA-336 F11	3	...	K11572	
Nozzle Flg	SA-336 F11	3	...	K11572	
Tubes	SA-213 T12	...	...	K11562	
Tubesheet	SA-336 F11	3	...	K11572	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Dish bottom	191.000	1.03	194.000	1.5000	N/A	No
channel shell	191.000	1.03	193.000	1.5000	N/A	No
shell 1	51.000	1.03	53.000	1.5000	N/A	No
shell 2	51.000	1.03	53.000	1.5000	N/A	No
Conical	51.000	1.03	53.000	1.5000	N/A	No
shell	51.000	1.03	55.000	1.5000	N/A	No

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Sphere	50.0	0.0	82.0	75.0	73.9	...	1.00	1.00
Cylinder	1550.0	1500.0	140.0	140.0	138.3	...	1.00	1.00
Cylinder	4650.0	3000.0	38.0	38.0	36.4	...	1.00	1.00
Cylinder	7650.0	3000.0	38.0	38.0	36.4	...	1.00	1.00
Conical	9150.0	1500.0	38.0	38.0	36.6	...	1.00	1.00
Cylinder	10650.0	1500.0	34.0	34.0	31.9	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	75	4.99976	1.03419	124.792
20	Tsht	140	8.38269	1.03419	170.278
Tsht	30	140	No Calc	1.03419	No Calc
30	40	38	12.0743	1.03419	19.3313
40	60	38	12.0743	1.03419	19.3313
60	70	38	7.45212	1.03419	32.6271
70	80	34	6.9598	1.03419	32.6004

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	Tsht	1904.8	25399974	No Calc	No Calc
Tsht	30	No Calc	No Calc	No Calc	No Calc
30	40	6000.03	25399974	No Calc	No Calc
40	60	6000.03	25399974	No Calc	No Calc
60	70	1500	1500	No Calc	No Calc
70	80	1500	25399974	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
B1-24 "	WRC-297	0.085	Shell	Passed

Weights:

Fabricated - Bare w/o removable internals	55349.3 kg.
Shop Test - Fabricated + water (full)	96681.5 kg.
Shipping - Fab. + rem. intls.+ shipping app.	55349.3 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	55349.3 kg.
Empty - Fab. + intls. + details + wghts.	55349.3 kg.
Operating - Empty + operating liquid (no ca)	55351.8 kg.
Field Test - Empty weight + test medium (full)	96681.5 kg.

**Nozzle Calculation Summary (bars & mm.):**

Description	MAWP	Ext	MAPNC	Tbl 4.5.2	Weld	Pass/Fail
M1-24"	152	OK	...	...	OK	Passed
A1-24"	145	OK	...	OK	OK	Passed
B1-24 "	145	OK	...	OK	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 145 Nozzle : A1-24" [Tubeside]

VIII-2 Nozzle to Gross Structural Discontinuity Check:

- *Error: Nozzle A1-24", was too close to a gross structural discontinuity.*
- *Error: Nozzle B1-24 ", was too close to a gross structural discontinuity.*

Please see ASME VIII-2 Section 4.5.13.1 for more information.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm.	Layout Angle deg	Inside Radius mm.
10	M1-24"	0.000	0.000	281.300
20	A1-24"	750.000	90.000	281.300
20	B1-24 "	750.000	270.000	281.300

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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**Nozzle Schedule:**

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
M1-24"	24.000 in	Actual	WNF	24.000	25.000	...	...	429.52	...
A1-24"	24.000 in	Actual	None	24.000	25.000	...	...	499.70	...
B1-24 "	24.000 in	Actual	None	24.000	25.000	...	...	499.70	...

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
M1-24"	SA-336 F11	20.000	10.000	...	...	...
A1-24"	SA-336 F11	38.000	10.000	...	...	...
B1-24 "	SA-336 F11	38.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
M1-24"	...	0.0	300.00	0.00	Dish bottom
A1-24"	800.000	90.0	300.00	0.00	channel shell
B1-24 "	800.000	270.0	300.00	0.00	channel shell

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	SPHERICAL HEAD: 82.0mm. THK X 2429.0mm. ID X 0.0mm.	SA-387 11
1	TUBESHEET: 2508.0mm X 100.0mm THK	SA-336 F11
673	TUBES: 5000.0mm X 30.8mm DIA X 2.5mm THK	SA-213 T12
1	24.0" BLIND FLANGE(S)	SA-336 F11
1	CYLINDRICAL SEGMENT 140.0mm. THK X 2955.3mm. X 1500.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 140.0mm. THK X 4044.1mm. X 1500.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 140.0mm. THK X 0.0mm. X 1500.0mm.	SA-387 11
2	CYLINDRICAL SEGMENT 38.0mm. THK X 2775.3mm. X 3000.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 38.0mm. THK X 4230.9mm. X 3000.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 38.0mm. THK X 19.4mm. X 3000.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 38.0mm. THK X 2755.9mm. X 3000.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 38.0mm. THK X 1455.6mm. X 3000.0mm.	SA-387 11
2	CON. CONICAL SEGMENT 38.0mm. THK X 2946.7mm. X 1500.0mm. X 2549.	SA-387 11
1	CON. CONICAL SEGMENT 38.0mm. THK X 1092.1mm. X 1500.0mm. X 944.8	SA-387 11
2	CYLINDRICAL SEGMENT 34.0mm. THK X 2562.7mm. X 1500.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 34.0mm. THK X 944.1mm. X 1500.0mm.	SA-387 11
1	NAMEPLATE	...

**Minimum Design Metal Temperature Results Summary (°C):**

Description	Notes	Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm.	E*
shell 1	[8]	A	25	14	Yes	0.791	38.000	1.00
shell 2	[8]	A	25	14	Yes	0.791	38.000	1.00
Conical	[8]	A	25	14	Yes	0.795	38.000	1.00
shell	[8]	A	23	11	Yes	0.772	34.000	1.00
Tubesheet: SS	[13]	A	25	25	Yes	1.000	38.000	1.00
Warmest MDMT:			25	25				
Dish bottom	[9]	A	-104	-104	Yes	0.070	75.000	1.00
channel shell	[8][17]	Impact Testing reqd			Yes			
M1-24"	[1]	A	-104	-104	Yes	0.101	25.000	1.00
A1-24"	[1]	A	-104	-104	Yes	0.030	80.000	1.00
B1-24 "	[1]	A	-104	-104	Yes	0.030	80.000	1.00
Nozl. Bolting	[22]		-48		No	1.000	64.000	1.00
Tubesheet: CS	[14]	A	42	42	Yes	1.000	100.000	1.00
Warmest MDMT:			42	42				

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	25.0	-10.0	Fail

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.

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**Units used in this Analysis (bars):**

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Boiler

(A150NRJ10EQ001)

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	51	bars
Channel Side Design Pressure	191	bars
Shell Side Design Temperature	286.0	°C
Channel Side Design Temperature	286.0	°C
Radiography, Shell Side	None	
Radiography, Channel Side	None	
Service Type, Shell Side	None	
Service Type, Channel Side	None	
MDMT (CET), Shell Side	-10.0	°C
MDMT (CET), Tube Side	-10.0	°C
User defined MAWP, Shell Side	0	bars
User defined MAWP, Channel Side	0	bars
User defined MAPnc, Shell Side	0	bars
User defined MAPnc, Channel Side	0	bars
User defined Test Pres., Shell Side	0	bars
User defined Test Pres., Channel Side	0	bars

Projection of Nozzle from Vessel Top	0	mm.
Projection of Nozzle from Vessel Bottom	0	mm.
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	N	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	Hydrostatic
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

**Field Pressure Test:**

Type of Pressure Test	Hydrostatic
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	0
Base Elevation	7500 mm.
Percent Wind for Hydrottest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.500
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact. [CAR]	1.000
Component strength factor [Rpo]	1.500
Overstrength factor [Omega0]	2.000
Response Acceleration Factor [SMS]	0.326
Response Acceleration Factor [SM1]	0.139
Building Importance Factor [Ie]	1.000
Site Latitude	0.000 deg
Site Longitude	0.000 deg

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N



Perform Blast Load Analysis No
Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes
Using Metric Material Databases, ASME II D Yes
Using ASME VIII-1 Appendix 46 No
Calculate B31.3 type stress for Nozzles with Loads Yes
Reduce the MDMT due to lower membrane stress Yes
Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/6	
Element From Node	10	
Element To Node	20	
Element Type	Spherical	
Description	Dish bottom	
Distance "FROM" to "TO"	0	mm.
Element Outside Diameter	2579	mm.
Element Thickness	75	mm.
Internal Corrosion Allowance	1.5	mm.
Nominal Thickness	82	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	191	bars
Design Temperature Internal Pressure	286	°C
Design External Pressure	1.0342	bars
Design Temperature External Pressure	286	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	
Allowable Stress, Ambient	207	N./mm ²
Allowable Stress, Operating	165.12	N./mm ²
Allowable Stress, Hydrotest	294.49	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	0	mm.
Yield Stress, Operating	247.68	N./mm ²
MDMT Curve	A	
External Pressure Chart Name	CS-3	
UNS Number	K11789	
Class / Thickness / Grade	2 ::	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Nozzle	
Detail ID	M1-24"	
Dist. from "FROM" Node / Offset dist	0	mm.
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F11	
Element From Node	10	
Detail Type	Weight	
Detail ID	SF	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	6890	Kgf
Offset from Element Centerline	0	mm.



Element#	2/6
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	channel shell
Distance "FROM" to "TO"	1500 mm.
Element Outside Diameter	2508 mm.
Element Thickness	140 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	140 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	191 bars
Design Temperature Internal Pressure	286 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	286 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Nozzle
Detail ID	A1-24"
Dist. from "FROM" Node / Offset dist	750 mm.
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	None
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	None
Nozzle Matl	SA-336 F11

Element From Node	20
Detail Type	Nozzle
Detail ID	B1-24 "
Dist. from "FROM" Node / Offset dist	750 mm.
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	None
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	None
Nozzle Matl	SA-336 F11

Element#	3/6
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	shell 1
Distance "FROM" to "TO"	3000 mm.
Element Outside Diameter	2300 mm.
Element Thickness	38 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	38 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	51 bars
Design Temperature Internal Pressure	286 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	286 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11

Boiler

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FileName : Boiler

Input Echo:

Step: 1 1:50pm Sep 17,2026

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Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element#	4/6
Element From Node	40
Element To Node	60
Element Type	Cylinder
Description	shell 2
Distance "FROM" to "TO"	3000 mm.
Element Outside Diameter	2300 mm.
Element Thickness	38 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	38 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	51 bars
Design Temperature Internal Pressure	286 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	286 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element#	5/6
Element From Node	60
Element To Node	70
Element Type	Conical
Description	Conical
Distance "FROM" to "TO"	1500 mm.
Element Outside Diameter	2300 mm.
Element Thickness	38 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	38 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	51 bars
Design Temperature Internal Pressure	286 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	286 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Cone Diameter at "To" End	2000 mm.
Design Length of Cone	1500 mm.
Half Apex Angle of Cone	5.7105947 degrees
Toriconical (Y/N)	N
Weld is pre-Heated	No

Element#	6/6
Element From Node	70
Element To Node	80
Element Type	Cylinder
Description	shell
Distance "FROM" to "TO"	1500 mm.
Element Outside Diameter	2000 mm.
Element Thickness	34 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	34 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	51 bars

Boiler

(A150NRJ10EQ001)

PV Elite 28 U2

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v28

Licensee: SL User

FileName : Boiler



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Input Echo:

Step: 1 1:50pm Sep 17,2026

Design Temperature Internal Pressure	286	°C
Design External Pressure	1.0342	bars
Design Temperature External Pressure	286	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

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XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
Dish bottom		...	...	...	...
channel shell		...	1500	...	1500
shell 1		...	4600	...	3000
shell 2		...	7600	...	3000
Conical		...	9100	...	1500
shell		...	10600	...	1500

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
Dish bottom		191	82	1.5	2429	165.12	1
channel shell		191	140	1.5	2228	165.12	2
shell 1		51	38	1.5	2224	165.12	3
shell 2		51	38	1.5	2224	165.12	4
Conical		51	38	1.5	2223.6	165.12	5
shell		51	34	1.5	1932	165.12	6

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
Dish bottom		191	193.798	248.058	75	73.9078	1
channel shell		191	193.237	245.03	140	138.302	2
shell 1		51	53.254	69.5503	38	36.4312	3
shell 2		51	53.254	69.5503	38	36.4312	4
Conical		51	53.0028	69.2225	38	36.5995	5
shell		51	54.5517	71.5988	34	31.8511	6

Summary of Heat Exchanger Maximum Allowable Working Pressures :

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

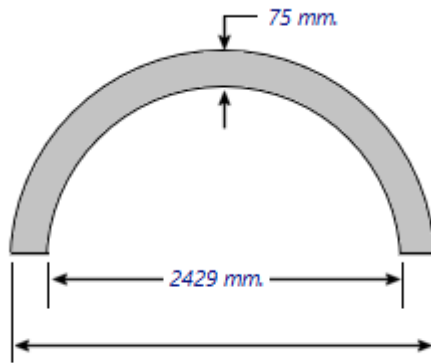
Shell Side MAWP = 17 bars, < design pressure: 51.000 bars
 Shell Side MAPnc = 32 bars
 Channel Side MAWP = 17 bars, < design pressure: 191.000 bars
 Channel Side MAPnc = 32 bars

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Dish bottom

Spherical Head Design Information:					
Design Pressure	P	191.000 bars	Int. Design Temperature		286 °C
Material		SA-387 11	Ext. Design Temperature		286 °C
Length		0.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	165.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	2429.000 mm.	Actual Stress		153.4 N./mm ²
Spec. Min. Thk t,e, or sa		75.0000 mm.	Spec. Nominal Thk.		82.000 mm.
Surface Area		105614.9 cm ²	SG of Contents		0.000
Element Volume		3752571.5 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		6296.7 Kgf	Operating Weight		6296.7 Kgf

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 30.362 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.181 mm.

Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:

$$= D/2(\exp(0.5P/(S \cdot E)) - 1) + ci + co$$

$$= 2432.0/2(\exp(0.5 \cdot 191.0/(165.119 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 72.4078 + 1.5 + 0.0 = 73.9078 \text{ mm.}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 2 \cdot 165.1 \cdot 1.0 \cdot \ln((2 \cdot 73.5 + 2432.0) / 2432.0) - 0.0$$

$$= 193.798 - 0.0 = 193.7977 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 2 \cdot 206.996 \cdot 1.0 \cdot \ln((2 \cdot 75.0 + 2429.0) / 2429.0)$$

$$= 248.0583 \text{ bars}$$



Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D^2 / (E (D_o^2 - D^2))$$

$$= 191.0 \cdot 2432.0^2 / (1.0 (2579.0^2 - 2432.0^2))$$

$$= 153.371 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= 0.5 \cdot P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 0.5 \cdot 191.0 / (1.0 \cdot \ln((2 \cdot 73.5 + 2432.0) / 2432.0))$$

$$= 162.735 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 4.898 %
 For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$ 41.191 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159)$$

$$= \text{square root}(4 \cdot 105614.9 / 3.14159)$$

$$= 3667.056 \text{ mm.}$$

MDMT Calculations in the Spherical Portion:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : -10 °C
 The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.070

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (5.109 \cdot 1.0) / (75.0 - 1.5 - 0.0)$$

$$= 0.070$$

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	17.356 bars
Governing Thk.	75.000 mm.	Min. Des. Mtl. Temp.	-10 °C
Stress/Thk. Ratio	0.070	Basic MDMT	-104 °C
		Temperature Difference	0 °C
		Computed Min. Temp.	-104 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 30.362 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.181 mm.

EXTERNAL PRESSURE RESULTS for: Dish bottom

Spherical Head From 10 to 20 Ext. Chart: CS-2 at 286 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 0.075 \cdot E_y (t / R_o)$$

$$= 0.075 \cdot 0.18730E+09 (73.5 / 1289.5)$$

$$= 800.739 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / S_y$$

$$= 800.7 / 247.7$$

$$= 3.233$$



Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 247.7/517.1$$

$$= 0.479$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 800.7/1.18730E+09$$

$$= 0.004275$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.479) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.160$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 247.7(1 + 0.002) / \ln(1 + 0.002)^{0.160318}$$

$$= 672.232 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.479)$$

$$= 0.313$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (517.1 \cdot \exp(0.313)) / (0.313^{0.313})$$

$$= 1016.803 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4791.5 - 0.5 \cdot 0.4792.5 - 0.4793.5$$

$$= 0.342$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	258	-1.764	0.311E-05	0.262E-05	0.775E-05	-0.580E-05	0.122E+08	0.170E-01
2	129	-4.571	0.825E-07	0.825E-07	0.169E-05	-0.169E-05	0.143E+09	0.337E-02
3	193	-3.167	0.690E-06	0.684E-06	0.412E-05	-0.406E-05	0.510E+08	0.473E-03
4	226	-2.466	0.155E-05	0.149E-05	0.578E-05	-0.544E-05	0.256E+08	0.456E-02
5	210	-2.816	0.105E-05	0.103E-05	0.491E-05	-0.477E-05	0.364E+08	0.150E-02
6	202	-2.992	0.854E-06	0.844E-06	0.450E-05	-0.442E-05	0.432E+08	0.405E-03
7	197	-3.080	0.769E-06	0.761E-06	0.431E-05	-0.424E-05	0.469E+08	0.583E-04
8	199	-3.036	0.810E-06	0.802E-06	0.440E-05	-0.433E-05	0.450E+08	0.167E-03
9	198	-3.058	0.789E-06	0.781E-06	0.436E-05	-0.428E-05	0.460E+08	0.527E-04
10	198	-3.069	0.779E-06	0.771E-06	0.433E-05	-0.426E-05	0.465E+08	0.316E-05
11	198	-3.063	0.784E-06	0.776E-06	0.434E-05	-0.427E-05	0.462E+08	0.247E-04
12	198	-3.066	0.781E-06	0.773E-06	0.434E-05	-0.427E-05	0.463E+08	0.107E-04
13	198	-3.067	0.780E-06	0.772E-06	0.433E-05	-0.427E-05	0.464E+08	0.378E-05
14	198	-3.068	0.779E-06	0.771E-06	0.433E-05	-0.426E-05	0.464E+08	0.307E-06
15	198	-3.068	0.779E-06	0.771E-06	0.433E-05	-0.426E-05	0.464E+08	0.143E-05
16	198	-3.068	0.779E-06	0.771E-06	0.433E-05	-0.426E-05	0.464E+08	0.561E-06
17	198	-3.068	0.779E-06	0.771E-06	0.433E-05	-0.426E-05	0.464E+08	0.127E-06

Factor of Safety [FS]:

Since $Fic > 0.55 F_y$ and $Fic < F_y$

$$= 2.407 - 0.741 \cdot Fic / F_y$$

$$= 2.407 - 0.741 \cdot 198.49/247.68$$

$$= 1.813$$

Allowable Circumferential Compressive Stress [Fha]:

$$= Fic/FS$$

$$= 198.495/1.813$$

Boiler

(A150NRJ10EQ001)

PV Elite 28 U2

Licensee: SL User

FileName : Boiler

Element: Dish bottom

PV Elite

v28



HEXAGON

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Step: 4 1:50pm Sep 17, 2026

= 109.475 N./mm²

Allowable External Pressure at the given Thickness [Pa]:

= 2 • Fha[t/Ro]

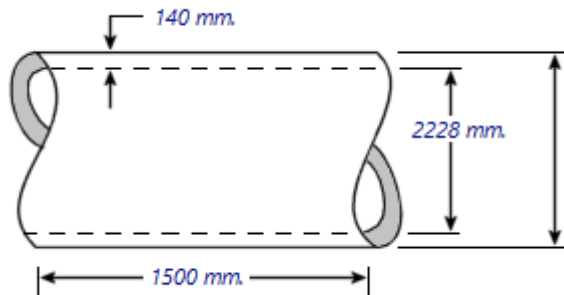
= 2 • 109.475[73.5/1289.5]

= 124.792 bars

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: channel shell

Cylindrical Shell Design Information:					
Design Pressure	P	191.000 bars	Int. Design Temperature	286 °C	
Material		SA-387 11	Ext. Design Temperature	286 °C	
Length		1500.000 mm.	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	165.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	2228.000 mm.	Actual Stress	153.8 N./mm ²	
Spec. Min. Thk t,e, or sa		140.0000 mm.	Spec. Nominal Thk.	140.000 mm.	
Surface Area		118186.7 cm ²	SG of Contents	0.000	
Element Volume		5849103.5 cm ³	Weight of Contents	0.0 Kg	
Empty Weight		12108.2 Kg	Operating Weight	12108.2 Kg	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2231.0 \cdot (\exp(191.0/(165.119 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 136.8023 + 1.5 + 0.0 = 138.3023 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 165.119 \cdot 1.0 \cdot \ln((2 \cdot 138.5 + 2231.0) / 2231.0) - 0.0$$

$$= 193.237 - 0.0 = 193.2369 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 206.996 \cdot 1.0 \cdot \ln((2 \cdot 140.0 + 2228.0) / 2228.0)$$

$$= 245.0305 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E \cdot (D_o - D))$$

$$= 191.0 \cdot 2231.0 / (1.0 \cdot (2508.0 - 2231.0))$$

$$= 153.8433 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 191.0 / (1.0 \cdot \ln((2 \cdot 138.5 + 2231.0) / 2231.0))$$



= 163.2075 N./mm²

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 5.912 %

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm..

This component must be impact tested.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	17.356 bars
		Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	140.000 mm.	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
		Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: channel shell

Cylindrical Shell From 20 to Tsht Ext. Chart: CS-2 at 286 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 1904.803 / \sqrt{(1254.0 \cdot 138.5)}$$

$$= 4.571$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 1.5 \text{ and } M_x < 13, Ch = 0.92 / (M_x - 0.579)$$

$$= 0.92 / (4.571 - 0.579)$$

$$= 0.2305$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y(t / D_o)$$

$$= 1.6 \cdot 0.23 \cdot 0.18730E+09(138.5/2508.0)$$

$$= 3814.547 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 3814.5/247.7$$

$$= 15.401$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_y / \sigma_{uts}$$

$$= 247.7/517.1$$

$$= 0.479$$

Ratio [AE]:

$$= Fhe / E$$

$$= 3814.5/0.18730E+09$$

$$= 0.020365$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / (\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.479) + (0.00002 - 0.002)) / (\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.160$$



Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m_1$$

$$= 247.7 (1 + 0.002) / \ln(1 + 0.002) 0.160318$$

$$= 672.232 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m_2 f(m_2^2 - R)$$

$$= 0.6 (1.0 - 0.479)$$

$$= 0.313$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$$

$$= (517.1 \cdot \exp(0.313)) / (0.313 0.313)$$

$$= 1016.803 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R_{1.5} - 0.5 \cdot R_{2.5} - R_{3.5}$$

$$= 1.5 \cdot 0.4791.5 - 0.5 \cdot 0.4792.5 - 0.4793.5$$

$$= 0.342$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	258	-1.764	0.311E-05	0.262E-05	0.775E-05	-0.580E-05	0.122E+08	0.893E-03
2	129	-4.571	0.825E-07	0.825E-07	0.169E-05	-0.169E-05	0.143E+09	0.195E-01
3	193	-3.167	0.690E-06	0.684E-06	0.412E-05	-0.406E-05	0.510E+08	0.166E-01
4	226	-2.466	0.155E-05	0.149E-05	0.578E-05	-0.544E-05	0.256E+08	0.115E-01
5	242	-2.115	0.222E-05	0.205E-05	0.672E-05	-0.590E-05	0.178E+08	0.673E-02
6	250	-1.939	0.264E-05	0.235E-05	0.723E-05	-0.596E-05	0.147E+08	0.337E-02
7	254	-1.851	0.287E-05	0.249E-05	0.749E-05	-0.591E-05	0.134E+08	0.137E-02
8	256	-1.807	0.299E-05	0.256E-05	0.762E-05	-0.586E-05	0.128E+08	0.270E-03
9	257	-1.786	0.305E-05	0.259E-05	0.768E-05	-0.583E-05	0.125E+08	0.303E-03
10	257	-1.797	0.302E-05	0.258E-05	0.765E-05	-0.585E-05	0.126E+08	0.142E-04
11	256	-1.802	0.300E-05	0.257E-05	0.763E-05	-0.586E-05	0.127E+08	0.128E-03
12	256	-1.799	0.301E-05	0.257E-05	0.764E-05	-0.585E-05	0.127E+08	0.573E-04
13	256	-1.798	0.301E-05	0.257E-05	0.765E-05	-0.585E-05	0.126E+08	0.216E-04
14	257	-1.797	0.302E-05	0.257E-05	0.765E-05	-0.585E-05	0.126E+08	0.369E-05
15	257	-1.797	0.302E-05	0.257E-05	0.765E-05	-0.585E-05	0.126E+08	0.525E-05
16	257	-1.797	0.302E-05	0.257E-05	0.765E-05	-0.585E-05	0.126E+08	0.778E-06
17	257	-1.797	0.302E-05	0.257E-05	0.765E-05	-0.585E-05	0.126E+08	0.146E-05

Factor of Safety [FS]:

$$\text{Since } F_{ic} [257.02] \geq F_y [247.68], FS = 1.667$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic} / FS$$

$$= 257.02 / 1.667$$

$$= 154.181 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

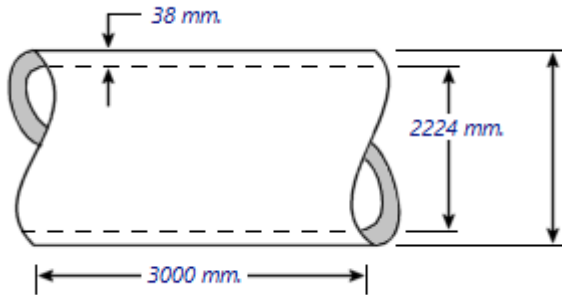
$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 1541.721 (138.5 / 2508.0)$$

$$= 170.278 \text{ bars}$$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: shell 1

Cylindrical Shell Design Information:			
Design Pressure P	51.000 bars	Int. Design Temperature	286 °C
Material	SA-387 11	Ext. Design Temperature	286 °C
Length	3000.000 mm.	External Pressure Chart	CS-3
		UNS Number	K11789
Int. Corr. All. c	1.5000 mm.	Allowable Stress (ope) S	165.1 N./mm ²
Ext. Corr. All. ce	0.0000 mm.	Allowable Stress (amb) Sa	207.0 N./mm ²
Inside Diameter Di	2224.000 mm.	Actual Stress	155.6 N./mm ²
Spec. Min. Thk t,e, or sa	38.0000 mm.	Spec. Nominal Thk.	38.000 mm.
Surface Area	216769.9 cm ²	SG of Contents	0.000
Element Volume	10151696.0 cm ³	Weight of Contents	0.0 Kg
Empty Weight	6278.8 Kg	Operating Weight	6278.8 Kg

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2227.0 \cdot (\exp(51.0/(165.119 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 34.9312 + 1.5 + 0.0 = 36.4312 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 165.119 \cdot 1.0 \cdot \ln((2 \cdot 36.5 + 2227.0) / 2227.0) - 0.0$$

$$= 53.254 - 0.0 = 53.254 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 206.996 \cdot 1.0 \cdot \ln((2 \cdot 38.0 + 2224.0) / 2224.0)$$

$$= 69.5503 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E \cdot (D_o - D))$$

$$= 51.0 \cdot 2227.0 / (1.0 \cdot (2300.0 - 2227.0))$$

$$= 155.5940 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 51.0 / (1.0 \cdot \ln((2 \cdot 36.5 + 2227.0) / 2227.0))$$

= 158.1302 N./mm²

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 1.680 %

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : 14 °C
 User entered Minimum Design Temperature : -10 °C
 The Curve for this Material is : A
 From Figure 3.8M- with PWHT, the MDMT is : 25 °C
 Temperature reduction per Graph 3.13 : 11 °C
 The computed stress ratio is : 0.791

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (28.882 \cdot 1.0) / (38.0 - 1.5 - 0.0)$
 $= 0.791$

Warning: Impact Testing Is Mandatory!

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	53.003 bars
Governing Thk.	38.000 mm.	Min. Des. Mtl. Temp.	-10 °C
Stress/Thk. Ratio	0.791	Basic MDMT	25 °C
		Temperature Difference	11 °C
		Computed Min. Temp.	14 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: shell 1

Cylindrical Shell From 30 to 40 Ext. Chart: CS-2 at 286 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:
 $= L / \sqrt{(R_o \cdot t)}$
 $= 6000.03 / \sqrt{(1150.0 \cdot 36.5)}$
 $= 29.286$

Compute the Factor [Ch]:
 Since $M_x > 13$ and $M_x < 2 \cdot (D_o/t)0.94$, $Ch = 1.12 \cdot M_x - 1.058$
 $= 1.12 \cdot 29.286 - 1.058$
 $= 0.0314$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:
 $= 1.6 \cdot Ch \cdot E_y (t / D_o)$
 $= 1.6 \cdot 0.031 \cdot 0.18730E+09 (36.5/2300.0)$
 $= 149.535 \text{ N./mm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:
 $= Fhe / F_y$
 $= 149.5/247.7$
 $= 0.604$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:
 $= \sigma_{ys} / \sigma_{uts}$
 $= 247.7/517.1$
 $= 0.479$



Ratio [AE]:

$$= F_{he} / E$$

$$= 149.5 / 18730E+09$$

$$= 0.000798$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / (\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.479) + (0.00002 - 0.002)) / (\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.160$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m1$$

$$= 247.7 (1 + 0.002) / \ln(1 + 0.002) 0.160318$$

$$= 672.232 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f / (m2f^2 - R)$$

$$= 0.6 (1.0 - 0.479)$$

$$= 0.313$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2 m2)$$

$$= (517.1 \cdot \exp(0.313)) / (0.313 0.313)$$

$$= 1016.803 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4791.5 - 0.5 \cdot 0.4792.5 - 0.4793.5$$

$$= 0.342$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	258	-1.764	0.311E-05	0.262E-05	0.775E-05	-0.580E-05	0.122E+08	0.205E-01
2	129	-4.571	0.825E-07	0.825E-07	0.169E-05	-0.169E-05	0.143E+09	0.106E-03
3	64	-5.975	0.219E-08	0.219E-08	0.368E-06	-0.368E-06	0.186E+09	0.450E-03
4	96	-5.273	0.183E-07	0.183E-07	0.897E-06	-0.897E-06	0.175E+09	0.245E-03
5	113	-4.922	0.410E-07	0.410E-07	0.126E-05	-0.126E-05	0.162E+09	0.101E-03
6	121	-4.747	0.588E-07	0.588E-07	0.146E-05	-0.146E-05	0.153E+09	0.796E-05
7	125	-4.659	0.699E-07	0.698E-07	0.157E-05	-0.157E-05	0.148E+09	0.462E-04
8	123	-4.703	0.642E-07	0.641E-07	0.152E-05	-0.152E-05	0.151E+09	0.184E-04
9	122	-4.725	0.615E-07	0.614E-07	0.149E-05	-0.149E-05	0.152E+09	0.506E-05
10	121	-4.736	0.601E-07	0.601E-07	0.148E-05	-0.148E-05	0.153E+09	0.149E-05
11	121	-4.730	0.608E-07	0.608E-07	0.148E-05	-0.148E-05	0.152E+09	0.177E-05
12	121	-4.733	0.605E-07	0.604E-07	0.148E-05	-0.148E-05	0.153E+09	0.138E-06
13	121	-4.734	0.603E-07	0.603E-07	0.148E-05	-0.148E-05	0.153E+09	0.678E-06
14	121	-4.734	0.604E-07	0.604E-07	0.148E-05	-0.148E-05	0.153E+09	0.270E-06
15	121	-4.733	0.604E-07	0.604E-07	0.148E-05	-0.148E-05	0.153E+09	0.665E-07
16	121	-4.733	0.604E-07	0.604E-07	0.148E-05	-0.148E-05	0.153E+09	0.356E-07
17	121	-4.733	0.604E-07	0.604E-07	0.148E-05	-0.148E-05	0.153E+09	0.155E-07

Factor of Safety [FS]:

$$\text{Since } F_{ic} [121.82] < 0.55 \cdot 247.68, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic} / FS$$

$$= 121.821 / 2.0$$

$$= 60.910 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

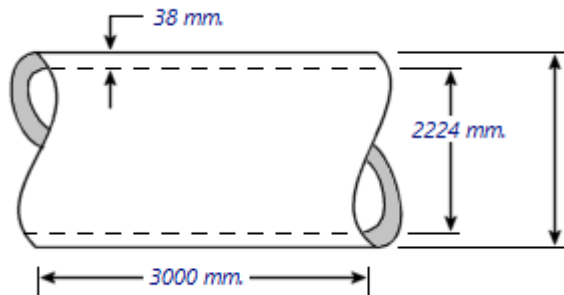
$$= 2 \cdot 609.068 (36.5 / 2300.0)$$

$$= 19.331 \text{ bars}$$



Internal Pressure Calculation Results:

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Internal Pressure Results for: shell 2

Cylindrical Shell Design Information:					
Design Pressure	P	51.000 bars	Int. Design Temperature		286 °C
Material		SA-387 11	Ext. Design Temperature		286 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	165.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	2224.000 mm.	Actual Stress		155.6 N./mm ²
Spec. Min. Thk t,e, or sa		38.0000 mm.	Spec. Nominal Thk.		38.000 mm.
Surface Area		216769.9 cm ²	SG of Contents		0.000
Element Volume		9588988.0 cm ³	Weight of Contents		0.0 Kg
Empty Weight		6278.8 Kg	Operating Weight		6278.8 Kg

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2227.0 \cdot (\exp(51.0/(165.119 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 34.9312 + 1.5 + 0.0 = 36.4312 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 165.119 \cdot 1.0 \cdot \ln((2 \cdot 36.5 + 2227.0) / 2227.0) - 0.0$$

$$= 53.254 - 0.0 = 53.254 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 206.996 \cdot 1.0 \cdot \ln((2 \cdot 38.0 + 2224.0) / 2224.0)$$

$$= 69.5503 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E(D_o - D))$$

$$= 51.0 \cdot 2227.0 / (1.0(2300.0 - 2227.0))$$

$$= 155.5940 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 51.0 / (1.0 \cdot \ln((2 \cdot 36.5 + 2227.0) / 2227.0))$$



$$= 158.1302 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o) \quad 1.680 \%$$

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	14 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A
From Figure 3.8M- with PWHT, the MDMT is	:	25 °C
Temperature reduction per Graph 3.13	:	11 °C
The computed stress ratio is	:	0.791

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (28.882 \cdot 1.0) / (38.0 - 1.5 - 0.0)$$

$$= 0.791$$

Warning: Impact Testing Is Mandatory!

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	53.003 bars
Governing Thk.	38.000 mm.	Min. Des. Mtl. Temp.	-10 °C
Stress/Thk. Ratio	0.791	Basic MDMT	25 °C
		Temperature Difference	11 °C
		Computed Min. Temp.	14 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: shell 2

Cylindrical Shell From 40 to 60 Ext. Chart: CS-2 at 286 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 6000.031 / \sqrt{(1150.0 \cdot 36.5)}$$

$$= 29.286$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 13 \text{ and } M_x < 2 \cdot (D_o/t)0.94, \text{ Ch} = 1.12 \cdot M_x - 1.058$$

$$= 1.12 \cdot 29.286 - 1.058$$

$$= 0.0314$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.031 \cdot 0.18730E+09 (36.5/2300.0)$$

$$= 149.535 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$

$$= 149.5/247.7$$

$$= 0.604$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 247.7/517.1$$

$$= 0.479$$



Ratio [AE]:

$$= F_{he} / E$$

$$= 149.5 / 18730E+09$$

$$= 0.000798$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / (\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.479) + (0.00002 - 0.002)) / (\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.160$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m1$$

$$= 247.7 (1 + 0.002) / \ln(1 + 0.002) 0.160318$$

$$= 672.232 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f / (m2f^2 - R)$$

$$= 0.6 (1.0 - 0.479)$$

$$= 0.313$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2 m2)$$

$$= (517.1 \cdot \exp(0.313)) / (0.313 0.313)$$

$$= 1016.803 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4791.5 - 0.5 \cdot 0.4792.5 - 0.4793.5$$

$$= 0.342$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	258	-1.764	0.311E-05	0.262E-05	0.775E-05	-0.580E-05	0.122E+08	0.205E-01
2	129	-4.571	0.825E-07	0.825E-07	0.169E-05	-0.169E-05	0.143E+09	0.106E-03
3	64	-5.975	0.219E-08	0.219E-08	0.368E-06	-0.368E-06	0.186E+09	0.450E-03
4	96	-5.273	0.183E-07	0.183E-07	0.897E-06	-0.897E-06	0.175E+09	0.245E-03
5	113	-4.922	0.410E-07	0.410E-07	0.126E-05	-0.126E-05	0.162E+09	0.101E-03
6	121	-4.747	0.588E-07	0.588E-07	0.146E-05	-0.146E-05	0.153E+09	0.796E-05
7	125	-4.659	0.699E-07	0.698E-07	0.157E-05	-0.157E-05	0.148E+09	0.462E-04
8	123	-4.703	0.642E-07	0.641E-07	0.152E-05	-0.152E-05	0.151E+09	0.184E-04
9	122	-4.725	0.615E-07	0.614E-07	0.149E-05	-0.149E-05	0.152E+09	0.506E-05
10	121	-4.736	0.601E-07	0.601E-07	0.148E-05	-0.148E-05	0.153E+09	0.149E-05
11	121	-4.730	0.608E-07	0.608E-07	0.148E-05	-0.148E-05	0.152E+09	0.177E-05
12	121	-4.733	0.605E-07	0.604E-07	0.148E-05	-0.148E-05	0.153E+09	0.138E-06
13	121	-4.734	0.603E-07	0.603E-07	0.148E-05	-0.148E-05	0.153E+09	0.678E-06
14	121	-4.734	0.604E-07	0.604E-07	0.148E-05	-0.148E-05	0.153E+09	0.270E-06
15	121	-4.733	0.604E-07	0.604E-07	0.148E-05	-0.148E-05	0.153E+09	0.664E-07
16	121	-4.733	0.604E-07	0.604E-07	0.148E-05	-0.148E-05	0.153E+09	0.356E-07
17	121	-4.733	0.604E-07	0.604E-07	0.148E-05	-0.148E-05	0.153E+09	0.154E-07

Factor of Safety [FS]:

$$\text{Since } F_{ic} [121.82] < 0.55 \cdot 247.68, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic} / FS$$

$$= 121.821 / 2.0$$

$$= 60.910 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

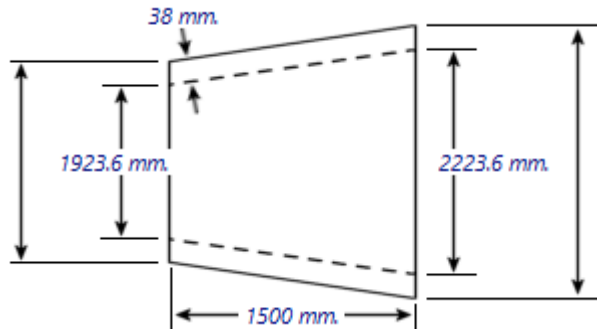
$$= 2 \cdot 609.068 (36.5 / 2300.0)$$

$$= 19.331 \text{ bars}$$



Internal Pressure Calculation Results:

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Internal Pressure Results for: Conical

Conical Head/Section Design Information:					
Design Pressure	P	51.000 bars	Int. Design Temperature		286 °C
Material		SA-387 11	Ext. Design Temperature		286 °C
Length		1500.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	165.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	2223.621 mm.	Actual Stress		156.3 N./mm ²
Spec. Min. Thk t,e, or sa		38.0000 mm.	Spec. Nominal Thk.		38.000 mm.
Surface Area		101821.7 cm ²	SG of Contents		0.000
Element Volume		5075449.5 cm ³	Weight of Contents		0.0 Kg
Empty Weight		2945.6 Kg	Operating Weight		2945.6 Kg
			To End Diameter		1923.621 mm.

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Thickness due to Internal Pressure: Equation 4.3.4.1 [tr]:

$$= 0.5 \cdot D_c / \cos(\alpha) \cdot (\exp(P / (S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2226.621 / \cos(5.711) \cdot (\exp(51.0 / (165.119 \cdot 1.0)) - 1) + c_i + c_o$$

$$= 35.0995 + 1.5 + 0.0 = 36.5995 \text{ mm.}$$

Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln(2 \cdot t_{cor} \cdot \cos(\alpha) + D_{cor}) / D_{cor}$$

$$= 165.119 \cdot 1.0 \cdot \ln(2 \cdot 36.5 \cdot \cos(5.711) + 2226.621) / 2226.621$$

$$= 53.0028 \text{ bars}$$

Maximum Allowable Pressure New and Cold [MAPnc]:

$$= S_a \cdot E \cdot \ln(2 \cdot t \cdot \cos(\alpha) + D) / D$$

$$= 206.996 \cdot 1.0 \cdot \ln(2 \cdot 38.0 \cdot \cos(5.711) + 2223.621) / 2223.621$$

$$= 69.2225 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D) \cos(\alpha))$$

$$= 51.0 \cdot 2226.621 / (1.0 (2299.621 - 2226.621) \cos(5.711))$$

$$= 156.343 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t_{cor} \cdot \cos(\alpha) + D_{cor}) / D_{cor}))$$

$$= 51.0 / (1.0 \cdot \ln((2 \cdot 36.5 \cdot \cos(5.711) + 2226.621) / 2226.621))$$

$$= 158.880 \text{ N./mm}^2$$

**Percent Fiber Elongation per Table 6.1**

Cones formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 1.937 %

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : 14 °C

User entered Minimum Design Temperature : -10 °C

The Curve for this Material is : A

From Figure 3.8M- with PWHT, the MDMT is : 25 °C

Temperature reduction per Graph 3.13 : 11 °C

The computed stress ratio is : 0.795

The computed thickness ratio [sr]:

$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$

$= (29.021 \cdot 1.0) / (38.0 - 1.5 - 0.0)$

= 0.795

Warning: Impact Testing Is Mandatory!

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	53.003 bars
Governing Thk.	38.000 mm.	Min. Des. Mtl. Temp.	-10 °C
Stress/Thk. Ratio	0.795	Basic MDMT	25 °C
		Temperature Difference	11 °C
		Computed Min. Temp.	14 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Conical**Cone From 60 to 70 Ext. Chart: CS-2 at 286 °C****Maximum External Pressure per ASME Division 2 Section 4.4:**

Determine Parameter [Mx]:

$= L / \sqrt{(R_o \cdot t)}$

$= 1507.481 / \sqrt{(1155.736 \cdot 36.5)}$

= 7.340

Compute the Factor [Ch]:

Since $M_x > 1.5$ and $M_x < 13$, $Ch = 0.92 / (M_x - 0.579)$

$= 0.92 / (7.34 - 0.579)$

= 0.1361

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$= 1.6 \cdot Ch \cdot E_y(t / D_o)$

$= 1.6 \cdot 0.136 \cdot 0.18730E+09(36.5/2311.471)$

= 644.001 N./mm²

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$= Fhe / F_y$

$= 644.0/247.7$

= 2.600

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$= \sigma_{ys} / \sigma_{uts}$

$= 247.7/517.1$

= 0.479

Ratio [AE]:

$= Fhe / E$



$$= 644.0/1.18730E+09$$

$$= 0.003438$$

Factor [m1]:

$$= (\ln(r) + (\text{epsp} - \text{epsys})) / (\ln(\ln(1 + \text{epsp})) / \ln(1 + \text{epsys}))$$

$$= (\ln(0.479) + (0.00002 - 0.002)) / (\ln(\ln(1 + 0.00002)) / \ln(1 + 0.002))$$

$$= 0.160$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \text{epsys}) / \ln(1 + \text{epsys}) m1$$

$$= 247.7 (1 + 0.002) / \ln(1 + 0.002) 0.160318$$

$$= 672.232 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.479)$$

$$= 0.313$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2 m2)$$

$$= (517.1 \cdot \exp(0.313)) / (0.313 0.313)$$

$$= 1016.803 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4791.5 - 0.5 \cdot 0.4792.5 - 0.4793.5$$

$$= 0.342$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	258	-1.764	0.311E-05	0.262E-05	0.775E-05	-0.580E-05	0.122E+08	0.178E-01
2	129	-4.571	0.825E-07	0.825E-07	0.169E-05	-0.169E-05	0.143E+09	0.253E-02
3	193	-3.167	0.690E-06	0.684E-06	0.412E-05	-0.406E-05	0.510E+08	0.364E-03
4	161	-3.869	0.265E-06	0.265E-06	0.276E-05	-0.275E-05	0.933E+08	0.171E-02
5	177	-3.518	0.437E-06	0.436E-06	0.340E-05	-0.338E-05	0.701E+08	0.901E-03
6	185	-3.343	0.552E-06	0.549E-06	0.375E-05	-0.372E-05	0.600E+08	0.339E-03
7	189	-3.255	0.618E-06	0.614E-06	0.393E-05	-0.389E-05	0.553E+08	0.710E-05
8	191	-3.211	0.653E-06	0.648E-06	0.402E-05	-0.398E-05	0.531E+08	0.173E-03
9	190	-3.233	0.635E-06	0.631E-06	0.398E-05	-0.393E-05	0.542E+08	0.818E-04
10	190	-3.244	0.626E-06	0.622E-06	0.395E-05	-0.391E-05	0.548E+08	0.371E-04
11	190	-3.250	0.622E-06	0.618E-06	0.394E-05	-0.390E-05	0.551E+08	0.149E-04
12	190	-3.252	0.620E-06	0.616E-06	0.394E-05	-0.389E-05	0.552E+08	0.388E-05
13	189	-3.254	0.619E-06	0.615E-06	0.393E-05	-0.389E-05	0.553E+08	0.162E-05
14	189	-3.253	0.619E-06	0.615E-06	0.393E-05	-0.389E-05	0.552E+08	0.113E-05
15	189	-3.253	0.619E-06	0.615E-06	0.393E-05	-0.389E-05	0.553E+08	0.243E-06
16	189	-3.253	0.619E-06	0.615E-06	0.393E-05	-0.389E-05	0.552E+08	0.444E-06
17	189	-3.253	0.619E-06	0.615E-06	0.393E-05	-0.389E-05	0.552E+08	0.100E-06

Factor of Safety [FS]:

$$\text{Since } F_{ic} > 0.55 F_y \text{ and } F_{ic} < F_y$$

$$= 2.407 - 0.741 \cdot F_{ic} / F_y$$

$$= 2.407 - 0.741 \cdot 189.96/247.68$$

$$= 1.839$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic}/FS$$

$$= 189.965/1.839$$

$$= 103.317 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 1033.105 (36.5/2311.471)$$

$$= 32.627 \text{ bars}$$

Boiler

(A150NRJ10EQ001)

PV Elite 28 U2

FileName : Boiler

Element: Conical

Elite®

v28

Licensee: SL User

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Step: 8 1:50pm Sep 17, 2026

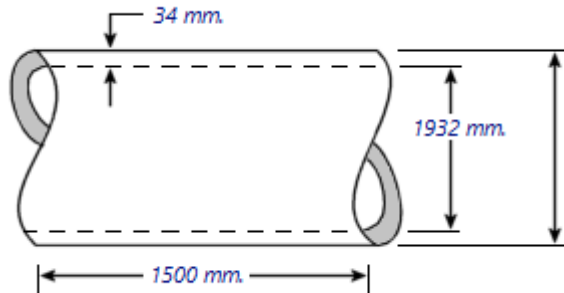


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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: shell

Cylindrical Shell Design Information:					
Design Pressure	P	51.000 bars	Int. Design Temperature	286 °C	
Material		SA-387 11	Ext. Design Temperature	286 °C	
Length		1500.000 mm.	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	165.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	1932.000 mm.	Actual Stress	151.8 N./mm ²	
Spec. Min. Thk t,e, or sa		34.0000 mm.	Spec. Nominal Thk.	34.000 mm.	
Surface Area		94247.8 cm ²	SG of Contents	0.000	
Element Volume		4398182.5 cm ³	Weight of Contents	0.0 Kgf	
Empty Weight		2441.4 Kgf	Operating Weight	2441.4 Kgf	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 1935.0 \cdot (\exp(51.0/(165.119 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 30.3511 + 1.5 + 0.0 = 31.8511 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 165.119 \cdot 1.0 \cdot \ln((2 \cdot 32.5 + 1935.0) / 1935.0) - 0.0$$

$$= 54.552 - 0.0 = 54.5517 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 206.996 \cdot 1.0 \cdot \ln((2 \cdot 34.0 + 1932.0) / 1932.0)$$

$$= 71.5988 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 51.0 \cdot 1935.0 / (1.0 (2000.0 - 1935.0))$$

$$= 151.8321 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 51.0 / (1.0 \cdot \ln((2 \cdot 32.5 + 1935.0) / 1935.0))$$



$$= 154.3685 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o) \quad 1.729 \%$$

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: 11 °C
User entered Minimum Design Temperature	: -10 °C
The Curve for this Material is	: A
From Figure 3.8M- with PWHT, the MDMT is	: 23 °C
Temperature reduction per Graph 3.13	: 12 °C
The computed stress ratio is	: 0.772

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (25.095 \cdot 1.0) / (34.0 - 1.5 - 0.0)$$

$$= 0.772$$

Warning: Impact Testing Is Mandatory!

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	53.003 bars
Governing Thk.	34.000 mm.	Min. Des. Mtl. Temp.	-10 °C
Stress/Thk. Ratio	0.772	Basic MDMT	23 °C
		Temperature Difference	12 °C
		Computed Min. Temp.	11 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: shell

Cylindrical Shell From 70 to 80 Ext. Chart: CS-2 at 286 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 1500.0 / \sqrt{(1000.0 \cdot 32.5)}$$

$$= 8.321$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 1.5 \text{ and } M_x < 13, Ch = 0.92 / (M_x - 0.579)$$

$$= 0.92 / (8.321 - 0.579)$$

$$= 0.1188$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.1188 \cdot 0.18730 \text{E}+09 (32.5/2000.0)$$

$$= 578.761 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$

$$= 578.8/247.7$$

$$= 2.337$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 247.7/517.1$$

$$= 0.479$$



Ratio [AE]:

= Fhe / E

= 578.8/18730E+09

= 0.003090

Factor [m1]:

= (ln(r) + (epsp - epsys)) / ln(ln(1 + epsp) / ln(1 + epsys))

= (ln(0.479) + (0.00002 - 0.002)) / ln(ln(1 + 0.00002) / ln(1 + 0.002))

= 0.160

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

= $\sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$

= 247.7(1 + 0.002) / ln(1 + 0.002) 0.160318

= 672.232 N./mm²

Curve Fitting Exponent for, Ferritic Steel [m2]:

= m2f(m2f2 - R)

= 0.6(1.0 - 0.479)

= 0.313

Curve Fitting Constant [A2]:

= ($\sigma_{uts} \cdot \exp(m2)$) / (m2 m2)

= (517.1 • exp(0.313)) / (0.313 0.313)

= 1016.803 N./mm²

Material Parameter for Stress-Strain Curve Model [K]:

= 1.5 • R1.5 - 0.5 • R2.5 - R3.5

= 1.5 • 0.4791.5 - 0.5 • 0.4792.5 - 0.4793.5

= 0.342

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	258	-1.764	0.311E-05	0.262E-05	0.775E-05	-0.580E-05	0.122E+08	0.182E-01
2	129	-4.571	0.825E-07	0.825E-07	0.169E-05	-0.169E-05	0.143E+09	0.219E-02
3	193	-3.167	0.690E-06	0.684E-06	0.412E-05	-0.406E-05	0.510E+08	0.712E-03
4	161	-3.869	0.265E-06	0.265E-06	0.276E-05	-0.275E-05	0.933E+08	0.136E-02
5	177	-3.518	0.437E-06	0.436E-06	0.340E-05	-0.338E-05	0.701E+08	0.553E-03
6	185	-3.343	0.552E-06	0.549E-06	0.375E-05	-0.372E-05	0.600E+08	0.921E-05
7	181	-3.431	0.492E-06	0.490E-06	0.357E-05	-0.355E-05	0.649E+08	0.288E-03
8	183	-3.387	0.521E-06	0.519E-06	0.366E-05	-0.363E-05	0.624E+08	0.143E-03
9	184	-3.365	0.536E-06	0.534E-06	0.370E-05	-0.367E-05	0.612E+08	0.681E-04
10	185	-3.354	0.544E-06	0.541E-06	0.373E-05	-0.369E-05	0.606E+08	0.297E-04
11	185	-3.348	0.548E-06	0.545E-06	0.374E-05	-0.371E-05	0.603E+08	0.103E-04
12	185	-3.346	0.550E-06	0.547E-06	0.374E-05	-0.371E-05	0.601E+08	0.578E-06
13	185	-3.344	0.551E-06	0.548E-06	0.375E-05	-0.371E-05	0.600E+08	0.431E-05
14	185	-3.345	0.551E-06	0.547E-06	0.374E-05	-0.371E-05	0.601E+08	0.187E-05
15	185	-3.345	0.550E-06	0.547E-06	0.374E-05	-0.371E-05	0.601E+08	0.644E-06
16	185	-3.345	0.550E-06	0.547E-06	0.374E-05	-0.371E-05	0.601E+08	0.332E-07
17	185	-3.346	0.550E-06	0.547E-06	0.374E-05	-0.371E-05	0.601E+08	0.272E-06

Factor of Safety [FS]:

Since Fic > 0.55 Fy and Fic < Fy

= 2.407 - 0.741 • Fic / Fy

= 2.407 - 0.741 • 185.72/247.68

= 1.851

Allowable Circumferential Compressive Stress [Fha]:

= Fic/FS

= 185.719/1.851

= 100.315 N./mm²

Allowable External Pressure at the given Thickness [Pa]:

= 2 • Fha(t / Do)

= 2 • 1003.089(32.5/2000.0)

= 32.600 bars

Boiler

(A150NRJ10EQ001)

PV Elite 28 U2

Licensee: SL User

FileName : Boiler

Element: shell

Elite

v28



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Step: 9 1:50pm Sep 17, 2026

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**Hydrostatic Test Pressure Results:**

Hydrotest Pressure Calculation [Shell Side]:

= Test Factor • ShellSide MAWP(St/S)

= 1.25 • 17.473(1.254)

= 27.380 bars

Hydrotest Pressure Calculation [Channel Side]:

= Test Factor • Channel Side MAWP(St/S)

= 1.25 • 17.356(1.254)

= 27.197 bars

Horizontal Test performed per: per VIII-2, Section 8.2

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Dish bottom	22.9	294.5	0.078	27.44
channel shell	23.2	294.5	0.079	27.42
shell 1	82.1	294.5	0.279	27.60
shell 2	82.1	294.5	0.279	27.60
Conical	82.5	294.5	0.280	27.60
shell	79.7	294.5	0.271	27.57

Theoretical Maximum Allowable test pressure Shellside is : 98.265 bars**Theoretical Maximum Allowable test pressure Tubeside : 348.388 bars****Stress ratios for Nozzle and Pad Materials (N./mm²):**

Description	Pad/Nozzle	Ambient	Operating	Ratio
M1-24"	Nozzle	207.00	165.12	1.254
A1-24"	Nozzle	207.00	165.12	1.254
B1-24 "	Nozzle	207.00	165.12	1.254

Minimum

1.254

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Dish bottom	207.00	165.12	1.254
channel shell	207.00	165.12	1.254
shell 1	207.00	165.12	1.254
shell 2	207.00	165.12	1.254
Conical	207.00	165.12	1.254
shell	207.00	165.12	1.254

Minimum

1.254

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	147.00	101.80	1.444
Tubesheet Material	207.00	121.46	1.704

Minimum

1.444

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):



Description	Ambient	Operating	Ratio
M1-24"	32.06	294.49	0.109
A1-24"	32.04	294.49	0.109
B1-24 "	32.04	294.49	0.109

Elements Suitable for Test Pressure.

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**Input Echo, Tubesheet Number 1, Description: Tube sheet****Shell Data:****Main Shell Description: shell 1**

Shell Test Pressure		27.38 bars
Shell Thickness	ts	38.0000 mm.
Shell Internal Corrosion Allowance	cas	0.0000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	2224.000 mm.
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-387 11

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K11789
Shell Allowable Stress at Temperature	Ss	207.00 N./mm ²
Shell Allowable Stress at Ambient		207.00 N./mm ²

Channel Data:**Channel Description: channel shell**

Channel Type:		Cylinder
Channel Test Pressure		27.20 bars
Channel Thickness	tc	140.0000 mm.
Channel Internal Corrosion Allowance	cac	0.0000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	2228.000 mm.
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-387 11

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K11789
Channel Allowable Stress at Temperature	Sc	207.00 N./mm ²
Channel Allowable Stress at Ambient		207.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	673
Tube Wall Thickness	et	2.5000 mm.
Tube Outside Diameter	D	30.8000 mm.
Total Straight Tube Length	Lt	5000.00 mm.
Straight Tube Length (bet. inner tubsht faces) L		4900.00 mm.
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-213 T12
Tube Material UNS Number		K11562
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		147.00 N./mm ²
Tube Allowable Stress At Ambient		147.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	220.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	75.0000 mm.
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm.
Groove Weld Leg	ag	2.1100 mm.
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		4.21.2
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	1014.580 mm.
Largest Center-to-Center Tube Distance	Ul	800.0000 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	0.0000 mm.
Area of Tube Layout	Ap	30933.3 cm ²
Perimeter of Tube Layout	Cp	6483.0 mm.

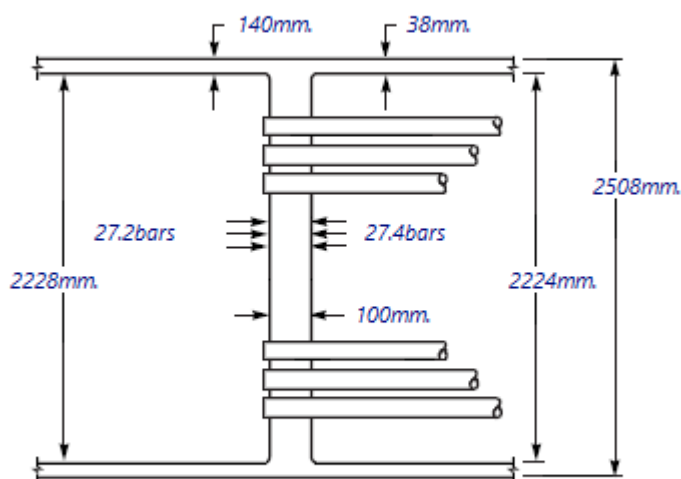
Tubesheet Data:



Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-336 F11
Tubesheet Material UNS Number		K11572
Tubesheet Allowable Stress at Temperature	S	207.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	207.00 N./mm ²
Thickness of Tubesheet	h	100.0000 mm.
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm.
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm.
Tubesheet Outside Diameter	A	2508.000 mm.
Area of the Untubed Lanes	AL	0.0 cm ²
Junction Stress Reduction option		Perform Plastic Calculation
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Material Group C

Elastic Mod. at Design Temperature	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Channel - TM-1 Material Group C

Elastic Mod. at Design Temperature	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Tubes - TM-1 Material Group C

Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.



TubeSheet - TM-1 Material Group C

Elastic Mod. at Design Temperature	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i$ Equation 4.3.1
 $= 0.5 \cdot 25.8 \cdot (\exp(27.197/(147.0 \cdot 1.0)) - 1) + 0$
 $= 0.2409 + 0$
 $= 0.2409 \text{ mm.}$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$= L / \sqrt{(R_o \cdot t)}$
 $= 5914.58 / \sqrt{(15.4 \cdot 2.5)}$
 $= 953.221$

Compute the Factor [Ch]:

Since $Mx > 2 \cdot (D_o/t)^{0.94}$, $Ch = 0.55 / (D_o/t)$
 $= 0.55 / (30.8/2.5)$
 $= 0.0446$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$= 1.6 \cdot Ch \cdot E_y(t / D_o)$
 $= 1.6 \cdot 0.045 \cdot 0.20408E+09(2.5/30.8)$
 $= 1183.279 \text{ N./mm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$= Fhe / F_y$
 $= 1183.3/220.0$
 $= 5.379$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$= \sigma_{ys} / \sigma_{uts}$
 $= 220.0/416.0$
 $= 0.529$

Ratio [AE]:

$= Fhe / E$
 $= 1183.3/20408E+09$
 $= 0.005798$

Factor [m1]:

$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$
 $= (\ln(0.529) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$
 $= 0.139$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$
 $= 220.0(1 + 0.002) / \ln(1 + 0.002)^{0.138786}$
 $= 522.303 \text{ N./mm}^2$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$= m2f(m2f2 - R)$
 $= 0.6(1.0 - 0.529)$
 $= 0.283$

Curve Fitting Constant [A2]:

$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$
 $= (416.0 \cdot \exp(0.283)) / (0.283^{0.283})$
 $= 788.754 \text{ N./mm}^2$

Material Parameter for Stress-Strain Curve Model [K]:



$$\begin{aligned}
 &= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5 \\
 &= 1.5 \cdot 0.5291.5 - 0.5 \cdot 0.5292.5 - 0.5293.5 \\
 &= 0.368
 \end{aligned}$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	207	-2.333	0.228E-05	0.217E-05	0.762E-05	-0.702E-05	0.181E+08	0.571E-02
2	103	-5.220	0.309E-07	0.309E-07	0.131E-05	-0.131E-05	0.181E+09	0.522E-02
3	155	-3.777	0.382E-06	0.381E-06	0.367E-05	-0.366E-05	0.790E+08	0.382E-02
4	181	-3.055	0.994E-06	0.984E-06	0.543E-05	-0.533E-05	0.391E+08	0.114E-02
5	194	-2.694	0.153E-05	0.149E-05	0.647E-05	-0.623E-05	0.267E+08	0.150E-02
6	188	-2.875	0.124E-05	0.122E-05	0.593E-05	-0.578E-05	0.323E+08	0.291E-04
7	185	-2.965	0.111E-05	0.110E-05	0.568E-05	-0.556E-05	0.356E+08	0.589E-03
8	186	-2.920	0.117E-05	0.115E-05	0.580E-05	-0.567E-05	0.339E+08	0.289E-03
9	187	-2.897	0.120E-05	0.119E-05	0.587E-05	-0.573E-05	0.331E+08	0.132E-03
10	188	-2.886	0.122E-05	0.120E-05	0.590E-05	-0.576E-05	0.327E+08	0.521E-04
11	188	-2.880	0.123E-05	0.121E-05	0.592E-05	-0.577E-05	0.325E+08	0.116E-04
12	188	-2.877	0.123E-05	0.121E-05	0.592E-05	-0.578E-05	0.324E+08	0.873E-05
13	188	-2.879	0.123E-05	0.121E-05	0.592E-05	-0.577E-05	0.325E+08	0.145E-05
14	188	-2.878	0.123E-05	0.121E-05	0.592E-05	-0.578E-05	0.325E+08	0.364E-05
15	188	-2.878	0.123E-05	0.121E-05	0.592E-05	-0.577E-05	0.325E+08	0.109E-05
16	188	-2.879	0.123E-05	0.121E-05	0.592E-05	-0.577E-05	0.325E+08	0.178E-06

Factor of Safety [FS]:

$$\begin{aligned}
 &\text{Since } F_{ic} > 0.55 F_y \text{ and } F_{ic} < F_y \\
 &= 2.407 - 0.741 \cdot F_{ic} / F_y \\
 &= 2.407 - 0.741 \cdot 188.34 / 220.0 \\
 &= 1.773
 \end{aligned}$$

Allowable Circumferential Compressive Stress [Fha]:

$$\begin{aligned}
 &= F_{ic} / FS \\
 &= 188.343 / 1.773 \\
 &= 106.251 \text{ N./mm}^2
 \end{aligned}$$

Allowable External Pressure at the given Thickness [Pa]:

$$\begin{aligned}
 &= 2 \cdot F_{ha} (t / D_o) \\
 &= 2 \cdot 1062.452 (2.5 / 30.8) \\
 &= 172.476 \text{ bars}
 \end{aligned}$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.7978 mm.
Allowable Internal Pressure at Corroded thickness	260.38 bars
Required Internal Design Pressure	27.20 bars
Allowable External Pressure at Corroded thickness	172.48 bars
Required External Design Pressure	27.38 bars
Required Thickness due to Shell Side pressure	0.7978 mm.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned}
 &= 2 \cdot r_o + d_t \\
 &= 2 \cdot 1014.58 + 30.8 \\
 &= 2059.960 \text{ mm.}
 \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$= (p - d_t) / p$$



$$= (75.0 - 30.8) / 75.0$$

$$= 0.589$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 2224.0 / 2059.9602$$

$$= 1.0796$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 2228.0 / 2059.9602$$

$$= 1.0816$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 0 / 100.0$$

$$= 0.000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2t_t(E_t T / E) \cdot (S_t T / S) \cdot (\rho); dt - 2t_t)$$

$$= \max(30.8 - 2 \cdot 2.5 (0.20408E+09 / 2.0408E+09) \cdot (146 / 206) \cdot (0); 30.8 - 2 \cdot 2.5)$$

$$= 30.8000 \text{ mm.}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 75.0 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 2059.96 \cdot 75.0) / (\pi \cdot 2059.96^2)) }$$

$$= 75.0000 \text{ mm.}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (75.0 - 30.8) / 75.0$$

$$= 0.5893$$

 $h/p = 1.333333$; $\mu^* = 0.589333$; $E^* = 139563104 \text{ KPa}$.

$$E^*/E = 0.683864$$
; $v^* = 0.297725$

ASME Section VIII-2 Part 4.18.7.4 Step 4:Shell coefficient [β_s]:

$$= ((12 (1 - v_s^2))^{1/4}) / \sqrt{ ((D_s + t_s) t_s) }$$

$$= ((12 (1 - 0.3^2))^{1/4}) / \sqrt{ ((2224.0 + 38.0) 38.0) }$$

$$= 0.0062 \text{ 1/mm.}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6 (1 - v_s^2))$$

$$= 0.006 \cdot 2040801 \cdot 38.0^3 / (6 (1 - 0.3^2))$$

$$= 127167592.0000 \text{ bars} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6 (D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6 (2224.0 + 38.0) \cdot 0.12717E+09 / 100.0^3 \cdot (1 + 100.0 \cdot 0.006 + (100.0 \cdot 0.006)^2 / 2)$$

$$= 3127813.0 \text{ bars}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2 (D_s + t_s)))$$

$$= 2224.0^2 / (4 \cdot 0.20408E+09 \cdot 38.0) \cdot (1 - 2224.0 \cdot 0.3 / (2 (2224.0 + 38.0)))$$

$$= 0.13593E+00 \text{ mm./N./mm}^2$$

Parameter [w_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.0796 \cdot 0.12717E+09 \cdot 0.0062 \cdot 0.135927 (1 + 100.0 \cdot 0.0062)$$

$$= 18746.6934 \text{ mm.}^2$$

Channel coefficient [β_c]:

$$= \left(\left(12 \left(1 - v_c^2 \right) \right)^{1/4} / \sqrt{\left(\left(D_c + t_c \right) t_c \right)} \right)$$

$$= \left(\left(12 \left(1 - 0.3^2 \right) \right)^{1/4} / \sqrt{\left(\left(2228.0 + 140.0 \right) 140.0 \right)} \right)$$

$$= 0.0032 \text{ 1/mm.}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / \left(6 \left(1 - v_s^2 \right) \right)$$

$$= 0.003 \cdot 2040801 \cdot 140.0^3 / \left(6 \left(1 - 0.3^2 \right) \right)$$

$$= 3238120960.0000 \text{ bars} \cdot \text{mm.}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6 \left(D_c + t_c / \cos(\theta) \right) \cdot k_c / h^3 \cdot \left(CmTheta + h \cdot \beta_c \cdot \left(CmDelta + CqTheta \right) / 2 + \left(h \cdot \beta_c \right)^2 / 2 \cdot CqDelta \right)$$

$$= 6 \left(2228.0 + 140.0 / \cos(0) \right) \cdot 0.32381E+1 / 100.0^3 \cdot \left(1.0 + 100.0 \cdot 0.003 \cdot \left(1.0 + 1.0 \right) / 2 + \left(100.0 \cdot 0.003 \right)^2 / 2 \cdot 1.0 \right)$$

$$= 62825552.0 \text{ bars}$$

Channel coefficient [δ_c]:

$$= D_c^2 / \left(4 \cdot E_c \cdot t_c \right) \cdot \left(1 - D_c \cdot v_c / \left(2 \left(D_c + t_c \right) \right) \right)$$

$$= 2228.0^2 / \left(4 \cdot 0.20408E+09 \cdot 140.0 \right) \cdot \left(1 - 2228.0 \cdot 0.3 / \left(2 \left(2228.0 + 140.0 \right) \right) \right)$$

$$= 0.37303E-01 \text{ mm./N./mm}^2$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c \left[\delta_c \left(CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2 \right) - \zeta_c \left(CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2 \right) \right]$$

$$= 1.082 \cdot 0.32381E+1 \left[0.37303 \left(1.0 + 0.003 \cdot 1.0 \cdot 100.0 \right) - 0 \left(1.0 / 100.0 + 1.0 \cdot 0.003 / 2 \right) \right]$$

$$= 54272.7148 \text{ mm.}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2508.0 / 2059.9602$$

$$= 1.2175$$

Coefficient [F]:

$$= \left(1 - v^* \right) / E^* \cdot \left(\lambda_s + \lambda_c + E \cdot \ln(K) \right)$$

$$= \left(1 - 0.3 \right) / .13956E+09 \cdot \left(3127813 + 62825552 + 0.20408E+09 \cdot \ln(1.22) \right)$$

$$= 33.3895$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot \left[\left(p_s - 1 \right) \cdot \left(p_s^2 + 1 \right) \cdot P_s - \left(p_c - 1 \right) \cdot \left(p_c^2 + 1 \right) \cdot P_t \right]$$

$$= 2059.96^2 / 16 \cdot \left[\left(1.08 - 1 \right) \cdot \left(1.08^2 + 1 \right) \cdot 27.38 - \left(1.082 - 1 \right) \cdot \left(1.082^2 + 1 \right) \cdot 27.197 \right]$$

$$= -24430.6465 \text{ bars} \cdot \text{mm.}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= -24430.646 + 54272.715 \cdot 27.197 - 18746.693 \cdot 27.38$$

$$= 938339.0000 \text{ bars} \cdot \text{mm.}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= \left((M^*) - D_o^2 / 32 \cdot F \left(P_s - P_t \right) \right) / \left(1 + F \right)$$

$$= \left((938339) - 2059.96^2 / 32 \cdot 33.39 \left(27.38 - 27.2 \right) \right) / \left(1 + 33.39 \right)$$

$$= 3726.6958 \text{ bars} \cdot \text{mm.}^2$$



Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2/64 \cdot (3 + \nu) \cdot (P_s - P_t)$$

$$= 3726.7 + 2059.96^2/64 \cdot (3 + 0.298) \cdot (27.38 - 27.2)$$

$$= 43735.5117 \text{ bars} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|M_p|; |M_o|)$$

$$= \max(|3726.696|; |43735.512|)$$

$$= 43735.5117 \text{ bars} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 43735.512 / (0.5893 \cdot (100.0 - 0)^2)$$

$$= 4.4530 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot \beta_T \cdot S_y$$

$$= 2 \cdot 0.95 \cdot 310.0$$

$$= 588.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 24255.777 / (0.5893 \cdot (6.4753 - 0)^2)$$

$$= 588.9891 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 6.4753 + 0 + 0$$

$$= 6.4753 \text{ mm.}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(6.4753; 0.0629)$$

$$= 6.4753 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$= |P_s - P_t|$$

$$= |27.38 - 27.2|$$

$$= 0.183 \text{ bars}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / D_o \cdot 0.8 \cdot \beta_T \cdot S_y$$

$$= 4 \cdot 0.589 \cdot 100.0 / 2059.96 \cdot 0.8 \cdot 0.95 \cdot 309.994$$

$$= 26.961 \text{ N./mm}^2$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot (0.8 \cdot \beta_T \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.589 \cdot 235.596) \cdot (4 \cdot 30933.3 / 6482.96) \cdot |27.38 - 27.2| + 0$$

$$= 0.0629 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 38.000mm.

Min. Shell len. adjacent to the tubesheet is: 523.277 mm.

Note: For a given Channel thickness of: 140.000 mm.

Min. Channel len. adjacent to the tubesheet is: 1005.296mm.

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (30.8 - 2.5) 2.5$$



= 2.223 cm²

Tube Weld Size Results per VIII-2 4.21.2:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.5 (30.8 - 2.5) 208.996$$

$$= 4736.812 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.11 (30.8 + 0.67 \cdot 2.11) 209.0$$

$$= 3868.1350 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 208.996 / 208.996$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.11 (30.8 + 0.67 \cdot 2.11) 209.0$$

$$= 2502.9114 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) \cdot f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 30.8)^2 + 1.07 \cdot 2.5 \cdot (30.8 - 2.5) \cdot 1.0 \cdot 1.0)} - 0.75 \cdot 30.8 \right)$$

$$= 3.169 \text{ mm.}$$

Minimum Required Fillet Weld Leg [afr]: 1.5843 mm.

Minimum Required Groove Weld Leg [agr]: 1.5843 mm.

Tube-Tubesheet Jt allowable, 4736.81 is $\geq$ tube strength 4736.81 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	4.5 $\leq$	589.0 N./mm ²	Ok
Tubesheet shear stress	0.1 $\leq$	235.6 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	6.475	100.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.584	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.584	2.110	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses				Case Type	Pass/Fail
			Bend	Allwd	Shear	Allwd		
D1uc	78.376	...	357	589	22	236	Ps+Pt D1	Ok
D2uc	78.488	...	357	589	22	236	Ps+Pt D2	Ok
D3uc	6.475	...	4	589	...	236	Ps+Pt D3	Ok
Max:	78.4883	... mm.	0.606			0.094 (Str. Ratio)		

**Load Case Definitions:**

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

-- Integral Cyl.	Reqd. Thk.	Shell	Stress	Channel	Stress	Pass	
Case#	Shell	Channel	Stress	Allwd	Stress	Allwd	Fail
D1uc	...	...	...	442	...	442	Ok
D2uc	...	...	...	442	...	442	Ok
D3uc	...	...	...	442	...	442	Ok
Max	...	...	...	...	...	...	...

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	78.488	100.000	Ok
Tube thickness :	0.798	2.500	Ok
Tube-tubesheet fillet weld leg :	1.584	2.110	Ok
Tube-tubesheet groove weld leg :	1.584	2.110	Ok

Given Shell Thickness: 38.0000 mm.

Given Channel Thickness: 140.0000 mm.

Min Shell length of thk, (38.000) adj. to tubesheet: 523.277 mm.

Min Channel length of thk, (140.000) adj. to tubesheet: 1005.296 mm.

Note: This is a full strength Tube to Tubesheet Joint.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 2, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	2579	73.5	No Calc	No Calc
20	Tsht	1904.8	2508	138.5	No Calc	No Calc
Tsht	30	No Calc	...	138.5	No Calc	No Calc
30	40	6000.03	2300	36.5	No Calc	No Calc
40	60	6000.03	2300	36.5	No Calc	No Calc
60	70	1500	2300	36.5	No Calc	No Calc
70	80	1500	2000	32.5	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	75	4.99976	1.03419	124.792
20	Tsht	140	8.38269	1.03419	170.278
Tsht	30	140	No Calc	1.03419	No Calc
30	40	38	12.0743	1.03419	19.3313
40	60	38	12.0743	1.03419	19.3313
60	70	38	7.45212	1.03419	32.6271
70	80	34	6.9598	1.03419	32.6004

Minimum

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External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	Tsht	1904.8	25399974	No Calc	No Calc
Tsht	30	No Calc	No Calc	No Calc	No Calc
30	40	6000.03	25399974	No Calc	No Calc
40	60	6000.03	25399974	No Calc	No Calc
60	70	1500	1500	No Calc	No Calc
70	80	1500	25399974	No Calc	No Calc

Elements Suitable for External Pressure.

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**Element and Detail Weights:**

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	6296.68	3752571	6188.8	3766492	...
20	30	12108.2	5849103	11986.1	5864865	...
30	40	6278.82	10151696	6034.97	10183164	...
40	60	6278.82	9588988	6034.97	9620454	...
60	70	2945.58	5075449	2831.33	5090193	...
70	80	2441.36	4398182	2335.44	4411852	...
Total		36349	38815988.00	35411	38937020.00	0



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
10	Nozl	646.465	...	-1327	...	M1-24"
10	Wght	6890	...	...	...	SF
20	Nozl	411.11	1264	750	...	A1-24"
20	Nozl	411.11	-1264	750	...	B1-24 "
20	FTsh	3440.28	...	1550	...	Tube sheet
20	Tube	7200.84	...	4000	...	Tubes

Total Weight of Each Detail Type:

Nozzles	1468.7
Weights	6890.0
Exchanger Components	10641.1
Liquid in Tubes	2.5

Sum of the Detail Weights 19002.3 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	36349.5	36349.5	36349.5	36349.5	36349.5	36349.5
Nozzles	1468.7	1468.7	1468.7	1468.7	1468.7	1468.7
Wld Weights	6890.0	6890.0	6890.0	6890.0	6890.0	6890.0
Exchanger	10641.1	10641.1	10641.1	10641.1	10641.1	10641.1
Tube Ope Lqd	...	...	...	...	...	2.5
Test Liquid	...	38792.3	...	...	...	...
Tube Tst Lqd	...	2539.9	...	...	...	...
Totals	55349.3	96681.5	55349.3	55349.3	55349.3	55351.8

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	55349.3 kg.
Shop Test Wt.	- Fabricated weight + water (full)	96681.5 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	55349.3 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	55349.3 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	55349.3 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	55351.8 kg.
Field Test Wt.	- Empty weight + test medium (full)	96681.5 kg.



Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	6239.6 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	16563.7 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	6503.2 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	2541444.5 cm ³
Weight of Ope Liq in tubes :	2.5 kg.
Weight of Water in tubes :	2539.9 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	105615	92677.6
20	30	118187	104992
30	40	216770	209607
40	60	216770	209607
60	70	101822	98204.5
70	80	94247.8	91043.4

Total	853410.9	806131.7 cm ²
Total	918.6	867.7 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	13833.1	13833.1	17583.4	...	13833.1
20	30	16370.7	16370.7	22216.3	...	16370.7
30	40	6278.82	6278.82	16424.3	...	6278.82
40	60	6278.82	6278.82	15861.9	...	6278.82
60	70	2945.58	2945.58	8017.93	...	2945.58
70	80	2441.36	2441.36	6836.86	...	2441.36

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	48148.5	48148.5	86940.8
20	30	34315.3	34315.3	69357.3
30	40	17944.6	17944.6	47141.1
40	60	11665.8	11665.8	30716.7
60	70	5386.94	5386.94	14854.8
70	80	2441.36	2441.36	6836.86

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	...	...	...
20	30	...	...	...
30	40	...	...	...
40	60	...	...	...

Boiler

(A150NRJ10EQ001)

PV Elite 28 U2

Licensee: SL User

FileName : Boiler

Element and Detail Weights:

Step: 12

1:50pm

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HEXAGON

60	70	...	...	...
70	80	...	...	...

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Nozzle Flange MAWP Results: (bars & °C)

Nozzle	Flange Rating		Des.			Equiv.	- - - -	Max Pressure	
Description	Ope.	Ambient	Temp	Class	Grade/Group	Press	UG-44	50%	DNV

Shellside Flange Rating

Channelside Flange Rating

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Conical Calculations per ASME VIII-2, 4.3.11

Input Values:

Conical

Design Internal Pressure, Small End	51.000	bars
Design Internal Pressure, Large End	51.000	bars
Temperature for Internal Pressure	286.0	°C
Design External Pressure	1.034	bars
Temperature for External Pressure	286.0	°C

CONE Properties:

Cone Material		SA-387 11	
Cone Allowable Stress at Temperature	Ss	165.12	N./mm ²
Cone Allowable Stress At Ambient		207.00	N./mm ²
Longitudinal Joint Efficiency of Cone	Ec,E2	1.00	
Circumferential Joint Efficiency of Cone		1.00	
Actual Thickness of Cone	Tc	38.0000	mm.
Corrosion Allowance for Cone	Cac	1.5000	mm.
Diameter Basis for Cone and Cylinders		ID	
Diameter of Small End of Cone	Ds	1932.000	mm.
Diameter of Large End of Cone	Dl	2224.000	mm.
Half Apex Angle for Cone		5.71	degrees
Axial Length of Cone	Lc	1500.000	mm.

SMALL-END Properties:

Attached Small End Material Specification		SA-387 11	
Operating Allowable Stress	Ss	165.12	N./mm ²
Ambient Allowable Stress		207.00	N./mm ²
Longitudinal Joint Efficiency	Es,E1	1.00	
Actual Thickness	Ts	34.0000	mm.
Corrosion Allowance	Cas	1.5000	mm.
Axial Length of Attached Element	Ls	1500.000	mm.

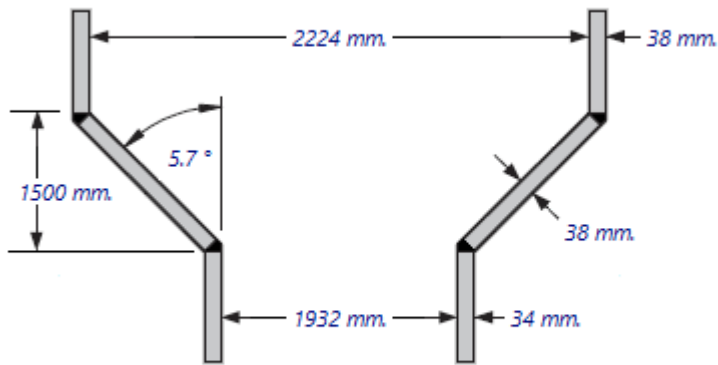
LARGE-END Properties:

Attached Large End Material Specification		SA-387 11	
Operating Allowable Stress	Sl	165.12	N./mm ²
Ambient Allowable Stress		207.00	N./mm ²
Longitudinal Joint Efficiency	El,E1	1.00	
Actual Thickness	Tl	38.0000	mm.
Corrosion Allowance	Cal	1.5000	mm.
Axial Length of Attached Element	Ll	6000.031	mm.

Type of Reinforcement at Large End of Cone: None

Type of Reinforcement at Small End of Cone: None

Warning: rtk -1.500mm., must be > Tl, 36.500mm., see 4.3.12



Cone Geometry

Calculated Values:

Elastic Modulus Data from ASME Section II Part D at 286 °C

Elastic Modulus of Cone Material	0.187E+09 KPa. at 286 °C
Elastic Modulus of Small Cylinder Material	0.187E+09 KPa. at 286 °C
Elastic Modulus of Large Cylinder Material	0.187E+09 KPa. at 286 °C
Elastic Modulus of Large End Reinforcement	0.187E+09 KPa. at 286 °C
Elastic Modulus of Small End Reinforcement	0.187E+09 KPa. at 286 °C

Elastic Modulus Data from ASME Section II Part D at 286 °C

Elastic Modulus of Cone Material	0.187E+09 KPa. at 286 °C
Elastic Modulus of Small Cylinder Material	0.187E+09 KPa. at 286 °C
Elastic Modulus of Large Cylinder Material	0.187E+09 KPa. at 286 °C
Elastic Modulus of Large End Reinforcement	0.187E+09 KPa. at 286 °C
Elastic Modulus of Small End Reinforcement	0.187E+09 KPa. at 286 °C

Forces and Moments on the ends of the Cone:

Since there are no supports, the forces and moments could not be resolved to the large and small ends of the cone.

Maximum Centroid Reinforcement Distance Large End	25.4000 mm.
Maximum Centroid Reinforcement Distance Small End	25.4000 mm.

Note: No ring was found close enough to the large end to be considered.

Note: No ring was found close enough to the small end to be considered.

Junction Calculations per 4.3.11:**Calculations for the Large End Cone/Shell Junction - Without Knuckle**

Fhe per 4.1.3 or 4.3.11.2 Step 6 [Fhe]:

$$= 0.4 \cdot E(tc/D)$$

$$= 0.4 \cdot 0.18730E+09(36.5/2300.0)$$

$$= 1189.0 \text{ N./mm}^2$$

Factor of Safety [FS]:

$$\text{Since } F_{ic} [247.68] \geq F_y [247.68], FS = 1.667$$

Allowable Stress per 4.3.11 Step 6 [Fha]:

$$= F_{ic}/FS$$

$$= 247.6785/1.667$$

$$= 148.577 \text{ N./mm}^2$$



Large End under Internal Pressure:

Cone Length along Cone surface per 4.3.46 [ConeMinLen]:

$$= 2 \cdot \sqrt{(RL \cdot Tc / \cos(\alpha)) + 1.4 \cdot \sqrt{(Rs \cdot tc / \cos(\alpha))}} \\ = 2 \cdot \sqrt{(1113.523 \cdot 36.5 / \cos(10.0)) + 1.4 \cdot \sqrt{(967.523 \cdot 36.5 / \cos(10.0))}} \\ = 671.412 \text{ mm.}$$

Results for the Large End Cylinder Without a Knuckle

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	77.799	247.678	Pass
Sigma sm-	77.799	247.678	Pass
Sigma thetam+	106.123	247.678	Pass
Sigma thetam-	106.123	247.678	Pass
Sigma sm+ + Sigma	-5.890	495.357	Pass
Sigma sm+ - Sigma	161.487	495.357	Pass
Sigma sm- + Sigma	-5.890	495.357	Pass
Sigma sm- - Sigma	161.487	495.357	Pass
Sigma thetam+ +	81.017	495.357	Pass
Sigma thetam+ -	131.230	495.357	Pass
Sigma thetam- +	81.017	495.357	Pass
Sigma thetam- -	131.230	495.357	Pass

Results for the Large End Cone Without a Knuckle

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	77.785	247.678	Pass
Sigma sm-	77.785	247.678	Pass
Sigma thetam+	105.623	247.678	Pass
Sigma thetam-	105.623	247.678	Pass
Sigma sm+ + Sigma	-5.903	495.357	Pass
Sigma sm+ - Sigma	161.473	495.357	Pass
Sigma sm- + Sigma	-5.903	495.357	Pass
Sigma sm- - Sigma	161.473	495.357	Pass
Sigma thetam+ +	80.516	495.357	Pass
Sigma thetam+ -	130.729	495.357	Pass
Sigma thetam- +	80.516	495.357	Pass
Sigma thetam- -	130.729	495.357	Pass

Cone Large end without a Knuckle Cylinder Section:

Net Axial Force ML +ve [XL+]:

$$= FL / (2 \cdot \pi \cdot RL) + ML / (\pi \cdot RL^2) \\ = 0.0 / (2 \cdot 3.141 \cdot 1113.523) + 0.0 / (3.142 \cdot 1113.523^2) \\ = 0.0 \text{ Kg/mm.}$$

Net Axial Force ML -ve [XL-]:

$$= FL / (2 \cdot \pi \cdot RL) - ML / (\pi \cdot RL^2) \\ = 0.0 / (2 \cdot 3.142 \cdot 1113.523) - 0.0 / (3.141 \cdot 1113.523^2) \\ = 0.0 \text{ Kg/mm.}$$

Thickness Ratio [n]:

$$= tc/tL = 36.5/36.5 = 1.0$$

Curve Fitting Geometry [H and B]:

$$H = \sqrt{0.5 \cdot D/tL} = \sqrt{0.5 \cdot 2227.046/36.5} = 5.5234 \\ B = \tan(10.0) = 0.1763$$

This analysis is preformed per Table 4.3.1 for the Large End:

Using Data from Table 4.3.3 and 4.3.4 knowing n
Table 4.3.3 - n: 1.0 < (n=1.0) < 1.25

Use n to construct a table: C1, C2, C3..... by interpolation

Table 4.3.3 [MsN1]:



Conical Section: Conical

Cone: 5 1:50pm Sep 17, 2026

C1 = -3.065534 C2 = 3.642747 C3 = 0.810048 C4 = -0.221192
 C5 = -0.081824 C6 = 0.035052 C7 = 0.025775 C8 = -0.015413
 C9 = 0.002102 C10 = -0.005587 C11 = 0.000000

Compute using Table 4.3.3 [MsN1]:

$$\begin{aligned}
 X &= C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot (\ln(H))^2 + C5 \cdot (\ln(B))^2 + C6 \cdot \ln(H) \cdot \ln(B) + \\
 &C7 \cdot (\ln(H))^3 + C8 \cdot (\ln(B))^3 + C9 \cdot \ln(H) \cdot (\ln(B))^2 + C10 \cdot (\ln(H))^2 \cdot \ln(B) \\
 &= C1 + C2 \cdot \ln(5.523) + C3 \cdot \ln(0.176) + C4 \cdot (\ln(5.523))^2 + C5 \cdot (\ln(0.176))^2 + C6 \cdot \ln(5.523) \cdot \ln(0.176) + \\
 &C7 \cdot (\ln(5.523))^3 + C8 \cdot (\ln(0.176))^3 + C9 \cdot \ln(5.523) \cdot (\ln(0.176))^2 + C10 \cdot (\ln(5.523))^2 \cdot \ln(0.176) \\
 X &= 1.006, \text{ MsN1} = -\exp(X) = -2.7347
 \end{aligned}$$

Table 4.3.3 [Qn1]:

C1 = -1.983852 C2 = 2.410703 C3 = 0.626443 C4 = -0.119151
 C5 = -0.115841 C6 = 0.122993 C7 = 0.012160 C8 = -0.016987
 C9 = 0.010919 C10 = -0.016653 C11 = 0.000000

Compute using Table 4.3.3 [Qn1]:

$$\begin{aligned}
 X &= C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot (\ln(H))^2 + C5 \cdot (\ln(B))^2 + C6 \cdot \ln(H) \cdot \ln(B) + \\
 &C7 \cdot (\ln(H))^3 + C8 \cdot (\ln(B))^3 + C9 \cdot \ln(H) \cdot (\ln(B))^2 + C10 \cdot (\ln(H))^2 \cdot \ln(B) \\
 &= C1 + C2 \cdot \ln(5.523) + C3 \cdot \ln(0.176) + C4 \cdot (\ln(5.523))^2 + C5 \cdot (\ln(0.176))^2 + C6 \cdot \ln(5.523) \cdot \ln(0.176) + \\
 &C7 \cdot (\ln(5.523))^3 + C8 \cdot (\ln(0.176))^3 + C9 \cdot \ln(5.523) \cdot (\ln(0.176))^2 + C10 \cdot (\ln(5.523))^2 \cdot \ln(0.176) \\
 X &= 0.2773, \text{ Qn} = -\exp(X) = -1.3196
 \end{aligned}$$

Table 4.3.4 MsN2:

C1 = -5.697151 C2 = 0.003838 C3 = 0.476317 C4 = -0.213157
 C5 = 2.233703 C6 = 0.000032 C7 = 0.002506 C8 = -0.001663
 C9 = -0.212965 C10 = 0.000138 C11 = -0.106203

Compute using Table 4.3.4 [MsN2]:

$$\begin{aligned}
 X1 &= C1 + C3 \cdot \ln(H^2) + C5 \cdot \ln(A) + C7 \cdot (\ln(H^2))^2 + C9 \cdot (\ln(A))^2 + C11 \cdot \ln(H^2) \cdot \ln(A) \\
 &= C1 + C3 \cdot \ln(5.523^2) + C5 \cdot \ln(0.175) + C7 \cdot (\ln(5.523^2))^2 + C9 \cdot (\ln(0.175))^2 + \\
 &C11 \cdot \ln(5.523^2) \cdot \ln(0.175) = -0.8615 \\
 X2 &= 1 + C2 \cdot \ln(H^2) + C4 \cdot \ln(AR) + C6 \cdot (\ln(H^2))^2 + C8 \cdot (\ln(AR))^2 + C10 \cdot \ln(H^2) \cdot \ln(AR) \\
 &= 1 + C2 \cdot \ln(5.523) + C4 \cdot \ln(0.175) + C6 \cdot (\ln(5.523^2))^2 + C8 \cdot (\ln(0.175))^2 + \\
 &C10 \cdot \ln(5.523^2) \cdot \ln(0.175) = 0.5149 \\
 \text{MsN2} &= \exp(X1/X2) = \exp(-0.862/0.515) = -0.1877
 \end{aligned}$$

Table 4.3.4 QN2:

C1 = -4.774616 C2 = 0.000461 C3 = -0.002831 C4 = -0.197117
 C5 = 1.982132 C6 = 0.000069 C7 = -0.000234 C8 = -0.003536
 C9 = -0.202493 C10 = -0.000088 C11 = 0.001365

Compute using Table 4.3.4 [Qn2]:

$$\begin{aligned}
 X3 &= C1 + C3 \cdot \ln(H^2) + C5 \cdot \ln(A) + C7 \cdot (\ln(H^2))^2 + C9 \cdot (\ln(A))^2 + C11 \cdot \ln(H^2) \cdot \ln(A) \\
 &= C1 + C3 \cdot \ln(5.523^2) + C5 \cdot \ln(0.175) + C7 \cdot (\ln(5.523^2))^2 + C9 \cdot (\ln(0.175))^2 + \\
 &C11 \cdot \ln(5.523^2) \cdot \ln(0.175) = -0.8615 \\
 X4 &= 1 + C2 \cdot \ln(H^2) + C4 \cdot \ln(AR) + C6 \cdot (\ln(H^2))^2 + C8 \cdot (\ln(AR))^2 + C10 \cdot \ln(H^2) \cdot \ln(AR) \\
 &= 1 + C2 \cdot \ln(5.523) + C4 \cdot \ln(0.175) + C6 \cdot (\ln(5.523^2))^2 + C8 \cdot (\ln(0.175))^2 + \\
 &C10 \cdot \ln(5.523^2) \cdot \ln(0.175) = 0.5149 \\
 \text{Qn2} &= \exp(X3/X4) = \exp(-1.286/0.529) = -0.088
 \end{aligned}$$

Stresses for the Large End Cylinder per Table 4.3.1

Stress Resultant Value [Msp]:

$$= P \cdot tL^2 \cdot \text{MsN1} = 51.0 \cdot 36.5^2 \cdot -2.735 = -1894.828 \text{ Kgf}$$

Compute with XL+ [Msx+]:

$$= XL \cdot tL \cdot \text{MsN2} = 0.0 \cdot 36.5 \cdot -0.188 = -0.0 \text{ Kgf}$$

Compute with XL- [Msx-]:

$$= XL \cdot tL \cdot \text{MsN2} = 0.0 \cdot 36.5 \cdot -0.188 = -0.0 \text{ Kgf}$$

Compute with XL+ [Ms+]:

$$= \text{MsP} + \text{Msx} = -1894.828 + -0.0 = -1894.828 \text{ Kgf}$$

Compute with XL- [Ms-]:

$$= \text{MsP} + \text{Msx} = -1894.828 + -0.0 = -1894.828 \text{ Kgf}$$



Compute [Qp]:

$$= P \cdot tL \cdot Q_{n1} = 51.0 \cdot 36.5 \cdot -1.32 = -25.0491 \text{ Kgf/mm.}$$

Compute with XL+ [Qx+]

$$= XL \cdot Q_{N2} = 0.0 \cdot -0.002 = -0.0 \text{ Kgf/mm.}$$

Compute with XL- [Qx-]

$$= XL \cdot Q_{N2} = 0.0 \cdot -0.002 = -0.0 \text{ Kgf/mm.}$$

Compute with XL+ [Q+]:

$$= Q_p + Q_x = -25.049 + -0.0 = -25.0491 \text{ Kgf/mm.}$$

Compute with XL- [Q-]:

$$= Q_p + Q_x = -25.049 + -0.0 = -25.0491 \text{ Kgf/mm.}$$

Compute [β_{cy}]:

$$= 3 \cdot (1 - v^2) / (RL^2 \cdot tL^2) \cdot 0.25$$

$$= (3 \cdot (1 - 0.3^2) / (1113.523^2 \cdot 36.5^2)) \cdot 0.25 = 0.0064 \text{ 1/mm.}$$

Compute with XL+ [Ns+]:

$$= P \cdot RL/2 + XL = 51.0 \cdot 1113.523/2 + 0.0 = 289.5583 \text{ Kgf/mm.}$$

Compute with XL- [Ns-]:

$$= P \cdot RL/2 + XL = 51.0 \cdot 1113.523/2 + 0.0 = 289.5583 \text{ Kgf/mm.}$$

Compute with XL+ [Ntheta+]:

$$= P \cdot RL + 2 \cdot \beta_{cy} \cdot RL \cdot (-M_s \cdot \beta_{cy} + Q)$$

$$= 51.0 \cdot 1113.523 + 2 \cdot 0.0064 \cdot 1113.523 \cdot$$

$$(-1894.828 \cdot 0.006 + -25.049) = 394.9805 \text{ Kgf/mm.}$$

Compute with XL- [Ntheta-]:

$$= P \cdot RL + 2 \cdot \beta_{cy} \cdot RL \cdot (-M_s \cdot \beta_{cy} + Q)$$

$$= 51.0 \cdot 1113.523 + 2 \cdot 0.0064 \cdot 1113.523 \cdot$$

$$(-1894.828 \cdot 0.006 + -25.049) = 394.9805 \text{ Kgf/mm.}$$

Note: Kpc has been set to 1.0

σ_{Sm} with XL+ [σ_{sm+}]:

$$= N_s / tL$$

$$= 289.558/36.5$$

$$= 77.799 \text{ N./mm}^2$$

σ_{Sm} with XL- [σ_{sm-}]:

$$= N_s / tL$$

$$= 289.558/36.5$$

$$= 77.799 \text{ N./mm}^2$$

σ_{Sb} with XL+ [σ_{sb+}]:

$$= 6 \cdot M_s / (tL^2 \cdot K_{pc})$$

$$= 6 \cdot -1894.828 / (36.5^2 \cdot 1.0)$$

$$= -83.688 \text{ N./mm}^2$$

σ_{Sb} with XL- [σ_{sb-}]:

$$= 6 \cdot M_s / (tL^2 \cdot K_{pc})$$

$$= 6 \cdot -1894.828 / (36.5^2 \cdot 1.0)$$

$$= -83.688 \text{ N./mm}^2$$

σ_{Thetam} with XL+ [$\sigma_{thetam+}$]:

$$= N_{theta} / tL$$

$$= 394.98/36.5 \text{ N./mm}^2$$

$$= 106.123 \text{ N./mm}^2$$

σ_{Thetam} with XL- [$\sigma_{thetam-}$]:

$$= N_{theta} / tL$$

$$= 394.98/36.5$$

$$= 106.123 \text{ N./mm}^2$$

σ_{Thetab} with XL+ [$\sigma_{thetab+}$]:



$$\begin{aligned}
 &= 6 \cdot v \cdot Ms / (tL^2 \cdot Kpc) \\
 &= 6 \cdot 0.3 \cdot -1894.828 / (36.5^2 \cdot 1.0) \\
 &= -25.106 \text{ N./mm}^2
 \end{aligned}$$

σ Thetab with XL- [σ thetab-]:

$$\begin{aligned}
 &= 6 \cdot v \cdot Ms / (tL^2 \cdot Kpc) \\
 &= 6 \cdot 0.3 \cdot -1894.828 / (36.5^2 \cdot 1.0) \\
 &= -25.106 \text{ N./mm}^2
 \end{aligned}$$

For the Large End Cone Junction per Table 4.3.1

Compute [McsP]:

$$= Msp = Msp = -1894.828 \text{ Kgf}$$

McsX with XL+ [McsX+]:

$$= MsX = -0.0 \text{ Kgf}$$

McsX with XL- [McsX-]:

$$= MsX = -0.0 \text{ Kgf}$$

Mcs with XL+ [Mcs+]:

$$\begin{aligned}
 &= McsP + MsX = -1894.828 + -0.0 \\
 &= -1894.828 \text{ Kgf}
 \end{aligned}$$

Mcs with XL- [Mcs-]:

$$\begin{aligned}
 &= McsP + MsX = -1894.828 + -0.0 \\
 &= -1894.828 \text{ Kgf}
 \end{aligned}$$

Compute with XL+ [McsX+]:

$$= MsX = -0.0 \text{ Kgf}$$

Compute with XL- [McsX-]:

$$= MsX = -0.0 \text{ Kgf}$$

Compute with XL+ [Qc+]:

$$\begin{aligned}
 &= Q \cdot \cos(a) + Ns \cdot \sin(a) \\
 &= -25.049 \cdot \cos(10.0) + 289.558 \cdot \sin(10.0) \\
 &= 25.613 \text{ Kgf/mm.}
 \end{aligned}$$

Compute with XL- [Qc-]:

$$\begin{aligned}
 &= Q \cdot \cos(a) + Ns \cdot \sin(a) \\
 &= -25.049 \cdot \cos(10.0) + 289.558 \cdot \sin(10.0) \\
 &= 25.613 \text{ Kgf/mm.}
 \end{aligned}$$

Compute [Rc]:

$$= RL/\cos(a) = 1113.523/\cos(10.0) = 1130.7012 \text{ mm.}$$

Compute [β -co]:

$$\begin{aligned}
 &= (3 \cdot (1 - v^2) / (Rc^2 \cdot tC^2))^{1/4} \\
 &= (3 \cdot (1 - 0.3^2) / (1130.701^2 \cdot 36.5^2))^{1/4} \\
 &= 0.006 \text{ 1/mm.}
 \end{aligned}$$

Compute with XL +ve [Ncs+]:

$$\begin{aligned}
 &= Ns \cdot \cos(a) - Q \cdot \sin(a) \\
 &= 289.558 \cdot \cos(10.0) - -25.049 \cdot \sin(10.0) \\
 &= 289.509 \text{ Kgf/mm.}
 \end{aligned}$$

Compute with XL -ve [Ncs-]:

$$\begin{aligned}
 &= Ns \cdot \cos(a) - Q \cdot \sin(a) \\
 &= 289.558 \cdot \cos(10.0) - -25.049 \cdot \sin(10.0) \\
 &= 289.509 \text{ Kgf/mm.}
 \end{aligned}$$

Compute with XL +ve [NcTheta+]:

$$\begin{aligned}
 &= P \cdot RL/\cos(a) + 2 \cdot \beta_{co} \cdot Rc(-Mcs \cdot \beta_{co} - Qc) \\
 &= 51.0 \cdot 1113.523/\cos(10.0) + 2 \cdot 0.006 \cdot 1130.701 \cdot \\
 &\quad (-1894.828 \cdot 0.006 - 25.613) \\
 &= 393.116 \text{ Kgf/mm.}
 \end{aligned}$$



Compute with XL -ve [NcTheta-]:

$$= P \cdot RL / \cos(A) + 2 \cdot \beta_{co} \cdot Rc \cdot (-Mcs \cdot \beta_{co} - Qc)$$

$$= 51.0 \cdot 1113.523 / \cos(10.0) + 2 \cdot 0.006 \cdot 1130.701 \cdot (-1894.828 \cdot 0.006 - 25.613)$$

$$= 393.116 \text{ Kg/mm.}$$

Note: Kcpc = 1.0

Compute with XL +ve [$\sigma_{sm}+$]:

$$= Ncs/tC = 289.509/36.5 = 77.7854 \text{ N./mm}^2$$

Compute with XL -ve [$\sigma_{sm}-$]:

$$= Ncs/tC = 289.509/36.5 = 777.8083 \text{ N./mm}^2$$

Compute with XL +ve [$\sigma_{sb}+$]:

$$= 6 \cdot Mcs / (tC^2 \cdot Kcpc)$$

$$= 6 \cdot -1894.828 / (36.5^2 \cdot 1.0)$$

$$= -83.688 \text{ N./mm}^2$$

Compute with XL -ve [$\sigma_{sb}-$]:

$$= 6 \cdot Mcs / (tC^2 \cdot Kcpc)$$

$$= 6 \cdot -1894.828 / (36.5^2 \cdot 1.0)$$

$$= -83.688 \text{ N./mm}^2$$

Compute with XL +ve [$\sigma_{Thetam}+$]:

$$= NCTheta/tC = 393.116/36.5 = 105.62 \text{ N./mm}^2$$

Compute with XL -ve [$\sigma_{Thetam}-$]:

$$= NCTheta/tC = 393.116/36.5 = 105.62 \text{ N./mm}^2$$

Compute with XL +ve [$\sigma_{thetab}+$]:

$$= 6 \cdot v \cdot Mcs / (tC^2 \cdot Kcpc)$$

$$= 6 \cdot 0.3 \cdot -1894.828 / (36.5^2 \cdot 1.0)$$

$$= -25.106 \text{ N./mm}^2$$

Compute with XL -ve [$\sigma_{thetab}-$]:

$$= 6 \cdot v \cdot Mcs / (tC^2 \cdot Kcpc)$$

$$= 6 \cdot 0.3 \cdot -1894.828 / (36.5^2 \cdot 1.0)$$

$$= -25.106 \text{ N./mm}^2$$

Large End under External Pressure:

Cone Length along Cone surface per 4.3.46 [ConeMinLen]:

$$= 2 \cdot \sqrt{(RL \cdot Tc / \cos(\alpha)) + 1.4 \cdot \sqrt{(Rs \cdot tc / \cos(\alpha))}}$$

$$= 2 \cdot \sqrt{(1113.523 \cdot 36.5 / \cos(10.0)) + 1.4 \cdot \sqrt{(967.523 \cdot 36.5 / \cos(10.0))}}$$

$$= 671.412 \text{ mm.}$$

Results for the Large End Cylinder Without a Knuckle

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	-1.578	247.678	Pass
Sigma sm-	1.578	247.678	Pass
Sigma thetam+	-2.152	247.678	Pass
Sigma thetam-	-2.152	247.678	Pass
Sigma sm+ + Sigm	0.119	495.357	Pass
Sigma sm+ - Sigm	-3.275	495.357	Pass
Sigma sm- + Sigm	0.119	495.357	Pass
Sigma sm- - Sigm	-3.275	495.357	Pass
Sigma thetam+ +	-1.643	495.357	Pass
Sigma thetam+ -	-2.661	495.357	Pass
Sigma thetam- +	-1.643	495.357	Pass
Sigma thetam- -	-2.661	495.357	Pass

Results for the Large End Cone Without a Knuckle

Stress Type	Stress	Allowable	Pass/Fail
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Sigma sm+	-1.577	247.678	Pass
Sigma sm-	1.577	247.678	Pass
Sigma thetam+	-2.142	247.678	Pass
Sigma thetam-	-2.142	247.678	Pass
Sigma sm+ + Sigm	0.120	495.357	Pass
Sigma sm+ - Sigm	-3.274	495.357	Pass
Sigma sm- + Sigm	0.120	495.357	Pass
Sigma sm- - Sigm	-3.274	495.357	Pass
Sigma thetam+ +	-1.633	495.357	Pass
Sigma thetam+ -	-2.651	495.357	Pass
Sigma thetam- +	-1.633	495.357	Pass
Sigma thetam- -	-2.651	495.357	Pass

Average Compressive Stresses and comparisons with Fha and Fxa:

Sigma _{smCyl} + + S	-1.577	-170.885	Pass
Sigma _{smCyl} - + S	-1.577	-170.885	Pass
Sigma _{thetamCyl} +	-2.147	-148.577	Pass
Sigma _{thetamCyl} -	-2.147	-148.577	Pass

Cone Large end without a Knuckle Cylinder Section:

Net Axial Force ML +ve [XL+]:

$$= FL / (2 \cdot \pi \cdot RL) + ML / (\pi \cdot RL^2)$$

$$= 0.0 / (2 \cdot 3.141 \cdot 1113.523) + 0.0 / (3.142 \cdot 1113.523^2)$$

$$= 0.0 \text{ Kg/mm.}$$

Net Axial Force ML -ve [XL-]:

$$= FL / (2 \cdot \pi \cdot RL) - ML / (\pi \cdot RL^2)$$

$$= 0.0 / (2 \cdot 3.142 \cdot 1113.523) - 0.0 / (3.141 \cdot 1113.523^2)$$

$$= 0.0 \text{ Kg/mm.}$$

Thickness Ratio [n]:

$$= t_c / t_L = 36.5 / 36.5 = 1.0$$

Curve Fitting Geometry [H and B]:

$$H = \text{Sqrt}(0.5 \cdot D / t_L) = \text{Sqrt}(0.5 \cdot 2227.046 / 36.5) = 5.5234$$

$$B = \tan(10.0) = 0.1763$$

This analysis is preformed per Table 4.3.1 for the Large End:

Using Data from Table 4.3.3 and 4.3.4 knowing n

Table 4.3.3 - n: 1.0 < (n=1.0) < 1.25

Use n to construct a table: C1, C2, C3..... by interpolation

Table 4.3.3 [MsN1]:

C1 =	-3.065534	C2 =	3.642747	C3 =	0.810048	C4 =	-0.221192
C5 =	-0.081824	C6 =	0.035052	C7 =	0.025775	C8 =	-0.015413
C9 =	0.002102	C10 =	-0.005587	C11 =	0.000000		

Compute using Table 4.3.3 [MsN1]:

$$X = C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot (\ln(H))^2 + C5 \cdot (\ln(B))^2 + C6 \cdot \ln(H) \cdot \ln(B) +$$

$$C7 \cdot (\ln(H))^3 + C8 \cdot (\ln(B))^3 + C9 \cdot \ln(H) \cdot (\ln(B))^2 + C10 \cdot (\ln(H))^2 \cdot \ln(B)$$

$$= C1 + C2 \cdot \ln(5.523) + C3 \cdot \ln(0.176) + C4 \cdot (\ln(5.523))^2 + C5 \cdot (\ln(0.176))^2 + C6 \cdot \ln(5.523) \cdot \ln(0.176) +$$

$$C7 \cdot (\ln(5.523))^3 + C8 \cdot (\ln(0.176))^3 + C9 \cdot \ln(5.523) \cdot (\ln(0.176))^2 + C10 \cdot (\ln(5.523))^2 \cdot \ln(0.176)$$

$$X = 1.006, \text{ MsN1} = -\exp(X) = -2.7347$$

Table 4.3.3 [Qn1]:

C1 =	-1.983852	C2 =	2.410703	C3 =	0.626443	C4 =	-0.119151
C5 =	-0.115841	C6 =	0.122993	C7 =	0.012160	C8 =	-0.016987
C9 =	0.010919	C10 =	-0.016653	C11 =	0.000000		

Compute using Table 4.3.3 [Qn1]:

$$X = C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot (\ln(H))^2 + C5 \cdot (\ln(B))^2 + C6 \cdot \ln(H) \cdot \ln(B) +$$

$$C7 \cdot (\ln(H))^3 + C8 \cdot (\ln(B))^3 + C9 \cdot \ln(H) \cdot (\ln(B))^2 + C10 \cdot (\ln(H))^2 \cdot \ln(B)$$

$$= C1 + C2 \cdot \ln(5.523) + C3 \cdot \ln(0.176) + C4 \cdot (\ln(5.523))^2 + C5 \cdot (\ln(0.176))^2 + C6 \cdot \ln(5.523) \cdot \ln(0.176) +$$

$$C7 \cdot (\ln(5.523))^3 + C8 \cdot (\ln(0.176))^3 + C9 \cdot \ln(5.523) \cdot (\ln(0.176))^2 + C10 \cdot (\ln(5.523))^2 \cdot \ln(0.176)$$

$$X = 0.2773, \text{ Qn} = -\exp(X) = -1.3196$$



Table 4.3.4 MsN2:

C1 = -5.697151 C2 = 0.003838 C3 = 0.476317 C4 = -0.213157
 C5 = 2.233703 C6 = 0.000032 C7 = 0.002506 C8 = -0.001663
 C9 = -0.212965 C10 = 0.000138 C11 = -0.106203

Compute using Table 4.3.4 [MsN2]:

$X1 = C1 + C3 \cdot \ln(H^2) + C5 \cdot \ln(A) + C7 \cdot (\ln(H^2))^2 + C9 \cdot (\ln(A))^2 + C11 \cdot \ln(H^2) \cdot \ln(A)$
 $= C1 + C3 \cdot \ln(5.523^2) + C5 \cdot \ln(0.175) + C7 \cdot (\ln(5.523^2))^2 + C9 \cdot (\ln(0.175))^2 +$
 $C11 \cdot \ln(5.523^2) \cdot \ln(0.175) = -0.8615$
 $X2 = 1 + C2 \cdot \ln(H^2) + C4 \cdot \ln(AR) + C6 \cdot (\ln(H^2))^2 + C8 \cdot (\ln(AR))^2 + C10 \cdot \ln(H^2) \cdot \ln(AR)$
 $= 1 + C2 \cdot \ln(5.523) + C4 \cdot \ln(0.175) + C6 \cdot (\ln(5.523^2))^2 + C8 \cdot (\ln(0.175))^2 +$
 $C10 \cdot \ln(5.523^2) \cdot \ln(0.17) = 0.5149$
 $MsN2 = \exp(X1/X2) = \exp(-0.862/0.515) = -0.1877$

Table 4.3.4 QN2:

C1 = -4.774616 C2 = 0.000461 C3 = -0.002831 C4 = -0.197117
 C5 = 1.982132 C6 = 0.000069 C7 = -0.000234 C8 = -0.003536
 C9 = -0.202493 C10 = -0.000088 C11 = 0.001365

Compute using Table 4.3.4 [Qn2]:

$X3 = C1 + C3 \cdot \ln(H^2) + C5 \cdot \ln(A) + C7 \cdot (\ln(H^2))^2 + C9 \cdot (\ln(A))^2 + C11 \cdot \ln(H^2) \cdot \ln(A)$
 $= C1 + C3 \cdot \ln(5.523^2) + C5 \cdot \ln(0.175) + C7 \cdot (\ln(5.523^2))^2 + C9 \cdot (\ln(0.175))^2 +$
 $C11 \cdot \ln(5.523^2) \cdot \ln(0.175) = -0.8615$
 $X4 = 1 + C2 \cdot \ln(H^2) + C4 \cdot \ln(AR) + C6 \cdot (\ln(H^2))^2 + C8 \cdot (\ln(AR))^2 + C10 \cdot \ln(H^2) \cdot \ln(AR)$
 $= 1 + C2 \cdot \ln(5.523) + C4 \cdot \ln(0.175) + C6 \cdot (\ln(5.523^2))^2 + C8 \cdot (\ln(0.175))^2 +$
 $C10 \cdot \ln(5.523^2) \cdot \ln(0.175) = 0.5149$
 $Qn2 = \exp(X3/X4) = \exp(-1.286/0.529) = -0.088$

Stresses for the Large End Cylinder per Table 4.3.1

Stress Resultant Value [Msp]:

$$= P \cdot tL^2 \cdot MsN1 = -1.034 \cdot 36.5^2 \cdot -2.735 = 38.4238 \text{ Kgf}$$

Compute with XL+ [Msx+]:

$$= XL \cdot tL \cdot MsN2 = 0.0 \cdot 36.5 \cdot -0.188 = -0.0 \text{ Kgf}$$

Compute with XL- [Msx-]:

$$= XL \cdot tL \cdot MsN2 = 0.0 \cdot 36.5 \cdot -0.188 = -0.0 \text{ Kgf}$$

Compute with XL+ [Ms+]:

$$= MsP + Msx = 38.424 + -0.0 = 38.4238 \text{ Kgf}$$

Compute with XL- [Ms-]:

$$= MsP + Msx = 38.424 + -0.0 = 38.4238 \text{ Kgf}$$

Compute [Qp]:

$$= P \cdot tL \cdot Qn1 = -1.034 \cdot 36.5 \cdot -1.32 = 0.508 \text{ Kgf/mm.}$$

Compute with XL+ [Qx+]:

$$= XL \cdot QN2 = 0.0 \cdot -0.002 = -0.0 \text{ Kgf/mm.}$$

Compute with XL- [Qx-]:

$$= XL \cdot QN2 = 0.0 \cdot -0.002 = -0.0 \text{ Kgf/mm.}$$

Compute with XL+ [Q+]:

$$= Qp + Qx = 0.508 + -0.0 = 0.508 \text{ Kgf/mm.}$$

Compute with XL- [Q-]:

$$= Qp + Qx = 0.508 + -0.0 = 0.508 \text{ Kgf/mm.}$$

Compute [β_{cy}]:

$$= 3 \cdot (1 - \nu^2) / (RL^2 \cdot tL^2) \cdot 0.25$$

$$= (3 \cdot (1 - 0.3^2) / (1113.523^2 \cdot 36.5^2)) \cdot 0.25 = 0.0064 \text{ 1/mm.}$$

Compute with XL+ [Ns+]:

$$= P \cdot RL/2 + XL = -1.034 \cdot 1113.523/2 + 0.0 = -5.8717 \text{ Kgf/mm.}$$



Compute with XL- [Ns-]:

$$= P \cdot RL/2 + XL = -1.034 \cdot 1113.523/2 + 0.0 = -5.8717 \text{ Kg/mm.}$$

Compute with XL+ [Ntheta+]:

$$= P \cdot RL + 2 \cdot \beta_{cy} \cdot RL \cdot (-Ms \cdot \beta_{cy} + Q) \\ = -1.034 \cdot 1113.523 + 2 \cdot 0.0064 \cdot 1113.523 \cdot \\ (-38.424 \cdot 0.006 + 0.508) = -8.0095 \text{ Kg/mm.}$$

Compute with XL- [Ntheta-]:

$$= P \cdot RL + 2 \cdot \beta_{cy} \cdot RL \cdot (-Ms \cdot \beta_{cy} + Q) \\ = -1.034 \cdot 1113.523 + 2 \cdot 0.0064 \cdot 1113.523 \cdot \\ (-38.424 \cdot 0.006 + 0.508) = -8.0095 \text{ Kg/mm.}$$

Note: Kpc has been set to 1.0

σ_{Sm} with XL+ [σ_{sm}]:

$$= N_s / t_L \\ = -5.872/36.5 \\ = -1.578 \text{ N./mm}^2$$

σ_{Sm} with XL- [σ_{sm}]:

$$= N_s / t_L \\ = -5.872/36.5 \\ = -1.578 \text{ N./mm}^2$$

σ_{Sb} with XL+ [σ_{sb}]:

$$= 6 \cdot Ms / (t_L^2 \cdot K_{pc}) \\ = 6 \cdot 38.424 / (36.5^2 \cdot 1.0) \\ = 1.697 \text{ N./mm}^2$$

σ_{Sb} with XL- [σ_{sb}]:

$$= 6 \cdot Ms / (t_L^2 \cdot K_{pc}) \\ = 6 \cdot 38.424 / (36.5^2 \cdot 1.0) \\ = 1.697 \text{ N./mm}^2$$

σ_{Thetam} with XL+ [σ_{thetam}]:

$$= N_{theta} / t_L \\ = -8.01/36.5 \text{ N./mm}^2 \\ = -2.152 \text{ N./mm}^2$$

σ_{Thetam} with XL- [σ_{thetam}]:

$$= N_{theta} / t_L \\ = -8.01/36.5 \\ = -2.152 \text{ N./mm}^2$$

σ_{Thetab} with XL+ [σ_{thetab}]:

$$= 6 \cdot v \cdot Ms / (t_L^2 \cdot K_{pc}) \\ = 6 \cdot 0.3 \cdot 38.424 / (36.5^2 \cdot 1.0) \\ = 0.509 \text{ N./mm}^2$$

σ_{Thetab} with XL- [σ_{thetab}]:

$$= 6 \cdot v \cdot Ms / (t_L^2 \cdot K_{pc}) \\ = 6 \cdot 0.3 \cdot 38.424 / (36.5^2 \cdot 1.0) \\ = 0.509 \text{ N./mm}^2$$

For the Large End Cone Junction per Table 4.3.1

Compute [McsP]:

$$= M_{sp} = M_{sp} = 38.4238 \text{ Kg}$$

McsX with XL+ [McsX+]:

$$= M_{sX} = -0.0 \text{ Kg}$$

McsX with XL- [McsX-]:

$$= M_{sX} = -0.0 \text{ Kg}$$

Mcs with XL+ [Mcs+]:

$$= M_{csP} + M_{sX} = 38.424 + -0.0$$



$$= 38.424 \text{ Kgf}$$

Mcs with XL- [Mcs-]:

$$= \text{McsP} + \text{MsX} = 38.424 + -0.0$$

$$= 38.424 \text{ Kgf}$$

Compute with XL+ [McsX+]:

$$= \text{MsX} = -0.0 \text{ Kgf}$$

Compute with XL- [McsX-]:

$$= \text{MsX} = -0.0 \text{ Kgf}$$

Compute with XL+ [Qc+]:

$$= Q \cdot \cos(a) + Ns \cdot \sin(a)$$

$$= 0.508 \cdot \cos(10.0) + -5.872 \cdot \sin(10.0)$$

$$= -0.519 \text{ Kgf/mm.}$$

Compute with XL- [Qc-]:

$$= Q \cdot \cos(a) + Ns \cdot \sin(a)$$

$$= 0.508 \cdot \cos(10.0) + -5.872 \cdot \sin(10.0)$$

$$= -0.519 \text{ Kgf/mm.}$$

Compute [Rc]:

$$= \text{RL}/\cos(a) = 1113.523/\cos(10.0) = 1130.7012 \text{ mm.}$$

Compute [β -co]:

$$= (3 \cdot (1 - v^2) / (Rc^2 \cdot tC^2))^{1/4}$$

$$= (3 \cdot (1 - 0.3^2) / (1130.701^2 \cdot 36.5^2))^{1/4}$$

$$= 0.006 \text{ 1/mm.}$$

Compute with XL +ve [Ncs+]:

$$= Ns \cdot \cos(a) - Q \cdot \sin(a)$$

$$= -5.872 \cdot \cos(10.0) - 0.508 \cdot \sin(10.0)$$

$$= -5.871 \text{ Kgf/mm.}$$

Compute with XL -ve [Ncs-]:

$$= Ns \cdot \cos(a) - Q \cdot \sin(a)$$

$$= -5.872 \cdot \cos(10.0) - 0.508 \cdot \sin(10.0)$$

$$= -5.871 \text{ Kgf/mm.}$$

Compute with XL +ve [NcTheta+]:

$$= P \cdot \text{RL}/\cos(a) + 2 \cdot \beta_{co} \cdot Rc \cdot (-\text{Mcs} \cdot \beta_{co} - Qc)$$

$$= -1.034 \cdot 1113.523/\cos(10.0) + 2 \cdot 0.006 \cdot 1130.701 \cdot$$

$$(-38.424 \cdot 0.006 - -0.519)$$

$$= -7.972 \text{ Kgf/mm.}$$

Compute with XL -ve [NcTheta-]:

$$= P \cdot \text{RL}/\cos(A) + 2 \cdot \beta_{co} \cdot Rc \cdot (-\text{Mcs} \cdot \beta_{co} - Qc)$$

$$= -1.034 \cdot 1113.523/\cos(10.0) + 2 \cdot 0.006 \cdot 1130.701 \cdot$$

$$(-38.424 \cdot 0.006 - -0.519)$$

$$= -7.972 \text{ Kgf/mm.}$$

Note: $Kcpc = 1.0$

Compute with XL +ve [σ_{sm}]:

$$= \text{Ncs}/tC = -5.871/36.5 = -1.5773 \text{ N./mm}^2$$

Compute with XL -ve [σ_{sm}]:

$$= \text{Ncs}/tC = -5.871/36.5 = -15.7726 \text{ N./mm}^2$$

Compute with XL +ve [σ_{sb}]:

$$= 6 \cdot \text{Mcs}/(tC^2 \cdot Kcpc)$$

$$= 6 \cdot 38.424/(36.5^2 \cdot 1.0)$$

$$= 1.697 \text{ N./mm}^2$$

Compute with XL -ve [σ_{sb}]:

$$= 6 \cdot \text{Mcs}/(tC^2 \cdot Kcpc)$$

$$= 6 \cdot 38.424/(36.5^2 \cdot 1.0)$$



$$= 1.697 \text{ N./mm}^2$$

Compute with XL +ve [$\sigma_{\text{Thetam+}}$]:
 $= \text{NCTheta/tC} = -7.972/36.5 = -2.14 \text{ N./mm}^2$

Compute with XL -ve [$\sigma_{\text{Thetam-}}$]:
 $= \text{NCTheta/tC} = -7.972/36.5 = -2.14 \text{ N./mm}^2$

Compute with XL +ve [$\sigma_{\text{thetab+}}$]:
 $= 6 \cdot v \cdot \text{Mcs} / (tC^2 \cdot Kcpc)$
 $= 6 \cdot 0.3 \cdot 38.424 / (36.5^2 \cdot 1.0)$
 $= 0.509 \text{ N./mm}^2$

Compute with XL -ve [$\sigma_{\text{thetab-}}$]:
 $= 6 \cdot v \cdot \text{Mcs} / (tC^2 \cdot Kcpc)$
 $= 6 \cdot 0.3 \cdot 38.424 / (36.5^2 \cdot 1.0)$
 $= 0.509 \text{ N./mm}^2$

Calculations for the Small End Cone/Shell Junction

Fhe per 4.1.3 or 4.3.11.2 Step 6 [Fhe]:
 $= 0.4 \cdot E(tc/D)$
 $= 0.4 \cdot 0.18730E+09(36.5/2000.0)$
 $= 1367.4 \text{ N./mm}^2$

Factor of Safety [FS]:
 Since $Fic [247.68] \geq Fy [247.68]$, FS = 1.667

Allowable Stress per 4.3.11 Step 6 [Fha]:
 $= Fic/FS$
 $= 247.6785/1.667$
 $= 148.577 \text{ N./mm}^2$

Small End under Internal Pressure:

Results for Cylinder Small End Without a Flare:

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	75.918	247.678	Pass
Sigma sm-	75.918	247.678	Pass
Sigma thetam+	190.062	247.678	Pass
Sigma thetam-	190.062	247.678	Pass
Sigma sm+ + Sigma	156.455	495.357	Pass
Sigma sm- + Sigma	156.455	495.357	Pass
Sigma sm+ - Sigma	-4.620	495.357	Pass
Sigma sm- - Sigma	-4.620	495.357	Pass
Sigma thetam+ +	214.223	495.357	Pass
Sigma thetam- +	214.223	495.357	Pass
Sigma thetam+ -	165.901	495.357	Pass
Sigma thetam- -	165.901	495.357	Pass

Results for the Cone Small End Without a Flare:

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	67.481	247.678	Pass
Sigma sm-	67.481	247.678	Pass
Sigma thetam+	189.818	247.678	Pass
Sigma thetam-	189.818	247.678	Pass
Sigma sm+ + Sigma	131.334	495.357	Pass
Sigma sm- + Sigma	131.334	495.357	Pass
Sigma sm+ - Sigma	3.629	495.357	Pass
Sigma sm- - Sigma	3.629	495.357	Pass
Sigma thetam+ +	208.973	495.357	Pass
Sigma thetam- +	208.973	495.357	Pass
Sigma thetam+ -	170.662	495.357	Pass



Sigma thetam- - 170.662 495.357 Pass

Small End Cylinder Without a Flare

Compute the ratio equation 4.3.60 [n]:
 $= t_c/t_s = 36.5/32.5 = 1.1231$

Compute the ratio equation 4.3.61 [H]:
 $= \sqrt{(R_s/t_s)} = \sqrt{(967.523/32.5)} = 5.4562$

Compute the ratio equation 4.3.62 [B]:
 $= \tan(a) = \tan(10.0) = 0.1763$

Compute Ms+ from equation 4.3.59 [Xs+]:
 $= F_s / (2 \cdot \pi \cdot R_s) + M_s / (\pi \cdot R_s^2)$
 $= 0.0 / (2 \cdot 3.142 \cdot 967.523) + 0.0 / (3.141 \cdot 967.523^2)$
 $= 0.0000 \text{ Kgf/mm.}$

Compute Ms- from equation 4.3.59 [Xs-]:
 $= F_s / (2 \cdot \pi \cdot R_s) - M_s / (\pi \cdot R_s^2) + X_s'$
 $= 0.0 / (2 \cdot 3.142 \cdot 967.523) - 0.0 / (3.141 \cdot 967.523^2) + 0.0$
 $= 0.0000 \text{ Kgf/mm.}$

From Table 4.3.5 MsN:

C1 = -9.861689 C2 = 1.805782 C3 = 4.078004 C4 = -0.048448
 C5 = -1.067551 C6 = -0.021232 C7 = 0.004183 C8 = 0.118384
 C9 = 0.014479 C10 = -0.006556 C11 = 0.000000

Compute from Table 4.3.5 [MsN]:
 $= \exp[C1 + C2 \cdot \ln(H^2) + C3 \cdot \ln(a) + C4 \cdot (\ln(H^2))^2 + C5 \cdot \ln(a)^2 + C6 \cdot \ln(H^2) \cdot \ln(a) +$
 $C7 \cdot \ln(H^2)^3 + C8 \cdot \ln(a)^3 + C9 \cdot \ln(H^2) \cdot \ln(a)^2 + C10 \cdot \ln(H^2)^2 \cdot \ln(a)]$
 $= \exp[-9.862 + 1.806 \cdot \ln(5.456^2) + 4.078 \cdot \ln(10.0) +$
 $-0.048 \cdot (\ln(5.456^2))^2 + 1.068 \cdot \ln(10.0)^2 +$
 $-0.021 \cdot \ln(5.456^2) \cdot \ln(10.0) + 0.004 \cdot \ln(5.456^2)^3$
 $0.118 \cdot \ln(10.0)^3 + 0.014 \cdot \ln(5.456^2) \cdot \ln(10.0)^2 +$
 $-0.007 \cdot \ln(5.456^2)^2 \cdot \ln(10.0)] = 2.6318$

Compute from Table 4.3.2 [Msp]:
 $= p \cdot t_L^2 \cdot M_{sn} = 51.0 \cdot 32.5^2 \cdot 2.63179 = 1445.7216 \text{ Kgf}$

From table 4.3.6 MsN:

C1 = -0.000381 C2 = -0.000479 C3 = -0.001205 C4 = 0.137325
 C5 = 0.010227 C6 = 0.000010 C7 = -0.000004 C8 = 0.009263
 C9 = 0.005879 C10 = -0.000018 C11 = 0.205203

Compute from Table 4.3.6 [MsN]:
 $= (C1 + C3 \cdot H + C5 \cdot B + C7 \cdot H^2 + C9 \cdot B^2 + C11 \cdot H \cdot B) /$
 $(1 + C2 \cdot H + C4 \cdot B + C6 \cdot H^2 + C8 \cdot B^2 + C10 \cdot H \cdot B)$
 $= (-0.00038 + -0.00121 \cdot 5.456 + 0.01023 \cdot 10.0 +$
 $-0.00048 \cdot 5.456^2 + 0.00588 \cdot 10.0^2 +$
 $0.2052 \cdot 5.456 \cdot 10.0) / (1 + -0.00048 \cdot 5.456 +$
 $0.13732 \cdot 10.0 + 0.00001 \cdot 5.456^2 + 0.00926 \cdot 10.0^2 +$
 $-0.00002 \cdot 5.456 \cdot 10.0)$
 $= 0.1882$

Compute from Table 4.3.2 [Msx+]:
 $= X_s \cdot t_s \cdot M_{sn} = 0.0 \cdot 32.5 \cdot 0.188 = 0.0 \text{ Kgf}$

Compute from Table 4.3.2 [Msx-]:
 $= X_s \cdot t_s \cdot M_{sn} = 0.0 \cdot 32.5 \cdot 0.188 = 0.0 \text{ Kgf}$

Compute from Table 4.3.2 [Ms+]:
 $= M_s + M_{sx+} = 1445.722 + 0.0 = 1445.7216 \text{ Kgf}$

Compute from Table 4.3.2 [Ms-]:
 $= M_s + M_{sx-} = 1445.722 + 0.0 = 1445.7216 \text{ Kgf}$

From table 4.3.5 Qn:



C1 = 0.116462 C2 = 0.000014 C3 = 0.002745 C4 = 0.002937
 C5 = 0.000409 C6 = 0.000000 C7 = 0.000000 C8 = -0.000121
 C9 = -0.000160 C10 = 0.000001 C11 = -0.004573

Compute from Table 4.3.5 [Qn]:

$$\begin{aligned}
 &= (C1 + C3 \cdot H^2 + C5 \cdot \alpha + C7 \cdot H^4 + C9 \cdot \alpha^2 + C11 \cdot H^2 \cdot \alpha) / \\
 &\quad (1 + C2 \cdot H^2 + C4 \cdot \alpha + C6 \cdot H^4 + C8 \cdot \alpha^2 + C10 \cdot H^2 \cdot \alpha) \\
 &= (0.11646 + 0.00275 \cdot 5.456^2 + 0.00041 \cdot 10.0 \\
 &\quad 0.5 \cdot 456^4 + -0.00016 \cdot 10.0^2 + -0.00457 \cdot 5.456^2 \\
 &\quad * 10.0) / (1 + 0.00001 \cdot 5.456^2 + 0.00294 \cdot 10.0 + \\
 &\quad 0.5 \cdot 456^4 + -0.00012 \cdot 10.0^2 + 0.000001 \cdot 5.456^2 \cdot 10.0) \\
 &= -1.154
 \end{aligned}$$

Compute from Table 4.3.2 [Qp]:

$$= P \cdot ts \cdot Qn = 51.0 \cdot 32.5 \cdot -1.154 = -19.5113 \text{ Kgf/mm.}$$

From table 4.3.6 Qn:

C1 = -0.452795 C2 = 0.002095 C3 = -0.388706 C4 = 0.000456
 C5 = -0.134850 C6 = 0.001363 C7 = -0.000123 C8 = -0.018113
 C9 = 0.000499 C10 = 0.000055 C11 = 0.000000

Compute from Table 4.3.6 [Qn]:

$$\begin{aligned}
 &= C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot \ln(H)^2 + C5 \cdot \ln(B)^2 + C6 \cdot \ln(H) \cdot \ln(B) + \\
 &\quad C7 \cdot \ln(H)^3 + C8 \cdot \ln(B)^3 + C9 \cdot \ln(H) \cdot \ln(B)^2 + C10 \cdot \ln(H)^2 \cdot \ln(B) \\
 &= -0.4528 + 0.0021 \cdot \ln(5.456) + -0.38871 \cdot \ln(0.176) + \\
 &\quad 0.00046 \cdot \ln(5.456)^2 + -0.13485 \cdot (\ln(0.176))^2 + \\
 &\quad 0.00136 \cdot \ln(5.456) \cdot \ln(0.176) + -0.00012 \cdot \ln(5.456)^3 + \\
 &\quad -0.01811 \cdot \ln(0.176)^3 + 0.0005 \cdot \ln(5.456) \cdot \ln(0.176)^2 + \\
 &\quad 0.00006 \cdot \ln(5.456)^2 \cdot \ln(0.176) \\
 &= -0.0872
 \end{aligned}$$

Compute with Xs+ [Qx+]:

$$= Xs+ \cdot Qn = 0.0 \cdot -0.087 = -0.0 \text{ Kgf/mm.}$$

Compute with Xs- [Qx-]:

$$= Xs- \cdot Qn = 0.0 \cdot -0.087 = -0.0 \text{ Kgf/mm.}$$

Compute with Xs+ [Q+]:

$$= Qp + Qx+ = -19.511 + -0.0 = -19.5113 \text{ Kgf/mm.}$$

Compute with Xs- [Q-]:

$$= Qp + Qx- = -19.511 + -0.0 = -19.5113 \text{ Kgf/mm.}$$

Compute angle [βcy]:

$$\begin{aligned}
 &= (3 \cdot (1 - v^2) / (Rs^2 \cdot ts^2))^{0.25} = (3 \cdot (1 - 0.3^2) / (967.523^2 \cdot 32.5^2))^{0.25} \\
 &= 0.0072 \text{ 1/mm.}
 \end{aligned}$$

Compute with Xs+ [Ns+]:

$$= P \cdot Rs / 2 + Xs+ = 51.0 \cdot 967.523 / 2 + 0.0+ = 251.5927 \text{ Kgf/mm.}$$

Compute with Xs- [Ns-]:

$$= P \cdot Rs / 2 + Xs- = 51.0 \cdot 967.523 / 2 + 0.0- = 251.5927 \text{ Kgf/mm.}$$

Compute with Xs+ [NTheta+]:

$$\begin{aligned}
 &= P \cdot Rs + 2 \cdot \beta_{cy} \cdot Rs \cdot (-Ms \cdot \beta_{cy} - Q) \\
 &= 51.0 \cdot 967.523 + 2 \cdot 0.007 \cdot 967.523 \cdot \\
 &\quad (-1445.722 \cdot 0.007 - 0.508) = 629.8683 \text{ Kgf/mm.}
 \end{aligned}$$

Compute with Xs- [NTheta-]:

$$\begin{aligned}
 &= P \cdot Rs + 2 \cdot \beta_{cy} \cdot Rs \cdot (-Ms \cdot \beta_{cy} - Q) \\
 &= 51.0 \cdot 967.523 + 2 \cdot 0.007 \cdot 967.523 \cdot \\
 &\quad (-1445.722 \cdot 0.007 - 0.508) = 629.8683 \text{ Kgf/mm.}
 \end{aligned}$$

Compute Stress with Xs+ [σsm+]:

$$= Ns(+) / ts = 251.593 / 32.5 = 75.9178 \text{ N./mm}^2$$

Compute Stress with Xs- [σsm-]:

$$= Ns(-) / ts = 251.593 / 32.5 = 75.9178 \text{ N./mm}^2$$



Compute stress with Xs+ [σ_{sb+}]:

$$= 6 \cdot M_s(+)/(t s^2 \cdot K_{pc}) = 6 \cdot 1445.722/(32.5^2 \cdot 1.0) = 80.5374 \text{ N./mm}^2$$

Compute stress with Xs- [σ_{sb-}]:

$$= 6 \cdot M_s(-)/(t s^2 \cdot K_{pc}) = 6 \cdot 1445.722/(32.5^2 \cdot 1.0) = 80.5374 \text{ N./mm}^2$$

Compute stress with Xs+ [$\sigma_{\theta_{etam+}}$]:

$$= N_{\theta_{etam}}(+)/t s = 629.868/32.5 = 190.0619 \text{ N./mm}^2$$

Compute stress with Xs- [$\sigma_{\theta_{etam-}}$]:

$$= N_{\theta_{etam}}(-)/t s = 629.868/32.5 = 190.0619 \text{ N./mm}^2$$

Compute stress with Xs+ [$\sigma_{\theta_{etab+}}$]:

$$= 6 \cdot v \cdot M_s(+)/(t s^2 \cdot K_{pc}) = 6 \cdot 0.3 \cdot 1445.722/(32.5^2 \cdot 1.0) = 24.1612 \text{ N./mm}^2$$

Compute stress with Xs- [$\sigma_{\theta_{etab-}}$]:

$$= 6 \cdot v \cdot M_s(-)/(t s^2 \cdot K_{pc}) = 6 \cdot 0.3 \cdot 1445.722/(32.5^2 \cdot 1.0) = 24.1612 \text{ N./mm}^2$$

Small End Cone Without a Flare

Compute with Xs+ [Q_{c+}]:

$$= Q \cdot \cos(a) + N_s \cdot \sin(a) = -19.5 \cdot \cos(10.0) + 251.6 \cdot \sin(10.0) \\ = 24.4738 \text{ Kg/mm.}$$

Compute with Xs- [Q_{c-}]:

$$= Q \cdot \cos(a) + N_s \cdot \sin(a) = -19.5 \cdot \cos(10.0) + 251.6 \cdot \sin(10.0) \\ = 24.4738 \text{ Kg/mm.}$$

Compute effective radius [R_c]:

$$= R_s/\cos(a) = 967.523/\cos(10.0) = 982.4489 \text{ mm.}$$

Compute the angle [β_{co}]:

$$= (3 \cdot (1 - v^2)/(R_c^2 \cdot t c^2))^{0.25} = (3 \cdot (1 - 0.3^2)/(982.449^2 \cdot 36.5^2))^{0.25} \\ = 0.0068 \text{ 1/mm.}$$

Compute with Ms+ [N_{cs+}]:

$$= N_s \cdot \cos(a) - Q \cdot \sin(a) \\ = 251.593 \cdot \cos(10.0) - 19.511 \cdot \sin(10.0) = 251.1586 \text{ Kg/mm.}$$

Compute with Ms- [N_{cs-}]:

$$= N_s \cdot \cos(a) - Q \cdot \sin(a) \\ = 251.593 \cdot \cos(10.0) - 19.511 \cdot \sin(10.0) = 251.1586 \text{ Kg/mm.}$$

Compute with Ms+ [$N_{\theta_{etab+}}$]:

$$= P \cdot R_s/\cos(a) + 2 \cdot \beta_{co} \cdot R_c(-M_{cs} \cdot \beta_{co} + Q_c) \\ = 51.0 \cdot 967.523/\cos(10.0) + 2 \cdot 0.0068 \cdot 982.449 \cdot \\ (-1445.722 \cdot 0.0068 + 24.474) = 706.4816 \text{ Kg/mm.}$$

Compute with Ms- [$N_{\theta_{etab-}}$]:

$$= P \cdot R_s/\cos(a) + 2 \cdot \beta_{co} \cdot R_c(-M_{cs} \cdot \beta_{co} + Q_c) \\ = 51.0 \cdot 967.523/\cos(10.0) + 2 \cdot 0.0068 \cdot \\ (-1445.722 \cdot 0.007 + 24.474) = 706.4816 \text{ Kg/mm.}$$

Compute the stress with Ms+ [σ_{sm+}]:

$$= N_{cs}/t c = 251.159/36.5 = 67.4814 \text{ N./mm}^2$$

Compute the stress with Ms- [σ_{sm-}]:

$$= N_{cs}/t c = 251.159/36.5 = 67.4814 \text{ N./mm}^2$$

Compute the stress with Ms+ [σ_{sb+}]:

$$= 6 \cdot M_{cs}/(t c^2 \cdot K_{cpc}) = 6 \cdot 1445.722/(36.5^2 \cdot 1.0) = 63.8526 \text{ N./mm}^2$$

Compute the stress with Ms- [σ_{sb-}]:

$$= 6 \cdot M_{cs}/(t c^2 \cdot K_{cpc}) = 6 \cdot 1445.722/(36.5^2 \cdot 1.0) = 63.8526 \text{ N./mm}^2$$

Compute stress with Ms+ [$\sigma_{\theta_{etam+}}$]:

$$= N_{\theta_{etam}}/t c = 706.482/36.5 = 189.818 \text{ N./mm}^2$$



Compute stress with Ms- [$\sigma_{\theta\theta\theta m-}$]:

$$= N_{\theta\theta\theta}/t_c = 706.482/36.5 = 189.818 \text{ N./mm}^2$$

Compute stress with Ms+ [$\sigma_{\theta\theta\theta b+}$]:

$$= 6 \cdot v \cdot M_{cs}/(t_c^2 \cdot K_{cpc}) = 6 \cdot 0.3 \cdot 1445.722/(36.5^2 \cdot 1.0) = 19.1558 \text{ N./mm}^2$$

Compute stress with Ms- [$\sigma_{\theta\theta\theta b-}$]:

$$= 6 \cdot v \cdot M_{cs}/(t_c^2 \cdot K_{cpc}) = 6 \cdot 0.3 \cdot 1445.722/(36.5^2 \cdot 1.0) = 19.1558 \text{ N./mm}^2$$

Small End under External Pressure:

Results for Cylinder Small End Without a Flare:

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	-1.539	247.678	Pass
Sigma sm-	-1.539	247.678	Pass
Sigma thetam+	-3.854	247.678	Pass
Sigma thetam-	-3.854	247.678	Pass
Sigma sm+ + Sigma	-3.173	495.357	Pass
Sigma sm- + Sigma	-3.173	495.357	Pass
Sigma sm+ - Sigma	0.094	495.357	Pass
Sigma sm- - Sigma	0.094	495.357	Pass
Sigma thetam+ +	-4.344	495.357	Pass
Sigma thetam- +	-4.344	495.357	Pass
Sigma thetam+ -	-3.364	495.357	Pass
Sigma thetam- -	-3.364	495.357	Pass

Results for the Cone Small End Without a Flare:

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	-1.368	247.678	Pass
Sigma sm-	-1.368	247.678	Pass
Sigma thetam+	-3.849	247.678	Pass
Sigma thetam-	-3.849	247.678	Pass
Sigma sm+ + Sigma	-2.663	495.357	Pass
Sigma sm- + Sigma	-2.663	495.357	Pass
Sigma sm+ - Sigma	-0.074	495.357	Pass
Sigma sm- - Sigma	-0.074	495.357	Pass
Sigma thetam+ +	-4.238	495.357	Pass
Sigma thetam- +	-4.238	495.357	Pass
Sigma thetam+ -	-3.461	495.357	Pass
Sigma thetam- -	-3.461	495.357	Pass

Average Compressive Stresses and comparisons with Fha and Fxa:

Sigma _{asmCyl} + + S	-1.454	-170.885	Pass
Sigma _{asmCyl} - + S	-1.454	-170.885	Pass
Sigma _{thetamCyl} +	-3.852	-148.577	Pass
Sigma _{thetamCyl} -	-3.852	-148.577	Pass

Small End Cylinder Without a Flare

Compute the ratio equation 4.3.60 [n]:

$$= t_c/t_s = 36.5/32.5 = 1.1231$$

Compute the ratio equation 4.3.61 [H]:

$$= \sqrt{(R_s/t_s)} = \sqrt{(967.523/32.5)} = 5.4562$$

Compute the ratio equation 4.3.62 [B]:

$$= \tan(a) = \tan(10.0) = 0.1763$$

Compute Ms+ from equation 4.3.59 [Xs+]:

$$= F_s / (2 \cdot \pi \cdot R_s) + M_s / (\pi \cdot R_s^2) \\ = 0.0 / (2 \cdot 3.142 \cdot 967.523) + 0.0 / (3.141 \cdot 967.523^2) \\ = 0.0000 \text{ Kg/mm.}$$



Compute Ms- from equation 4.3.59 [Xs-]:

$$= F_s / (2 \cdot \pi \cdot R_s) - M_s / (\pi \cdot R_s^2) + X_s'$$

$$= 0.0 / (2 \cdot 3.142 \cdot 967.523) - 0.0 / (3.141 \cdot 967.523^2) + 0.0$$

$$= 0.0000 \text{ Kgf/mm.}$$

From Table 4.3.5 MsN:

$$\begin{aligned} C1 &= -9.861689 & C2 &= 1.805782 & C3 &= 4.078004 & C4 &= -0.048448 \\ C5 &= -1.067551 & C6 &= -0.021232 & C7 &= 0.004183 & C8 &= 0.118384 \\ C9 &= 0.014479 & C10 &= -0.006556 & C11 &= 0.000000 \end{aligned}$$

Compute from Table 4.3.5 [MsN]:

$$= \exp[C1 + C2 \cdot \ln(H^2) + C3 \cdot \ln(a) + C4 \cdot (\ln(H^2))^2 + C5 \cdot \ln(a)^2 + C6 \cdot \ln(H^2) \cdot \ln(a) + C7 \cdot \ln(H^2)^3 + C8 \cdot \ln(a)^3 + C9 \cdot \ln(H^2) \cdot \ln(a)^2 + C10 \cdot \ln(H^2)^2 \cdot \ln(a)]$$

$$= \exp[-9.862 + 1.806 \cdot \ln(5.456^2) + 4.078 \cdot \ln(10.0) +$$

$$-0.048 \cdot (\ln(5.456^2))^2 - 1.068 \cdot \ln(10.0)^2 +$$

$$-0.021 \cdot \ln(5.456^2) \cdot \ln(10.0) + 0.004 \cdot \ln(5.456^2)^3$$

$$+ 0.118 \cdot \ln(10.0)^3 + 0.014 \cdot \ln(5.456^2) \cdot \ln(10.0)^2 +$$

$$-0.007 \cdot \ln(5.456^2)^2 \cdot \ln(10.0)] = 2.6318$$

Compute from Table 4.3.2 [Msp]:

$$= p \cdot tL^2 \cdot M_{sn} = -1.034 \cdot 32.5^2 \cdot 2.63179 = -29.3167 \text{ Kgf}$$

From table 4.3.6 MsN:

$$\begin{aligned} C1 &= -0.000381 & C2 &= -0.000479 & C3 &= -0.001205 & C4 &= 0.137325 \\ C5 &= 0.010227 & C6 &= 0.000010 & C7 &= -0.000004 & C8 &= 0.009263 \\ C9 &= 0.005879 & C10 &= -0.000018 & C11 &= 0.205203 \end{aligned}$$

Compute from Table 4.3.6 [MsN]:

$$= (C1 + C3 \cdot H + C5 \cdot B + C7 \cdot H^2 + C9 \cdot B^2 + C11 \cdot H \cdot B) /$$

$$(1 + C2 \cdot H + C4 \cdot B + C6 \cdot H^2 + C8 \cdot B^2 + C10 \cdot H \cdot B)$$

$$= (-0.00038 + -0.00121 \cdot 5.456 + 0.01023 \cdot 10.0 +$$

$$-0.005456^2 + 0.00588 \cdot 10.0^2 +$$

$$0.2052 \cdot 5.456 \cdot 10.0) / (1 + -0.00048 \cdot 5.456 +$$

$$0.13732 \cdot 10.0 + 0.00001 \cdot 5.456^2 + 0.00926 \cdot 10.0^2 +$$

$$-0.00002 \cdot 5.456 \cdot 10.0)$$

$$= 0.1882$$

Compute from Table 4.3.2 [Msx+]:

$$= X_s \cdot t_s \cdot M_{sN} = 0.0 \cdot 32.5 \cdot 0.188 = 0.0 \text{ Kgf}$$

Compute from Table 4.3.2 [Msx-]:

$$= X_s \cdot t_s \cdot M_{sN} = 0.0 \cdot 32.5 \cdot 0.188 = 0.0 \text{ Kgf}$$

Compute from Table 4.3.2 [Ms+]:

$$= M_s + M_{sx+} = -29.317 + 0.0 = -29.3167 \text{ Kgf}$$

Compute from Table 4.3.2 [Ms-]:

$$= M_s + M_{sx-} = -29.317 + 0.0 = -29.3167 \text{ Kgf}$$

From table 4.3.5 Qn:

$$\begin{aligned} C1 &= 0.116462 & C2 &= 0.000014 & C3 &= 0.002745 & C4 &= 0.002937 \\ C5 &= 0.000409 & C6 &= 0.000000 & C7 &= 0.000000 & C8 &= -0.000121 \\ C9 &= -0.000160 & C10 &= 0.000001 & C11 &= -0.004573 \end{aligned}$$

Compute from Table 4.3.5 [Qn]:

$$= (C1 + C3 \cdot H^2 + C5 \cdot \alpha + C7 \cdot H^4 + C9 \cdot \alpha^2 + C11 \cdot H^2 \cdot \alpha) /$$

$$(1 + C2 \cdot H^2 + C4 \cdot \alpha + C6 \cdot H^4 + C8 \cdot \alpha^2 + C10 \cdot H^2 \cdot \alpha)$$

$$= (0.11646 + 0.00275 \cdot 5.456^2 + 0.00041 \cdot 10.0$$

$$+ 0.00000 \cdot 5.456^4 + -0.00016 \cdot 10.0^2 + -0.00457 \cdot 5.456^2$$

$$+ 0.00001 \cdot 5.456^2 \cdot 10.0) / (1 + 0.00001 \cdot 5.456^2 + 0.00294 \cdot 10.0 +$$

$$0.00000 \cdot 5.456^4 + -0.00012 \cdot 10.0^2 + 0.00001 \cdot 5.456^2 \cdot 10.0)$$

$$= -1.154$$

Compute from Table 4.3.2 [Qp]:

$$= P \cdot t_s \cdot Q_n = -1.034 \cdot 32.5 \cdot -1.154 = 0.3957 \text{ Kgf/mm.}$$

From table 4.3.6 Qn:



C1 = -0.452795 C2 = 0.002095 C3 = -0.388706 C4 = 0.000456
 C5 = -0.134850 C6 = 0.001363 C7 = -0.000123 C8 = -0.018113
 C9 = 0.000499 C10 = 0.000055 C11 = 0.000000

Compute from Table 4.3.6 [Qn]:

$$\begin{aligned}
 &= C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot \ln(H)^2 + C5 \cdot \ln(B)^2 + C6 \cdot \ln(H) \cdot \ln(B) + \\
 &\quad C7 \cdot \ln(H)^3 + C8 \cdot \ln(B)^3 + C9 \cdot \ln(H) \cdot \ln(B)^2 + C10 \cdot \ln(H)^2 \cdot \ln(B) \\
 &= -0.4528 + 0.0021 \cdot \ln(5.456) + -0.38871 \cdot \ln(0.176) + \\
 &\quad 0.00046 \cdot \ln(5.456)^2 + -0.13485 \cdot (\ln(0.176))^2 + \\
 &\quad 0.00136 \cdot \ln(5.456) \cdot \ln(0.176) + -0.00012 \cdot \ln(5.456)^3 + \\
 &\quad -0.01811 \cdot \ln(0.176)^3 + 0.0005 \cdot \ln(5.456) \cdot \ln(0.176)^2 + \\
 &\quad 0.00006 \cdot \ln(5.456)^2 \cdot \ln(0.176) \\
 &= -0.0872
 \end{aligned}$$

Compute with Xs+ [Qx+]:

$$= Xs+ \cdot Qn = 0.0 \cdot -0.087 = -0.0 \text{ Kg/mm.}$$

Compute with Xs- [Qx-]:

$$= Xs- \cdot Qn = 0.0 \cdot -0.087 = -0.0 \text{ Kg/mm.}$$

Compute with Xs+ [Q+]:

$$= Qp + Qx+ = 0.396 + -0.0 = 0.3957 \text{ Kg/mm.}$$

Compute with Xs- [Q-]:

$$= Qp + Qx- = 0.396 + -0.0 = 0.3957 \text{ Kg/mm.}$$

Compute angle [βcy]:

$$\begin{aligned}
 &= (3 \cdot (1 - v^2) / (Rs^2 \cdot ts^2))^{0.25} = (3 \cdot (1 - 0.3^2) / (967.523^2 \cdot 32.5^2))^{0.25} \\
 &= 0.0072 \text{ 1/mm.}
 \end{aligned}$$

Compute with Xs+ [Ns+]:

$$= P \cdot Rs / 2 + Xs+ = -1.034 \cdot 967.523 / 2 + 0.0+ = -5.1019 \text{ Kg/mm.}$$

Compute with Xs- [Ns-]:

$$= P \cdot Rs / 2 + Xs- = -1.034 \cdot 967.523 / 2 + 0.0- = -5.1019 \text{ Kg/mm.}$$

Compute with Xs+ [NTheta+]:

$$\begin{aligned}
 &= P \cdot Rs + 2 \cdot \beta_{cy} \cdot Rs \cdot (-Ms \cdot \beta_{cy} - Q) \\
 &= -1.034 \cdot 967.523 + 2 \cdot 0.007 \cdot 967.523 \cdot \\
 &\quad (-29.317 \cdot 0.007 - 0.508) = -12.7726 \text{ Kg/mm.}
 \end{aligned}$$

Compute with Xs- [NTheta-]:

$$\begin{aligned}
 &= P \cdot Rs + 2 \cdot \beta_{cy} \cdot Rs \cdot (-Ms \cdot \beta_{cy} - Q) \\
 &= -1.034 \cdot 967.523 + 2 \cdot 0.007 \cdot 967.523 \cdot \\
 &\quad (-29.317 \cdot 0.007 - 0.508) = -12.7726 \text{ Kg/mm.}
 \end{aligned}$$

Compute Stress with Xs+ [σsm+]:

$$= Ns(+) / ts = -5.102 / 32.5 = -1.5395 \text{ N./mm}^2$$

Compute Stress with Xs- [σsm-]:

$$= Ns(-) / ts = -5.102 / 32.5 = -1.5395 \text{ N./mm}^2$$

Compute stress with Xs+ [σsb+]:

$$= 6 \cdot Ms(+) / (ts^2 \cdot Kpc) = 6 \cdot -29.317 / (32.5^2 \cdot 1.0) = -1.6332 \text{ N./mm}^2$$

Compute stress with Xs- [σsb-]:

$$= 6 \cdot Ms(-) / (ts^2 \cdot Kpc) = 6 \cdot -29.317 / (32.5^2 \cdot 1.0) = -1.6332 \text{ N./mm}^2$$

Compute stress with Xs+ [σthetam+]:

$$= N_{\theta}(+) / ts = -12.773 / 32.5 = -3.8541 \text{ N./mm}^2$$

Compute stress with Xs- [σthetam-]:

$$= N_{\theta}(-) / ts = -12.773 / 32.5 = -3.8541 \text{ N./mm}^2$$

Compute stress with Xs+ [σthetab+]:

$$= 6 \cdot v \cdot Ms(+) / (ts^2 \cdot Kpc) = 6 \cdot 0.3 \cdot -29.317 / (32.5^2 \cdot 1.0) = -0.4899 \text{ N./mm}^2$$

Compute stress with Xs- [σthetab-]:



$$= 6 \cdot v \cdot Ms(-) / (ts^2 \cdot Kpc) = 6 \cdot 0.3 \cdot -29.317 / (32.5^2 \cdot 1.0) = -0.4899 \text{ N./mm}^2$$

Small End Cone Without a Flare

Compute with Xs+ [Qc+]:

$$= Q \cdot \cos(a) + Ns \cdot \sin(a) = 0.4 \cdot \cos(10.0) + -5.1 \cdot \sin(10.0) \\ = -0.4963 \text{ Kg/mm.}$$

Compute with Xs- [Qc-]:

$$= Q \cdot \cos(a) + Ns \cdot \sin(a) = 0.4 \cdot \cos(10.0) + -5.1 \cdot \sin(10.0) \\ = -0.4963 \text{ Kg/mm.}$$

Compute effective radius [Rc]:

$$= Rs / \cos(a) = 967.523 / \cos(10.0) = 982.4489 \text{ mm.}$$

Compute the angle [βco]:

$$= (3 \cdot (1 - v^2) / (Rc^2 \cdot tc^2))^{0.25} = (3 \cdot (1 - 0.3^2) / (982.449^2 \cdot 36.5^2))^{0.25} \\ = 0.0068 \text{ 1/mm.}$$

Compute with Ms+ [Ncs+]:

$$= Ns \cdot \cos(a) - Q \cdot \sin(a) \\ = -5.102 \cdot \cos(10.0) - 0.396 \cdot \sin(10.0) = -5.0931 \text{ Kg/mm.}$$

Compute with Ms- [Ncs-]:

$$= Ns \cdot \cos(a) - Q \cdot \sin(a) \\ = -5.102 \cdot \cos(10.0) - 0.396 \cdot \sin(10.0) = -5.0931 \text{ Kg/mm.}$$

Compute with Ms+ [Nctheta+]:

$$= P \cdot Rs / \cos(a) + 2 \cdot \beta_{co} \cdot Rc \cdot (-Mcs \cdot \beta_{co} + Qc) \\ = -1.034 \cdot 967.523 / \cos(10.0) + 2 \cdot 0.0068 \cdot 982.449 \cdot \\ (-29.317 \cdot 0.0068 + -0.496) = -14.3262 \text{ Kg/mm.}$$

Compute with Ms- [Nctheta-]:

$$= P \cdot Rs / \cos(a) + 2 \cdot \beta_{co} \cdot Rc \cdot (-Mcs \cdot \beta_{co} + Qc) \\ = -1.034 \cdot 967.523 / \cos(10.0) + 2 \cdot 0.0068 \cdot \\ (-29.317 \cdot 0.007 + -0.496) = -14.3262 \text{ Kg/mm.}$$

Compute the stress with Ms+ [σsm+]:

$$= Ncs / tc = -5.093 / 36.5 = -1.3684 \text{ N./mm}^2$$

Compute the stress with Ms- [σsm-]:

$$= Ncs / tc = -5.093 / 36.5 = -1.3684 \text{ N./mm}^2$$

Compute the stress with Ms+ [σsb+]:

$$= 6 \cdot Mcs / (tc^2 \cdot Kcpc) = 6 \cdot -29.317 / (36.5^2 \cdot 1.0) = -1.2948 \text{ N./mm}^2$$

Compute the stress with Ms- [σsb-]:

$$= 6 \cdot Mcs / (tc^2 \cdot Kcpc) = 6 \cdot -29.317 / (36.5^2 \cdot 1.0) = -1.2948 \text{ N./mm}^2$$

Compute stress with Ms+ [σthetam+]:

$$= Nctheta / tc = -14.326 / 36.5 = -3.849 \text{ N./mm}^2$$

Compute stress with Ms- [σthetam-]:

$$= Nctheta / tc = -14.326 / 36.5 = -3.849 \text{ N./mm}^2$$

Compute stress with Ms+ [σthetab+]:

$$= 6 \cdot v \cdot Mcs / (tc^2 \cdot Kcpc) = 6 \cdot 0.3 \cdot -29.317 / (36.5^2 \cdot 1.0) = -0.3884 \text{ N./mm}^2$$

Compute stress with Ms- [σthetab-]:

$$= 6 \cdot v \cdot Mcs / (tc^2 \cdot Kcpc) = 6 \cdot 0.3 \cdot -29.317 / (36.5^2 \cdot 1.0) = -0.3884 \text{ N./mm}^2$$



Shop/Field Installation Options :

Note : The CG is computed from the first Element From Node

Center of Gravity of the Nozzles	-164.224 mm.
Center of Gravity of the Added Weights (Operating)	0.000 mm.
Center of Gravity of the Added Weights (Empty)	0.000 mm.
Center of Gravity of the Tubesheet(s)	1550.000 mm.
Center of Gravity of the Tubes	4000.000 mm.
Center of Gravity of Bare Shell New and Cold	3067.281 mm.
Center of Gravity of Bare Shell Corroded	3028.142 mm.
Vessel CG in the Operating Condition	2593.749 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	2626.749 mm.

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**Input, Nozzle Desc: M1-24"****From: 10**

Pressure for Reinforcement Calculations	P	17.356	bars
Temperature for Internal Pressure	Temp	286	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	286	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.12	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Hemispherical Head	D	2429.00	mm.
Head Finished (Minimum) Thickness	t	75.0000	mm.
Head Internal Corrosion Allowance	cas	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

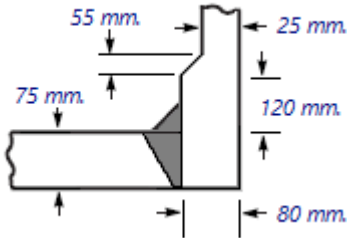
Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	165.12	N./mm ²
Allowable Stress At Ambient	Sna	207.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		22.0315	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	25.0000	mm.
Flange Material		SA-336 F11	
Hub Height of Integral Nozzle	h	120.0000	mm.
Height of Beveled Transition	L'	55.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: M1-24"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	22.031	in.
Actual Thickness Used in Calculation	0.984	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 562.6 \cdot (\exp(17.356/(165.119 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 2.9726 + 1.5 + 0.0 = 4.4725 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 165.119 \cdot 1.0 \cdot \ln((2 \cdot 23.5 + 562.6) / 562.6) - 0.0$$

$$= 132.474 - 0.0 = 132.474 \text{ bars}$$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[165.1/165.1, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 80.0 - 1.5$$

$$= 78.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 25.0 - 1.5$$

$$= 23.500 \text{ mm.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2429.0/2 + 1.5$$

$$= 1216.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1216.0 \cdot 73.5}, 2 \cdot 281.3)$$

$$= 298.958 \text{ mm.}$$

Intermediate value [Cn]:

$$= \min(((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min(((73.5 + 0.0)/78.5)^{0.35}, 1)$$

$$= \min(0.977, 1)$$

$$= 0.977$$

Intermediate value [Cp]:

$$= 9999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t + t_e + F_p \cdot \sqrt{R_n \cdot t_n}, L_{pr1} + t)$$

$$= \min(73.5 + 0.0 + 0.977 \cdot \sqrt{281.3 \cdot 78.5}, 300.0 + 73.5)$$

$$= 218.716 \text{ mm.}$$



Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 73.5 + ((0.0 \cdot 1.0) / 298.958)$$

$$= 73.500 \text{ mm.}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$

$$= 73.5 \cdot 298.958$$

$$= 219.734 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$

$$= 151.898 + 16.616$$

$$= 168.513 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg41^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$

$$= 219.734 + 1.0(168.513 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 388.747 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn]$$

$$= 78.5 / \ln[1 + 78.5/281.3]$$

$$= 318.941 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$

$$= 73.5 / \ln[1 + 73.5/1216.0]$$

$$= 1252.390 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 165.119 \cdot 1.0$$

$$= 247.7 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH$$

$$= 17.356 \cdot 318.941 \cdot 218.716$$

$$= 12346.2 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs \cdot (LR + tn) / 2$$

$$= 17.356 \cdot 1252.39 \cdot (298.958 + 78.5) / 2$$

$$= 41833.1 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= (P \cdot Rxs \cdot Rnc) / 2$$

$$= (17.356 \cdot 1252.39 \cdot 281.3) / 2$$

$$= 31176.0 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (41833.1 + 31176.0 + 12346.2) / 17.356$$

$$= 4822.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/(2 \cdot teff))$$

$$= 247.678 / (2 \cdot 4822.688/388.747 - 1252.39/(2 \cdot 73.5))$$

$$= 152.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:



$$\begin{aligned}
 &= 2 \cdot S[t/Rxs] \\
 &= 2 \cdot 165.119[73.5/1252.39] \\
 &= 193.8 \text{ bars}
 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned}
 &= \min(Pmax1, Pmax2) \\
 &= \min(152.018, 193.798) \\
 &= 152.018 \text{ bars}
 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned}
 &= (fN + fS + fY) / AT \\
 &= (12346.218 + 41833.082 + 31176.023) / 388.747 \\
 &= 21.532 \text{ N./mm}^2
 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned}
 &= P \cdot Rxs / (2 \cdot teff) \\
 &= 17.356 \cdot 1252.39 / (2 \cdot 73.5) \\
 &= 14.8 \text{ N./mm}^2
 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned}
 &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\
 &= \max(2 \cdot 21.532 - 14.787, 14.787) \\
 &= 28.3 \text{ N./mm}^2
 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	28.28	N./mm ²
Maximum Allowable Working Pressure	Pmax	152.02	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned}
 &= (Rnc + tn) / Rnc \\
 &= (281.3 + 78.5) / 281.3 \\
 &= 1.279 \text{ For set-in Nozzles}
 \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned}
 &= \pi/2 \cdot (Rn + tn) \\
 &= \pi/2 \cdot (281.3 + 78.5) \\
 &= 565.173 \text{ mm.}
 \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 78.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned}
 &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\
 &= \min(31176 \cdot 1.28, 1.5 \cdot 165.1(170.259 + 0.0), \pi/4 \cdot 17.4 \cdot 281.3^2 \cdot 1.28^2) \\
 &= 17995.076 \text{ Kg}
 \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned}
 &= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T)) \\
 &= 17995.076 / (565.173(0.49 \cdot 7.071 + 0.6 \cdot 20.0 + 0.49 \cdot 0.0)) \\
 &= 20.191 < \text{ or } = \text{ to } 165.119 \text{ Weld Stress Passed}
 \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned}
 &= tn / \ln[1 + tn/Rn] \\
 &= 78.5 / \ln[1 + 78.5/281.3] \\
 &= 318.941 \text{ mm.}
 \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned}
 &= teff / \ln[1 + teff/Reff] \\
 &= 73.5 / \ln[1 + 73.5/1216.0] \\
 &= 1252.390 \text{ mm.}
 \end{aligned}$$



Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 165.119$$

$$= 247.7 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.034 \cdot 318.941 \cdot 218.716$$

$$= 735.7 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (L_R + t_n)/2$$

$$= 1.034 \cdot 1252.39 \cdot (298.958 + 78.5)/2$$

$$= 2492.7 \text{ Kg}$$

Discontinuity Force from External Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc})/2$$

$$= (1.034 \cdot 1252.39 \cdot 281.3)/2$$

$$= 1857.7 \text{ Kg}$$

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N)/P$$

$$= (2492.7 + 1857.7 + 735.7)/1.034$$

$$= 4822.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot A_p / AT - R_{xs} / (2 \cdot t_{eff}))$$

$$= 247.678 / (2 \cdot 4822.688 / 388.747 - 1252.39 / (2 \cdot 73.5))$$

$$= 152.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S / [t / R_{xs}]$$

$$= 2 \cdot 165.119 [73.5 / 1252.39]$$

$$= 193.8 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(152.018, 193.798)$$

$$= 152.018 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / AT$$

$$= (735.681 + 2492.73 + 1857.702) / 388.747$$

$$= 1.283 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / (2 \cdot t_{eff})$$

$$= 1.034 \cdot 1252.39 / (2 \cdot 73.5)$$

$$= 0.9 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 1.283 - 0.881, 0.881)$$

$$= 1.7 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	1.68	N./mm ²
Maximum Allowable External Pressure	Pmax	152.02	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-10 °C

Boiler

(A150NRJ10EQ001)

PV Elite 28 U2 Licensee: SL User

FileName : Boiler

Nozzle Calcs.: M1-24"

Elite v28



HEXAGON

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Nozl: 13 1:50pm Sep 17,2026

The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C

The computed stress ratio is : 0.101

The computed thickness ratio [sr]:

= (tr • E*)/(tnom - ci - cext)

= (2.369 • 1.0)/(25.0 - 1.5 - 0.0)

= 0.101

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -104 °C

User entered Minimum Design Temperature : -10 °C

The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C

The computed stress ratio is : 0.070

The computed thickness ratio [sr]:

= (tr • E*)/(tnom - ci - cext)

= (5.109 • 1.0)/(75.0 - 1.5 - 0.0)

= 0.070

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

The Drop for this Nozzle is : 54.5197 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 429.5197 mm.

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**Input, Nozzle Desc: A1-24"****From: 20**

Pressure for Reinforcement Calculations	P	17.356	bars
Temperature for Internal Pressure	Temp	286	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	286	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.12	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2228.00	mm.
Design Length of Section	L	1904.8029	mm.
Shell Finished (Minimum) Thickness	t	140.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		750.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

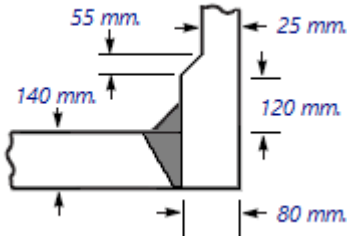
Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	165.12	N./mm ²
Allowable Stress At Ambient	Sna	207.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		22.0315	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	25.0000	mm.
Hub Height of Integral Nozzle	h	120.0000	mm.
Height of Beveled Transition	L'	55.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	38.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: A1-24"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	22.031	in.
Actual Thickness Used in Calculation	0.984	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 562.6 \cdot (\exp(17.356/(165.119 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 2.9726 + 1.5 + 0.0 = 4.4725 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 165.119 \cdot 1.0 \cdot \ln((2 \cdot 23.5 + 562.6) / 562.6) - 0.0$$

$$= 132.474 - 0.0 = 132.474 \text{ bars}$$

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 214.16 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 752.86 mm.

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	9.840 mm
Min. required shell thickness, internal p + c :	13.288 mm.
Min. required shell thickness, external p + c :	9.883 mm.

Nozzle neck minimum required thickness : 9.840 mm.

Minimum nozzle neck thickness 24.994 mm., is greater than required thickness 9.840 mm.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[165.1/165.1, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 80.0 - 1.5$$

$$= 78.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 25.0 - 1.5$$

$$= 23.500 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2231.0/138.5$$

$$= 16.108 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2228.0/2 + 1.5$$



$$= 1115.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{R_{eff} \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1115.5 \cdot 138.5}, 2 \cdot 281.3)$$

$$= 393.061 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 138.5, 0.0] + \sqrt{281.3 \cdot 78.5}$$

$$= 148.600 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 300.0$$

$$= 300.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(138.5 + 0.0)$$

$$= 1108.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$

$$= \min[148.6, 300.0, 1108.0] + 138.5$$

$$= 287.100 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 138.5 + ((0.0 \cdot 1.0) / 393.061)$$

$$= 138.500 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 281.3 + 78.5) / (\sqrt{(2231.0 + 138.5) \cdot 138.5}))$$

$$= 1.119$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 138.5 \cdot 393.061 \cdot \max((1.119/5)^{0.85}, 1)$$

$$= 544.389 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$

$$= 202.923 + 18.361$$

$$= 221.284 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 544.389 + 1.0(221.284 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 766.173 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 78.5 / \ln[1 + 78.5/281.3]$$

$$= 318.941 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 138.5 / \ln[1 + 138.5/1115.5]$$

$$= 1183.399 \text{ mm.}$$



Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned}
 &= 1.5 \cdot S \cdot E \\
 &= 1.5 \cdot 165.119 \cdot 1.0 \\
 &= 247.7 \text{ N./mm}^2
 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned}
 &= P \cdot R_{xn} \cdot L_H \\
 &= 17.356 \cdot 318.941 \cdot 287.1 \\
 &= 16206.4 \text{ Kgf}
 \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned}
 &= P \cdot R_{xs} (L_R + t_n) \\
 &= 17.356 \cdot 1183.399 (393.061 + 78.5) \\
 &= 98766.7 \text{ Kgf}
 \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned}
 &= P \cdot R_{xs} \cdot R_{nc} \\
 &= 17.356 \cdot 1183.399 \cdot 281.3 \\
 &= 58917.3 \text{ Kgf}
 \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned}
 &= (f_S + f_Y + f_N) / P \\
 &= (98766.7 + 58917.3 + 16206.4) / 17.356 \\
 &= 9825.0 \text{ cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned}
 &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\
 &= 247.678 / (2 \cdot 9825.032 / 766.173 - 1183.399 / 138.5) \\
 &= 144.8 \text{ bars}
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned}
 &= S / [t / R_{xs}] \\
 &= 165.119 [138.5 / 1183.399] \\
 &= 193.2 \text{ bars}
 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned}
 &= \min(P_{max1}, P_{max2}) \\
 &= \min(144.81, 193.237) \\
 &= 144.810 \text{ bars}
 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned}
 &= (f_N + f_S + f_Y) / A_T \\
 &= (16206.384 + 98766.688 + 58917.262) / 766.173 \\
 &= 22.258 \text{ N./mm}^2
 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned}
 &= P \cdot R_{xs} / t_{eff} \\
 &= 17.356 \cdot 1183.399 / 138.5 \\
 &= 14.8 \text{ N./mm}^2
 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned}
 &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\
 &= \max(2 \cdot 22.258 - 14.83, 14.83) \\
 &= 29.7 \text{ N./mm}^2
 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	29.68	N./mm ²
Maximum Allowable Working Pressure	Pmax	144.81	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned}
 &= (R_{nc} + t_n) / R_{nc} \\
 &= (281.3 + 78.5) / 281.3 \\
 &= 1.279 \text{ For set-in Nozzles}
 \end{aligned}$$



Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (281.3 + 78.5)$$

$$= 565.173 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 78.500, 6.000) \text{ Passed}$$

Weld Load Value [fwlds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(58917 \cdot 1.28, 1.5 \cdot 165.1(221.284 + 0.0), \pi/4 \cdot 17.4 \cdot 281.3^2 \cdot 1.28^2)$$

$$= 17995.076 \text{ Kg}$$

Weld Stress Value [τ]:

$$= fwlds / (L_t(0.49 \cdot L41T + 0.6 \cdot t_w1 + 0.49 \cdot L43T))$$

$$= 17995.076 / (565.173(0.49 \cdot 7.071 + 0.6 \cdot 38.0 + 0.49 \cdot 0.0))$$

$$= 11.889 < \text{or } = \text{to } 165.119 \text{ Weld Stress Passed}$$

Nozzle/Structural Discontinuity Proximity Warning:

This nozzle's diameter limit intersects a major structural discontinuity. Reduce the diameter limit in the nozzle dialog so that there is no interference or move the nozzle farther away from the intersecting weld.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 78.5 / \ln[1 + 78.5/281.3]$$

$$= 318.941 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 138.5 / \ln[1 + 138.5/1115.5]$$

$$= 1183.399 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 165.119$$

$$= 247.7 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.034 \cdot 318.941 \cdot 287.1$$

$$= 965.7 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs}(L_R + t_n)$$

$$= 1.034 \cdot 1183.399(393.061 + 78.5)$$

$$= 5885.3 \text{ Kg}$$

Discontinuity Force from External Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.034 \cdot 1183.399 \cdot 281.3$$

$$= 3510.7 \text{ Kg}$$

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N)/P$$

$$= (5885.3 + 3510.7 + 965.7)/1.034$$

$$= 9825.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:



$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$
 $= 247.678 / (2 \cdot 9825.032/766.173 - 1183.399/138.5)$
 $= 144.8 \text{ bars}$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$= S[t/Rxs]$
 $= 165.119 [138.5/1183.399]$
 $= 193.2 \text{ bars}$

Maximum Allowable Working Pressure [Pmax]:

$= \min(Pmax1, Pmax2)$
 $= \min(144.81, 193.237)$
 $= 144.810 \text{ bars}$

Average Primary Membrane Stress [σ_{Avg}]:

$= (fN + fS + fY) / AT$
 $= (965.698 + 5885.263 + 3510.735) / 766.173$
 $= 1.326 \text{ N./mm}^2$

General Primary Membrane Stress [σ_{Circ}]:

$= P \cdot Rxs / teff$
 $= 1.034 \cdot 1183.399/138.5$
 $= 0.9 \text{ N./mm}^2$

Maximum Local Primary Membrane Stress [PL]:

$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
 $= \max(2 \cdot 1.326 - 0.884, 0.884)$
 $= 1.8 \text{ N./mm}^2$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	1.77	N./mm ²
Maximum Allowable External Pressure	Pmax	144.81	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : -10 °C
 The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.030

The computed thickness ratio [sr]:

$= (tr \cdot E*) / (tnom - ci - cext)$
 $= (2.369 \cdot 1.0) / (80.0 - 1.5 - 0.0)$
 $= 0.030$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

The Drop for this Nozzle is : 59.7040 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 499.7040 mm.

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**Input, Nozzle Desc: B1-24 "****From: 20**

Pressure for Reinforcement Calculations	P	17.356	bars
Temperature for Internal Pressure	Temp	286	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	286	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.12	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2228.00	mm.
Design Length of Section	L	1904.8029	mm.
Shell Finished (Minimum) Thickness	t	140.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		750.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

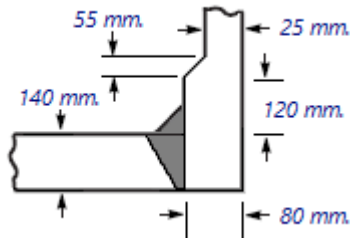
Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	165.12	N./mm ²
Allowable Stress At Ambient	Sna	207.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		22.0315	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	25.0000	mm.
Hub Height of Integral Nozzle	h	120.0000	mm.
Height of Beveled Transition	L'	55.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	38.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: B1-24 "

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	22.031	in.
Actual Thickness Used in Calculation	0.984	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 562.6 \cdot (\exp(17.356/(165.119 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 2.9726 + 1.5 + 0.0 = 4.4725 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 165.119 \cdot 1.0 \cdot \ln((2 \cdot 23.5 + 562.6) / 562.6) - 0.0$$

$$= 132.474 - 0.0 = 132.474 \text{ bars}$$

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 330.97 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 752.86 mm.

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	9.840 mm
Min. required shell thickness, internal p + c :	13.288 mm.
Min. required shell thickness, external p + c :	9.883 mm.

Nozzle neck minimum required thickness : 9.840 mm.

Minimum nozzle neck thickness 24.994 mm., is greater than required thickness 9.840 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	39.2,	Allowable :	165.1 N./mm ²	Passed
Expansion :	0.0,	Allowable :	426.0 N./mm ²	Passed
Occasional :	10.0,	Allowable :	219.6 N./mm ²	Passed
Shear :	18.5,	Allowable :	115.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[165.1/165.1, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 80.0 - 1.5$$

$$= 78.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$



$$= 25.0 - 1.5$$

$$= 23.500 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2231.0/138.5$$

$$= 16.108 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2228.0/2 + 1.5$$

$$= 1115.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1115.5 \cdot 138.5}, 2 \cdot 281.3)$$

$$= 393.061 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 138.5, 0.0] + \sqrt{(281.3 \cdot 78.5)}$$

$$= 148.600 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 300.0$$

$$= 300.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(138.5 + 0.0)$$

$$= 1108.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[148.6, 300.0, 1108.0] + 138.5$$

$$= 287.100 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 138.5 + ((0.0 \cdot 1.0) / 393.061)$$

$$= 138.500 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + teff) \cdot teff}))$$

$$= \min(12, (2 \cdot 281.3 + 78.5) / (\sqrt{(2231.0 + 138.5) \cdot 138.5}))$$

$$= 1.119$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 138.5 \cdot 393.061 \cdot \max((1.119/5)^{0.85}, 1)$$

$$= 544.389 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$

$$= 202.923 + 18.361$$

$$= 221.284 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$

$$= 544.389 + 1.0(221.284 + 0.0) + 0.5 + 0.0 + 0.0$$



$$= 766.173 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 78.5 / \ln[1 + 78.5/281.3]$$

$$= 318.941 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 138.5 / \ln[1 + 138.5/1115.5]$$

$$= 1183.399 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 165.119 \cdot 1.0$$

$$= 247.7 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 17.356 \cdot 318.941 \cdot 287.1$$

$$= 16206.4 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 17.356 \cdot 1183.399 (393.061 + 78.5)$$

$$= 98766.7 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 17.356 \cdot 1183.399 \cdot 281.3$$

$$= 58917.3 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (98766.7 + 58917.3 + 16206.4) / 17.356$$

$$= 9825.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 247.678 / (2 \cdot 9825.032 / 766.173 - 1183.399 / 138.5)$$

$$= 144.8 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 165.119 [138.5 / 1183.399]$$

$$= 193.2 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(144.81, 193.237)$$

$$= 144.810 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (16206.384 + 98766.688 + 58917.262) / 766.173$$

$$= 22.258 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 17.356 \cdot 1183.399 / 138.5$$

$$= 14.8 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 22.258 - 14.83, 14.83)$$

$$= 29.7 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:



Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	29.68	N./mm ²
Maximum Allowable Working Pressure	Pmax	144.81	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (281.3 + 78.5) / 281.3$$

$$= 1.279 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (281.3 + 78.5)$$

$$= 565.173 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 78.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(58917 \cdot 1.28, 1.5 \cdot 165.1(221.284 + 0.0), \pi/4 \cdot 17.4 \cdot 281.3^2 \cdot 1.28^2)$$

$$= 17995.076 \text{ Kg}$$

Weld Stress Value [τ]:

$$= fwelds / (L_T(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 17995.076 / (565.173(0.49 \cdot 7.071 + 0.6 \cdot 38.0 + 0.49 \cdot 0.0))$$

$$= 11.889 < \text{ or } = \text{ to } 165.119 \text{ Weld Stress Passed}$$

Nozzle/Structural Discontinuity Proximity Warning:

This nozzle's diameter limit intersects a major structural discontinuity. Reduce the diameter limit in the nozzle dialog so that there is no interference or move the nozzle farther away from the intersecting weld.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n / R_n]$$

$$= 78.5 / \ln[1 + 78.5 / 281.3]$$

$$= 318.941 \text{ mm.}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff} / R_{eff}]$$

$$= 138.5 / \ln[1 + 138.5 / 1115.5]$$

$$= 1183.399 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 165.119$$

$$= 247.7 \text{ N./mm}^2$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.034 \cdot 318.941 \cdot 287.1$$

$$= 965.7 \text{ Kg}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs}(L_R + t_n)$$

$$= 1.034 \cdot 1183.399(393.061 + 78.5)$$

$$= 5885.3 \text{ Kg}$$



Discontinuity Force from External Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.034 \cdot 1183.399 \cdot 281.3$$

$$= 3510.7 \text{ Kgf}$$

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (5885.3 + 3510.7 + 965.7) / 1.034$$

$$= 9825.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 247.678 / (2 \cdot 9825.032 / 766.173 - 1183.399 / 138.5)$$

$$= 144.8 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [R_{xs}]$$

$$= 165.119 [138.5 / 1183.399]$$

$$= 193.2 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(144.81, 193.237)$$

$$= 144.810 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (965.698 + 5885.263 + 3510.735) / 766.173$$

$$= 1.326 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 1.034 \cdot 1183.399 / 138.5$$

$$= 0.9 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 1.326 - 0.884, 0.884)$$

$$= 1.8 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	1.77	N./mm ²
Maximum Allowable External Pressure	Pmax	144.81	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A

As the Stress Ratio < 0.240, the MDMT is	:	-104 °C
The computed stress ratio is	:	0.030

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (2.369 \cdot 1.0) / (80.0 - 1.5 - 0.0)$$

$$= 0.030$$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

The Drop for this Nozzle is : 59.7040 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 499.7040 mm.

Input Echo, WRC297 Item 1, Description: B1-24 " :



Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2228.000	mm.
Shell Thickness	140.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 207.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 165.12	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	559.600	mm.
Nozzle Thickness	80.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 207.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 165.12	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	17.36	bars
Radial Load	P	-8810.95	Kgf
Circumferential Shear	Vc	8810.95	Kgf
Longitudinal Shear	Vl	6608.18	Kgf
Circumferential Moment	Mc	13739.75	Kg-m.
Longitudinal Moment	Ml	13739.75	Kg-m.
Torsional Moment	Mt	15853.58	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_v/T_v (17.914) is < 20
check the Limitations of WRC bulletin 297.

Warning - The ratio, D_n/T_n (8.995) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 719.6 + 2 \cdot 1.65 \cdot \sqrt{(1184.75(140.0 - 1.5))}$$

$$= 2056.356 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2369.500	mm.
Nozzle Outside Diameter	(d) =	719.600	mm.
Vessel Thickness used	(T) =	138.500	mm.
Nozzle Thickness used	(t) =	78.500	mm.
T / t	=	1.764	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.256$$

Nr/P	=	0.133
Mr/P	=	0.132
M0/P	=	0.063



NØ/P = 0.146
 MrD/Mc = 0.224
 NrDL/Mc = 0.181
 MØd/Mc = 0.138
 NØDL/Mc = 0.045
 MrD/Ml = 0.164
 NrDL/Ml = 0.137
 MØD/Ml = 0.102
 NØDL/Ml = 0.155

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	1	-1	1	-1
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-1	-1	1	1
Outplane Bending (ML)	-5	5	5	-5
Normal Pressure Stress	13	14	13	14
Outplane Stress Summary	8	18	23	9

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-9	9	9	-9
Inplane Pressure Stress	6	6	6	6
Inplane Stress Summary	0	11	21	-4

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	9	18	23	14

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	1	-1	1	-1
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-8	8	8	-8
Outplane Membrane (ML)	0	0	0	0



Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	6	6	6	6

Outplane Stress Summary	0	13	17	-2
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-13	13	13	-13
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	13	14	13	14

Inplane Stress Summary	2	23	32	0
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	1	1
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	2	23	32	4
------------------------	---	----	----	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-1	-1	1	1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	8	6	8

Hoop Stress Summary	5	7	8	10
---------------------	---	---	---	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-5	-5	5	5
Axial Bending (ML)	-17	17	17	-17
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-13	9	32	-13
----------------------	-----	---	----	-----

**Nozzle Stresses (N./mm²):**

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	B1
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	3	3	1	1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au	Al	Bu	B1
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	20	12	32	24

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	D1
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	8	6	8
Hoop Stress Summary	6	8	7	9

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	D1
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	-5	-5	5	5
Axial Bending (Mc)	-24	24	24	-24
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-21	16	39	-20

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	D1
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	1	1	3	3

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	D1
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	27	17	39	30

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int.	Stress Values at
------------------------	------------------



Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	13	14	13	14	13	14	13	14
Circ. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Circ. Q (SUS)	-4	4	6	-6	-10	10	16	-16
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	-1	-1	1	1	0	0	0	0
Long. Q (SUS)	-6	6	12	-12	-7	7	9	-9
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	13.0	14.0	13.0	14.0	13.0	14.0	13.0	14.0
Pm+Pl (SUS)	12.0	13.0	14.0	15.0	12.0	13.0	14.0	15.0
Pm+Pl+Q (Total)	9.2	17.2	20.6	14.1	3.6	23.1	30.1	3.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	14.00	165.12	Passed
Pm+Pl (SUS)	15.00	247.68	Passed
Pm+Pl+Q (TOTAL)	30.07	495.36	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:
Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	8	6	8	6	8	6	8
Circ. Pl (SUS)	-1	-1	1	1	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-5	-5	5	5	-5	-5	5	5
Long. Q (SUS)	-12	12	22	-22	-19	19	29	-29
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	6.0	8.0	6.0	8.0	6.0	8.0	6.0	8.0
Pm+Pl (SUS)	7.0	9.0	8.0	9.0	8.0	10.0	8.0	8.0
Pm+Pl+Q (Total)	19.4	11.0	30.2	23.3	27.3	17.4	37.1	29.3

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	8.00	165.12	Passed
Pm+Pl (SUS)	10.00	247.68	Passed
Pm+Pl+Q (TOTAL)	37.13	495.36	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

Boiler

(A150NRJ10EQ001)

PV Elite 28 U2

Licensee: SL User

FileName : Boiler

Nozzle Calcs.: B1-24 "

Elite

v28



HEXAGON

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Nozl: 15 1:50pm Sep 17,2026

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**Input Echo, Tubesheet Number 1, Description: Tube sheet****Shell Data:****Main Shell Description: shell 1**

Shell Maximum Design Pressure	Psd,max	51.00 bars
Shell Maximum Operating Pressure	Psox,max	32.50 bars
Shell Minimum Operating Pressure	Psox,min	0.00 bars
Shell Thickness	ts	38.0000 mm.
Shell Internal Corrosion Allowance	cas	1.5000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	2224.000 mm.
Shell Temperature for Internal Pressure	Ts	286.00 °C
Shell Material		SA-387 11

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K11789
Shell Allowable Stress at Temperature	Ss	165.12 N./mm ²
Shell Allowable Stress at Ambient		207.00 N./mm ²

Channel Data:**Channel Description: channel shell**

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	191.00 bars
Channel Maximum Operating Pressure	Ptox,max	25.00 bars
Channel Minimum Operating Pressure	Ptox,min	0.00 bars
Channel Thickness	tc	140.0000 mm.
Channel Internal Corrosion Allowance	cac	1.5000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	2228.000 mm.
Channel Design Temperature	TEMPC	310.00 °C
Channel Flange Matl		SA-387 11

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K11789
Channel Allowable Stress at Temperature	Sc	162.80 N./mm ²
Channel Allowable Stress at Ambient		207.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	673
Tube Wall Thickness	et	2.5000 mm.
Tube Outside Diameter	D	30.8000 mm.
Total Straight Tube Length	Lt	5000.00 mm.
Straight Tube Length (bet. inner tubsht faces) L		4900.00 mm.
Design Temperature of the Tubes		465.00 °C
Tube Material		SA-213 T12
Tube Material UNS Number		K11562
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		101.80 N./mm ²
Tube Allowable Stress At Ambient		147.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	152.60 N./mm ²
Tube Pitch (Center to Center Spacing)	P	75.0000 mm.
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm.
Groove Weld Leg	ag	2.1100 mm.
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		4.21.2
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	1014.580 mm.
Largest Center-to-Center Tube Distance	Ul	800.0000 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	0.0000 mm.
Area of Tube Layout	Ap	30933.3 cm ²



Perimeter of Tube Layout Cp 6483.0 mm.

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature T 465.00 °C
Tubesheet Material SA-336 F11

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

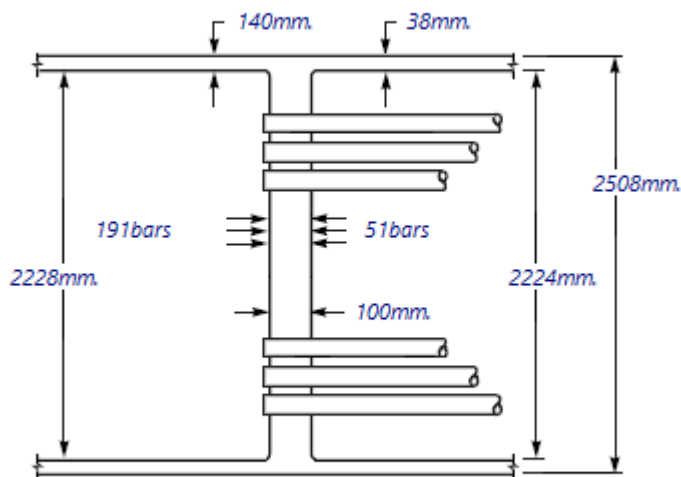
Tubesheet Material UNS Number K11572
Tubesheet Allowable Stress at Temperature S 121.46 N./mm²
Tubesheet Allowable Stress at Ambient Tt 207.00 N./mm²
Thickness of Tubesheet h 100.0000 mm.
Tubesheet Corr. Allowance (Shell side) Cats 1.5000 mm.
Tubesheet Corr. Allowance (Channel side) Catc 1.5000 mm.
Tubesheet Outside Diameter A 2508.000 mm.

Area of the Untubed Lanes AL 0.0 cm²

Junction Stress Reduction option Perform Plastic Calculation
Perform Differential Pressure Design No
Run Multiple Load Cases Yes

Tubesheet Integral with Both
Tubesheet Extended as Flange No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Material Group C

Elastic Mod. at Design Temperature 286.0 °C 0.18730E+09 KPa.
Elastic Mod. at Ambient Temperature 21.1 °C 0.20408E+09 KPa.

Channel - TM-1 Material Group C

Elastic Mod. at Design Temperature 310.0 °C 0.18581E+09 KPa.



Elastic Mod. at Ambient Temperature 21.1 °C 0.20408E+09 KPa.

Tubes - TM-1 Material Group C

Elastic Mod. at Tubsht. Design Temp. 465.0 °C 0.17270E+09 KPa.

Elastic Mod. at Ambient Temperature 21.1 °C 0.20408E+09 KPa.

TubeSheet - TM-1 Material Group C

Elastic Mod. at Design Temperature 465.0 °C 0.17270E+09 KPa.

Elastic Mod. at Ambient Temperature 21.1 °C 0.20408E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:

= $0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i$ Equation 4.3.1

= $0.5 \cdot 25.8 \cdot (\exp(191.0/(101.8 \cdot 1.0)) - 1) + 0$

= 2.6624 + 0

= 2.6624 mm.

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

= $L / \sqrt{R_o \cdot t}$

= $5914.58 / \sqrt{15.4 \cdot 2.5}$

= 953.221

Compute the Factor [Ch]:

Since $M_x > 2 \cdot (D_o/t)^{0.94}$, $Ch = 0.55 / (D_o/t)$

= $0.55 / (30.8/2.5)$

= 0.0446

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

= $1.6 \cdot Ch \cdot E_y(t / D_o)$

= $1.6 \cdot 0.045 \cdot 0.17270E+09(2.5/30.8)$

= 1001.332 N./mm²

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

= Fhe / F_y

= $1001.3/152.6$

= 6.562

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

= $\sigma_{ys} / \sigma_{uts}$

= $152.6/393.7$

= 0.388

Ratio [AE]:

= Fhe / E

= $1001.3/17270E+09$

= 0.005798

Factor [m1]:

= $(\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$

= $(\ln(0.388) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$

= 0.206

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

= $\sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m1$

= $152.6(1 + 0.002) / \ln(1 + 0.002) 0.206255$

= 551.059 N./mm²

Curve Fitting Exponent for, Ferritic Steel [m2]:

= $m2f(m2f2 - R)$

= $0.6(1.0 - 0.388)$

= 0.367



Curve Fitting Constant [A2]:

$$= (\text{outs} \cdot \exp(m2)) / (m2 \cdot m2)$$

$$= (393.7 \cdot \exp(0.367)) / (0.367 \cdot 0.367)$$

$$= 821.219 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.3881.5 - 0.5 \cdot 0.3882.5 - 0.3883.5$$

$$= 0.279$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	196	-0.685	0.837E-05	-0.156E-05	0.142E-04	0.113E-04	0.305E+07	0.588E-01
2	98	-3.611	0.581E-06	0.579E-06	0.430E-05	-0.428E-05	0.568E+08	0.407E-02
3	147	-2.148	0.277E-05	0.256E-05	0.863E-05	-0.766E-05	0.145E+08	0.435E-02
4	123	-2.880	0.137E-05	0.135E-05	0.631E-05	-0.616E-05	0.290E+08	0.156E-02
5	135	-2.514	0.198E-05	0.191E-05	0.743E-05	-0.705E-05	0.206E+08	0.766E-03
6	129	-2.697	0.166E-05	0.162E-05	0.686E-05	-0.662E-05	0.245E+08	0.519E-03
7	132	-2.605	0.181E-05	0.176E-05	0.714E-05	-0.684E-05	0.225E+08	0.893E-04
8	130	-2.651	0.173E-05	0.169E-05	0.700E-05	-0.673E-05	0.234E+08	0.223E-03
9	131	-2.628	0.177E-05	0.172E-05	0.707E-05	-0.679E-05	0.229E+08	0.689E-04
10	131	-2.617	0.179E-05	0.174E-05	0.711E-05	-0.682E-05	0.227E+08	0.965E-05
11	131	-2.623	0.178E-05	0.173E-05	0.709E-05	-0.680E-05	0.228E+08	0.298E-04
12	131	-2.620	0.179E-05	0.174E-05	0.710E-05	-0.681E-05	0.228E+08	0.101E-04
13	131	-2.618	0.179E-05	0.174E-05	0.710E-05	-0.681E-05	0.227E+08	0.228E-06
14	131	-2.618	0.179E-05	0.174E-05	0.711E-05	-0.681E-05	0.227E+08	0.471E-05
15	131	-2.618	0.179E-05	0.174E-05	0.710E-05	-0.681E-05	0.227E+08	0.224E-05
16	131	-2.618	0.179E-05	0.174E-05	0.710E-05	-0.681E-05	0.227E+08	0.101E-05

Factor of Safety [FS]:

Since Fic > 0.55 Fy and Fic < Fy

$$= 2.407 - 0.741 \cdot \text{Fic} / \text{Fy}$$

$$= 2.407 - 0.741 \cdot 131.82/152.6$$

$$= 1.767$$

Allowable Circumferential Compressive Stress [Fha]:

$$= \text{Fic} / \text{FS}$$

$$= 131.82 / 1.767$$

$$= 74.605 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot \text{Fha} (t / \text{Do})$$

$$= 2 \cdot 746.003 (2.5/30.8)$$

$$= 121.104 \text{ bars}$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	2.6624 mm.
Allowable Internal Pressure at Corroded thickness	180.32 bars
Required Internal Design Pressure	191.00 bars
Allowable External Pressure at Corroded thickness	121.10 bars
Required External Design Pressure	51.00 bars
Required Thickness due to Shell Side pressure	1.4670 mm.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:



Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned}
 &= 2 \cdot r_o + dt \\
 &= 2 \cdot 1014.58 + 30.8 \\
 &= 2059.960 \text{ mm.}
 \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned}
 &= (p - dt) / p \\
 &= (75.0 - 30.8) / 75.0 \\
 &= 0.589
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$$\begin{aligned}
 &= D_s / D_o \text{ (Configurations a, b, c)} \\
 &= 2224.0 / 2059.9602 \\
 &= 1.0796
 \end{aligned}$$

Ratio [ρ_c]:

$$\begin{aligned}
 &= D_c / D_o \text{ (Configurations a, e, f)} \\
 &= 2228.0 / 2059.9602 \\
 &= 1.0816
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$\begin{aligned}
 &= l_{tx} / h \\
 &= 0 / 100.0 \\
 &= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}
 \end{aligned}$$

Effective tube hole Diameter [d^*]:

$$\begin{aligned}
 &= \max(dt - 2t(E_t/E) \cdot (S_t/S) \cdot (\rho); dt - 2t) \\
 &= \max(30.8 - 2 \cdot 2.5 (0.17270E+09 / 1.7270E+09) \cdot \\
 &\quad (101/121) \cdot (0); 30.8 - 2 \cdot 2.5) \\
 &= 30.8000 \text{ mm.}
 \end{aligned}$$

Effective tube pitch [p^*]:

$$\begin{aligned}
 &= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) } \\
 &= 75.0 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 2059.96 \cdot 75.0) / (\pi \cdot 2059.96^2)) } \\
 &= 75.0000 \text{ mm.}
 \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned}
 &= (p^* - d^*) / p^* \\
 &= (75.0 - 30.8) / 75.0 \\
 &= 0.5893
 \end{aligned}$$

$h/p = 1.333333$; $\mu^* = 0.589333$; $E^* = 118103176 \text{ KPa.}$

$E^*/E = 0.683864$; $v^* = 0.297725$

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned}
 &= ((12(1 - v_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s } \\
 &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (2224.0 + 38.0) 38.0 } \\
 &= 0.0062 \text{ 1/mm.}
 \end{aligned}$$

Shell coefficient [k_s]:

$$\begin{aligned}
 &= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - v_s^2)) \\
 &= 0.006 \cdot 1873002 \cdot 38.0^3 / (6(1 - 0.3^2)) \\
 &= 116711568.0000 \text{ bars} \cdot \text{mm.}^2
 \end{aligned}$$

Shell coefficient [λ_s]:

$$\begin{aligned}
 &= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2) \\
 &= 6(2224.0 + 38.0) \cdot 0.11671E+09 / 100.0^3 \cdot (1 + 100.0 \cdot 0.006 + (100.0 \cdot 0.006)^2 / 2) \\
 &= 2870636.5 \text{ bars}
 \end{aligned}$$

Shell coefficient [δ_s]:

$$\begin{aligned}
 &= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s))) \\
 &= 2224.0^2 / (4 \cdot 0.18730E+09 \cdot 38.0) \cdot (1 - 2224.0 \cdot 0.3 / (2(2224.0 + 38.0)))
 \end{aligned}$$



$$= 0.14810E+00 \text{ mm./N./mm}^2$$

Parameter [ws]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.0796 \cdot 0.11671E+09 \cdot 0.0062 \cdot 0.148104 (1 + 100.0 \cdot 0.0062)$$

$$= 18746.6914 \text{ mm.}^2$$

Channel coefficient [βc]:

$$= \left(\left(12(1 - \nu_c^2) \right)^{1/4} / \sqrt{\left(\left(D_c + t_c \right) t_c \right)} \right)$$

$$= \left(\left(12(1 - 0.3^2) \right)^{1/4} / \sqrt{\left(\left(2228.0 + 140.0 \right) 140.0 \right)} \right)$$

$$= 0.0032 \text{ 1/mm.}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2))$$

$$= 0.003 \cdot 1858110 \cdot 140.0^3 / (6(1 - 0.3^2))$$

$$= 2948247040.0000 \text{ bars} \cdot \text{mm.}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(2228.0 + 140.0 / \cos(0)) \cdot 0.29482E+1 / 100.0^3 \cdot (1.0 + 100.0 \cdot 0.003 \cdot (1.0 + 1.0) / 2 + (100.0 \cdot 0.003)^2 / 2 \cdot 1.0)$$

$$= 57201456.0 \text{ bars}$$

Channel coefficient [δc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 2228.0^2 / (4 \cdot 0.18581E+09 \cdot 140.0) \cdot (1 - 2228.0 \cdot 0.3 / (2(2228.0 + 140.0)))$$

$$= 0.40971E-01 \text{ mm./N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ωc]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.082 \cdot 0.29482E+1 [0.040971 (1.0 + 0.003 \cdot 1.0 \cdot 100.0) - 0 (1.0 / 100.0 + 1.0 \cdot 0.003 / 2)]$$

$$= 54272.7109 \text{ mm.}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2508.0 / 2059.9602$$

$$= 1.2175$$

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.3) / 1.1810E+09 \cdot (2870636 + 57201456 + 0.17270E+09 \cdot \ln(1.22))$$

$$= 35.9227$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(\rho_s - 1) \cdot (\rho_s^2 + 1) \cdot P_s - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot P_t]$$

$$= 2059.96^2 / 16 \cdot [(1.08 - 1) \cdot (1.08^2 + 1) \cdot 51.0 - (1.082 - 1) \cdot (1.082^2 + 1) \cdot 191.0]$$

$$= -6633535.0000 \text{ bars} \cdot \text{mm.}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= -6633535.0 + 54272.711 \cdot 191.0 - 18746.691 \cdot 51.0$$

$$= 2776471.5000 \text{ bars} \cdot \text{mm.}^2$$

**ASME Section VIII-2 Part 4.18.7.4 Step 7:**

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M \bullet) - D_o^2/32 \bullet F (P_s - P_t)) / (1 + F)$$

$$= ((2776471) - 2059.96^2/32 \bullet 35.923 (51.0 - 191.0)) / (1 + 35.92)$$

$$= 18137420.0000 \text{ bars} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2/64 \bullet (3 + \nu \bullet) \bullet (P_s - P_t)$$

$$= 18137420 + 2059.96^2/64 \bullet (3 + 0.298) \bullet (51.0 - 191.0)$$

$$= -12473765.0000 \text{ bars} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| M_p | ; | M_o |)$$

$$= \max(| 18137420 | ; | -0.1E+08 |)$$

$$= 18137420.0000 \text{ bars} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \bullet M / (\mu \bullet (h - h_g')^2)$$

$$= 6 \bullet 18137420 / (0.5893 (100.0 - 0)^2)$$

$$= 1846.6771 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \bullet S$$

$$= 2 \bullet 121.46$$

$$= 242.92 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \bullet M / (\mu \bullet (h - h_g')^2)$$

$$= 6 \bullet 15890961 / (0.5893 (258.2836 - 0)^2)$$

$$= 242.5336 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 258.2836 + 0 + 0$$

$$= 258.2836 \text{ mm.}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB} ; H_{reqS})$$

$$= \max(258.2836 ; 116.6589)$$

$$= 258.2836 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:Absolute pressure differential [$| P_s - P_t |$]:

$$= | P_s - P_t |$$

$$= | 51.0 - 191.0 |$$

$$= 140.000 \text{ bars}$$

Shear Stress check [τ_{limit}]:

$$= 4 \bullet \mu \bullet h / D_o \bullet \min(0.8 \bullet S ; 0.533 \bullet S_y)$$

$$= 4 \bullet 0.589 \bullet 100.0 / 2059.96 \bullet \min(0.8 \bullet 121.461 ; 0.533 \bullet 215.405)$$

$$= 11.120 \text{ N./mm}^2$$

Note: As $P_s - P_t > \tau_{limit}$ & A_p, C_p are given, the more accurate formula for Tau is used.

Average shear stress at the outer edge of perforated region [τ]:

$$= (1 / (4 \bullet \mu)) \bullet (1 / h \{ (4 \bullet A_p) / C_p \}) \bullet | P_s - P_t |$$

$$= (1 / (4 \bullet 0.589)) \bullet (1 / 100.0 \{ (4 \bullet 30933.3) / 6483.0 \}) \bullet | 51.0 - 191.0 |$$

$$= 113.3562 \text{ N./mm}^2$$

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \bullet \mu \bullet \min(0.8 \bullet S ; 0.533 \bullet S_y)) \bullet (4 \bullet A_p / C_p) \bullet | P_s - P_t | + C_{ats} + C_{atc}$$

$$= 1 / (4 \bullet 0.589 \bullet 97.169) \bullet (4 \bullet 30933.3 / 6482.96) \bullet | 51.0 - 191.0 | + 0$$

$$= 116.6589 \text{ mm.}$$

**ASME Section VIII-2 Part 4.18.7.4 Step 10:**

Note: For a given Shell thickness of: 38.000mm.

Min. Shell len. adjacent to the tubesheet is: 523.277 mm.

Axial Shell Membrane Stress [σ_s, m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 2224.0^2 / (4 \cdot 38.0 (2224.0 + 38.0)) \cdot 51.0$$

$$= 73.3717 \text{ N./mm}^2$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E^* \cdot$$

$$Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 2623840 / 38.0^2 \cdot [0.006 \cdot 0.7531 + 6(1 - 0.298) / 11810E+09 \cdot$$

$$2059.96 / 100.0^3 \cdot (1 + 100.0 \cdot 0.006 / 2) \cdot (18137420 + 2059.96^2 / 32 \cdot$$

$$(51.0 - 191.0))]$$

$$= 27.478 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= | \sigma_{s,m} | + | \sigma_{s,b} | \text{ (Should be } \leq 1.5 \cdot S_s)$$

$$= 100.850 \text{ N./mm}^2 \text{ (Should be } \leq 247.68)$$

Note: For a given Channel thickness of: 140.000 mm.

Min. Channel len. adjacent to the tubesheet is: 1005.296mm.

Axial Channel Membrane Stress [σ_c, m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c)) \cdot \cos(\theta) \cdot P_t$$

$$= 2228.0^2 / (4 \cdot 140.0 (2228.0 + 140.0)) \cdot \cos(0) \cdot 191.0$$

$$= 71.502 \text{ N./mm}^2$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t -$$

$$6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (C_m \theta +$$

$$h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 66280748 / 140.0^2 \cdot [(0.040971 \cdot 1.0 \cdot 0.003 - 0 \cdot 1.0) \cdot 191.0 -$$

$$6((1 - 0.298) / 11810E+09) \cdot 2059.96 / 100.0^3 \cdot (1.0 +$$

$$0.003 \cdot 100.0 / 2 \cdot 1.0) \cdot (18137420 + 2059.96^2 / 32 \cdot (51.0 - 191.0))]$$

$$= 551.437 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 622.940 \text{ N./mm}^2 \text{ (Should be } \leq 244.20)$$

Step 11, The Cylinder-to-Tubesheet Junction is Overstressed.

Note: As Channel stress is > SPS,c, 622.94 > 488.83

simplified Elastic-Plastic calcs. cannot be performed. Reconsider the design.

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (30.8 - 2.5) 2.5$$

$$= 2.223 \text{ cm}^2$$

Tube Weld Size Results per VIII-2 4.21.2:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.5 (30.8 - 2.5) 101.801$$

$$= 2307.282 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.11 (30.8 + 0.67 \cdot 2.11) 101.8$$

$$= 1884.1536 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$



$$= 101.801/101.801$$

$$= 1.000$$

Design Strength Ratio [fd]:
= 1 for full strength welds

Fillet Weld Strength [Ff]:
= $0.55 \cdot \pi \cdot af \cdot (do + 0.67 \cdot af) Sw$ (but not $> Ft$)
= $0.55 \cdot \pi \cdot 2.11 \cdot (30.8 + 0.67 \cdot 2.11) 101.8$
= 1219.1584 Kgf

Minimum Length of Weld [ar]:
= $2 \cdot (\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t)fw} + fd) - 0.75 \cdot do$
= $2 \cdot (\sqrt{(0.75 \cdot 30.8)^2 + 1.07 \cdot 2.5 \cdot (30.8 - 2.5) 1.0 \cdot 1.0} - 0.75 \cdot 30.8)$
= 3.169 mm.

Minimum Required Fillet Weld Leg [afr]: 1.5843 mm.
Minimum Required Groove Weld Leg [agr]: 1.5843 mm.

Tube-Tubesheet Jt allowable, 2307.28 is $\geq$ tube strength 2307.28 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
<i>Tubesheet bend. stress</i>	1846.7	$\leq 242.9 \text{ N./mm}^2$	FAIL
<i>Tubesheet shear stress</i>	113.4	$\leq 97.2 \text{ N./mm}^2$	FAIL
Stress in Shell at Tubesheet	100.8	$\leq 247.7 \text{ N./mm}^2$	Ok
<i>Stress in Channel at Tubesheet</i>	622.9	$\leq 244.2 \text{ N./mm}^2$	FAIL

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
<i>Tubesheet Thickness :</i>	258.284	100.000	Fail
Tube-Tubesheet Fillet Weld Leg :	1.584	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.584	2.110	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend	Stresses Allwd	Shear	Allwd	Case Type	Pass/ Fail
D1uc	301.465	...	2513	243	155	97	Ps+Pt D1	Fail
D2uc	160.081	...	666	243	41	97	Ps+Pt D2	Fail
D3uc	258.284	...	1847	243	113	97	Ps+Pt D3	Fail
D1c	305.745	...	2673	243	159	97	Ps+Pt-c D1	Fail
D2c	162.824	...	709	243	43	97	Ps+Pt-c D2	Fail
D3c	260.653	...	1964	243	117	97	Ps+Pt-c D3	Fail
Max:	305.7448	...	mm.	11.004		1.641	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:
Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psdx,min, Ptox,max etc. per the
Load Case Tables

[c]:
With or Without Corrosion Allowance

[D1, D2, D3]:
Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

--	Integral Cyl.	Reqd. Thk.	Shell	Stress	Channel	Stress	Pass
Case#	Shell	Channel	Stress	Allwd	Stress	Allwd	Fail
D1uc	...	...	303	495	792	244	Fail
D2uc	...	...	403	495	169	244	Ok
D3uc	...	...	101	248	623	244	Fail
D1c	...	...	312	495	818	244	Fail
D2c	...	...	419	495	176	244	Ok
D3c	...	...	108	248	642	244	Fail
Max	...	...	0.846		3.349		

Note:
Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	305.745	100.000	Fail
Tube thickness :	2.662	2.500	Fail
Tube-tubesheet fillet weld leg :	1.584	2.110	Ok
Tube-tubesheet groove weld leg :	1.584	2.110	Ok

Given Shell Thickness: 38.0000 mm.
Given Channel Thickness: 140.0000 mm.

Min Shell length of thk, (38.000) adj. to tubesheet: 523.277 mm.

Min Channel length of thk, (140.000) adj. to tubesheet: 1005.296 mm.

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	17.356	1.000	17.475	1.000
Tubesheet Shear Stress	116.407	1.000	116.407	1.000
Shell Stress (Axial, Junction)	116.407	0.767	17.475	0.580
Tubesheet-Channel Junction	17.356	0.304	17.475	0.247
Tube Pressure Stress	180.319	1.000	121.103	1.000
Minimum MAWP	17.356		17.475	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	31.528	1.000	31.774	1.000
Tubesheet Shear Stress	204.062	1.000	204.062	1.000
Shell Stress (Axial, Junction)	204.062	0.515	31.774	0.807
Tubesheet-Channel Junction	31.528	0.421	31.774	0.339
Tube Pressure Stress	260.375	1.000	172.504	1.000



Minimum MAPnc

31.528

31.774

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure 3.10:

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness}))$$

$$= \max(100.0/4, \min(100.0, 38.0))$$

$$= 38.000 \text{ mm.}$$

MDMT reduction ratio:

$$= \min(\text{ps}/\text{Shellside MAPnc}, 1)$$

$$= \min(51.0/31.77, 1)$$

$$= 1.000$$

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : 25 °C
 User entered Minimum Design Temperature : -10 °C
 The Curve for this Material is : A
 From Figure 3.8M- with PWHT, the MDMT is : 25 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

Warning: Impact Testing Is Mandatory!

Channel Side MDMT calculation:

Governing thickness on the shell side per figure 3.10:

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness}))$$

$$= \max(100.0/4, \min(100.0, 140.0))$$

$$= 100.000 \text{ mm.}$$

MDMT reduction ratio:

$$= \min(\text{pt}/\text{Tubeside MAPnc}, 1)$$

$$= \min(191.0/31.53, 1)$$

$$= 1.000$$

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : 42 °C
 User entered Minimum Design Temperature : -10 °C
 The Curve for this Material is : A
 From Figure 3.8M- with PWHT, the MDMT is : 42 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

Warning: Impact Testing Is Mandatory!

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PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: